

GENIFY

LECTURE 12

12-August-2025

TOPICS COVERED:

RIS

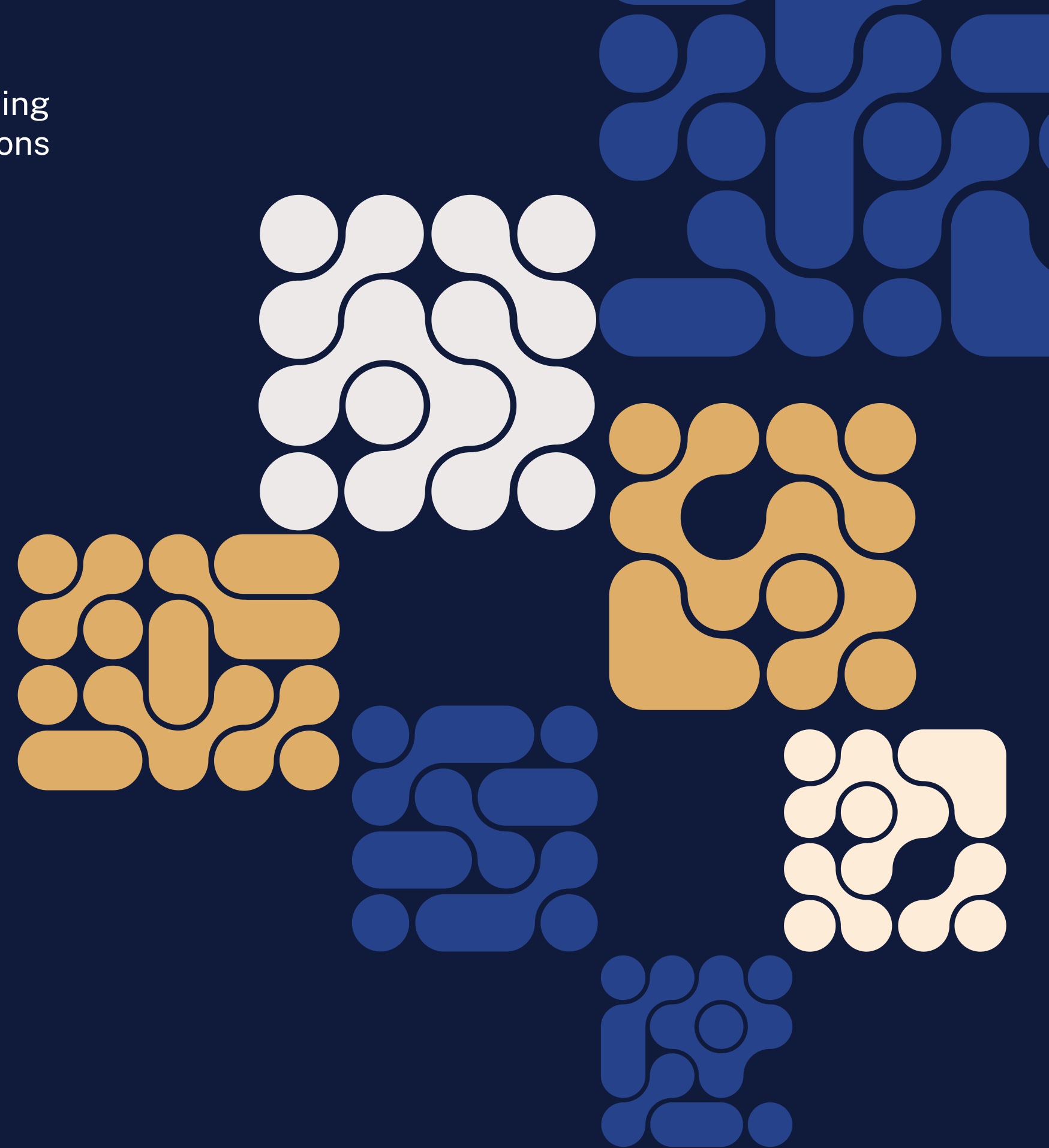
- RECURRENT NEURAL NETWORKS (RNN)

Presenter

Shahzaib Kashif

Co-Presenter

Shayan Hassan Baig



RECAP QUIZ TIME

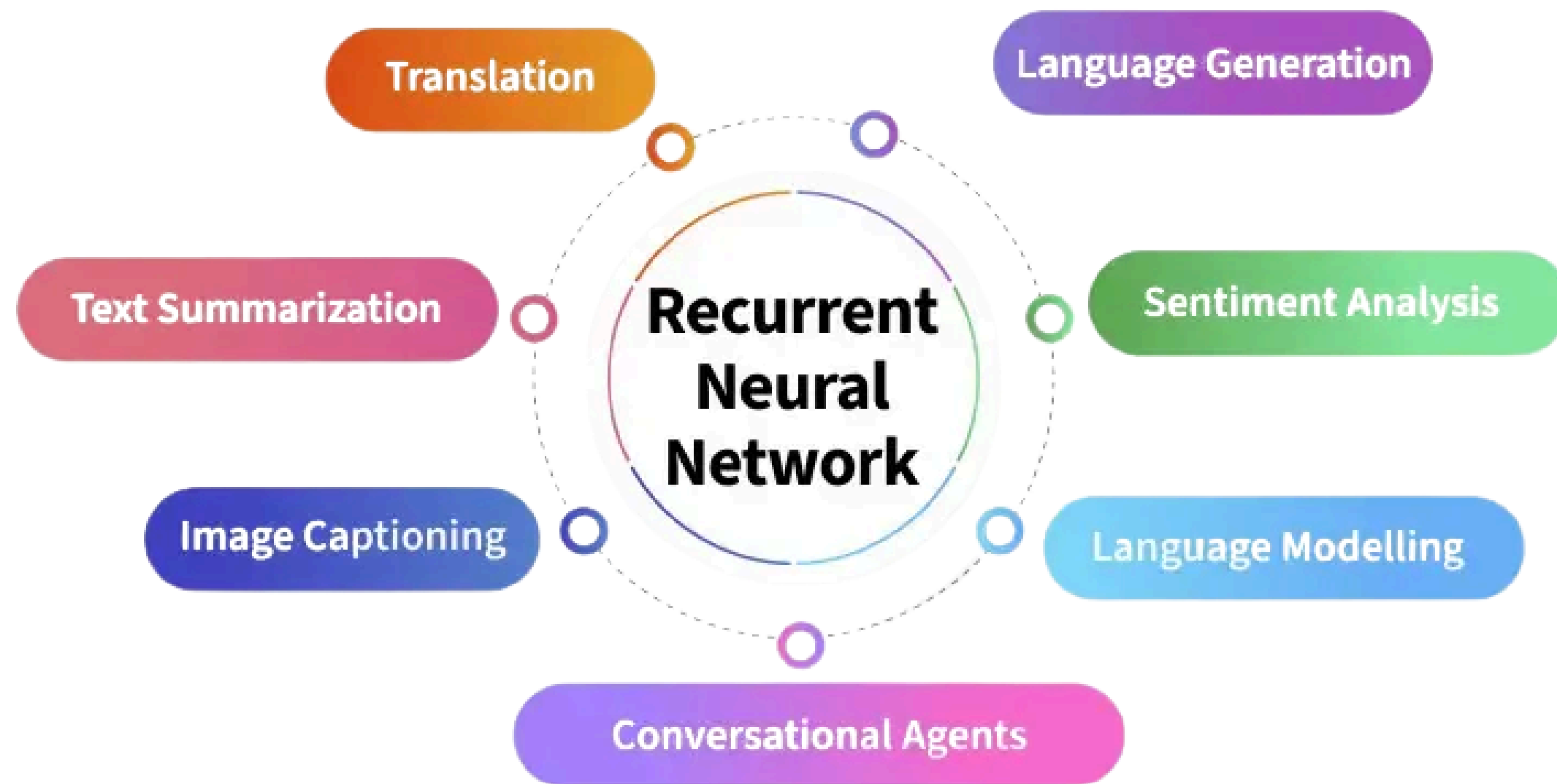
ASSIGNMENT DISCUSSION



RECURRENT NEURAL NETWORKS

Why do we even need RNNs

- ANNs work in Feed forward manner.
- There can be cases where we'd need to remember some context or previous layer's information tied with the next layer.



Why do we even need RNNs

New Message

Recipients

Subject

Hello, h

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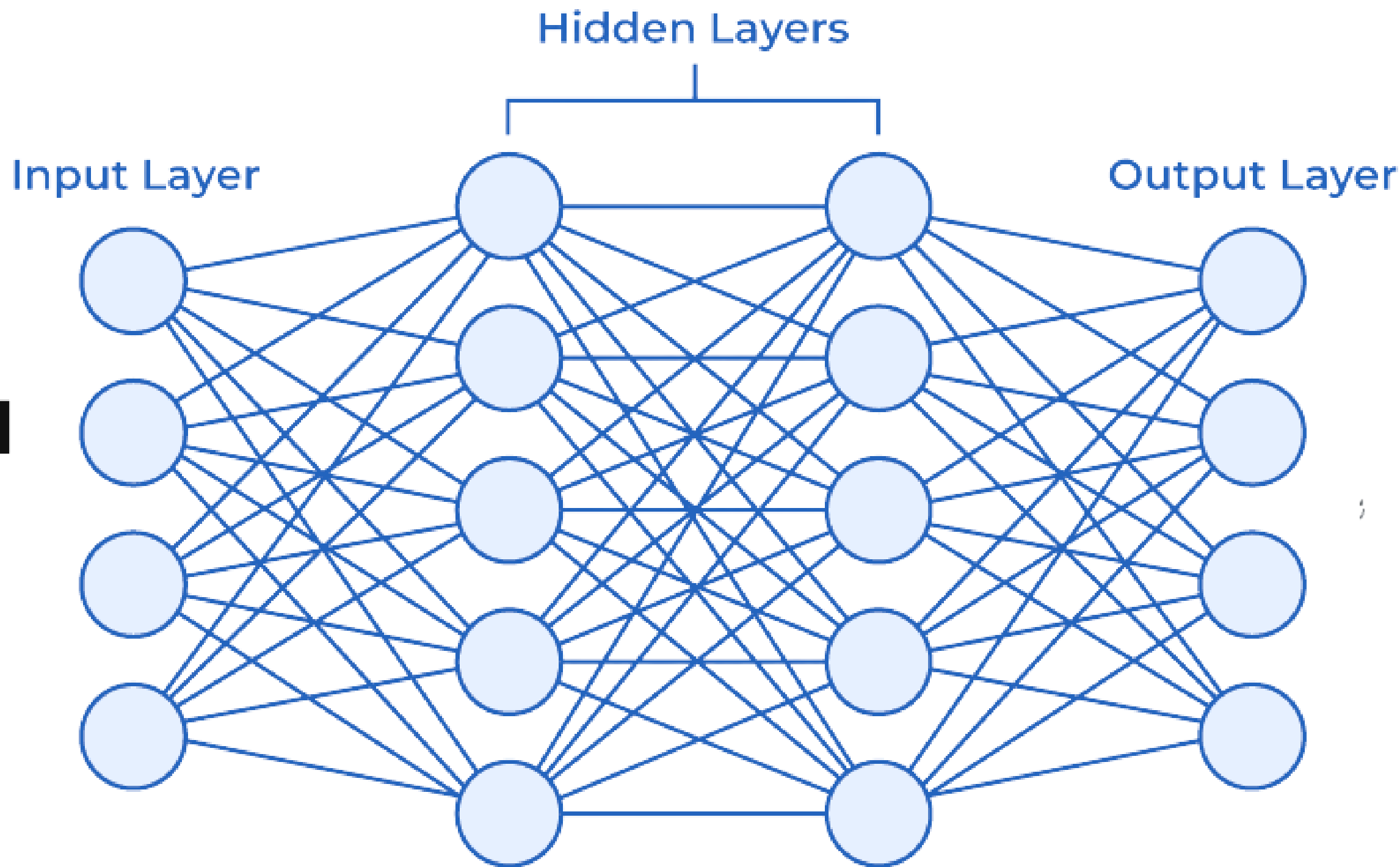
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Problems in ANN

Problem 01: Fixed Number of Input and Output Neurons

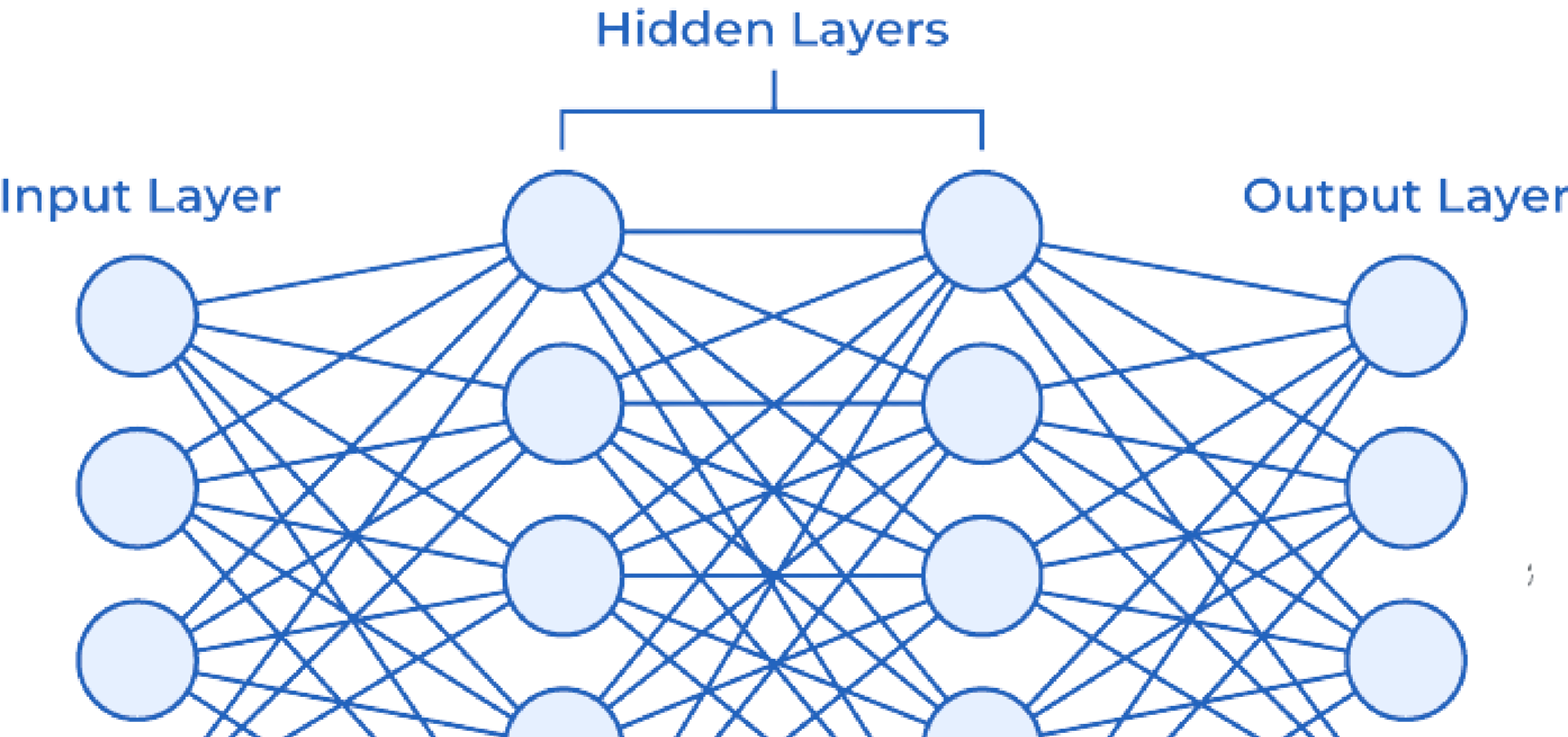


Hello, Good morning

ہیلو، صبح بخیر

Problems in ANN

Problem 01: Fixed Number of Input and Output Neurons



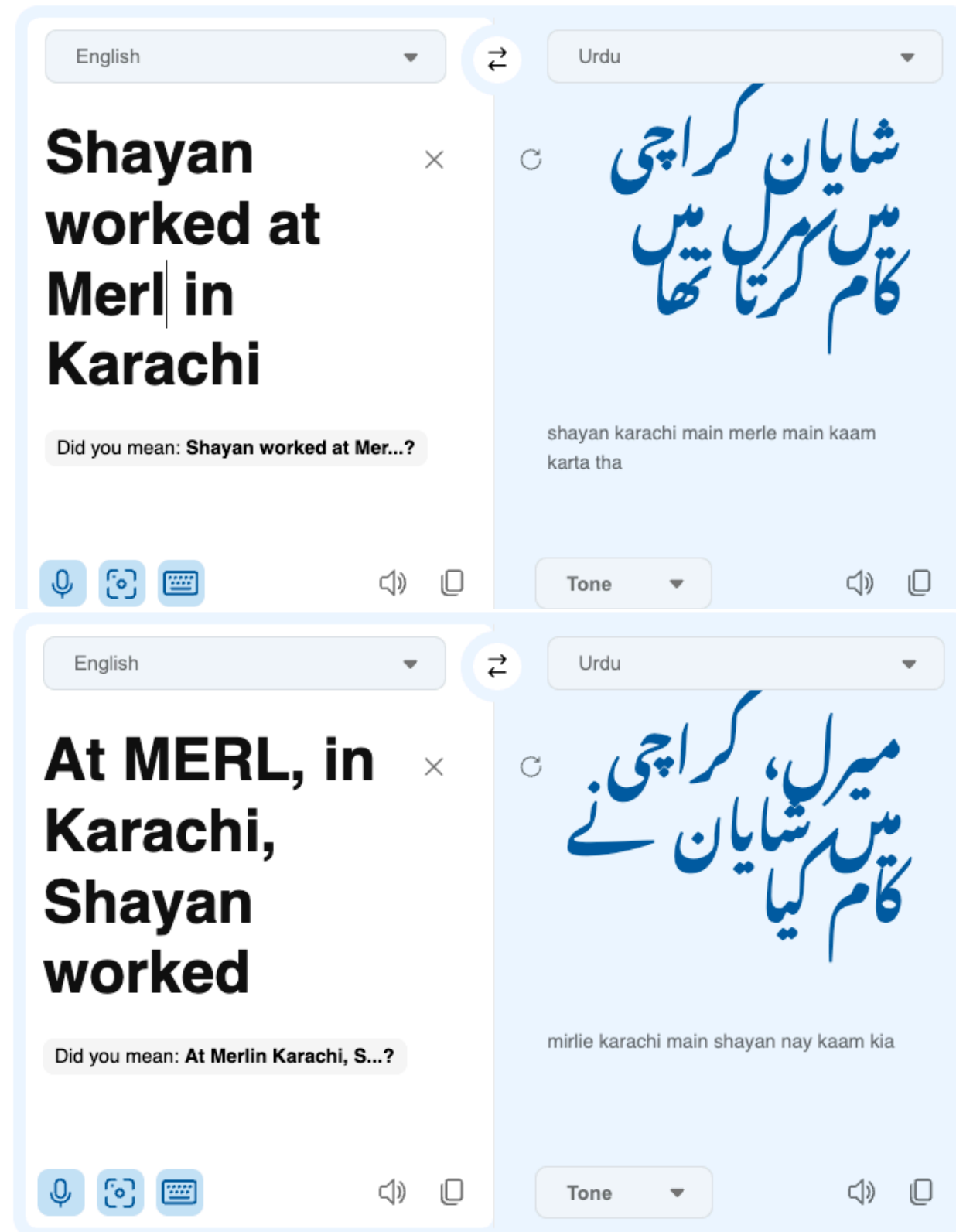
Hello, Good morning

ہیلو، صبح بخیر

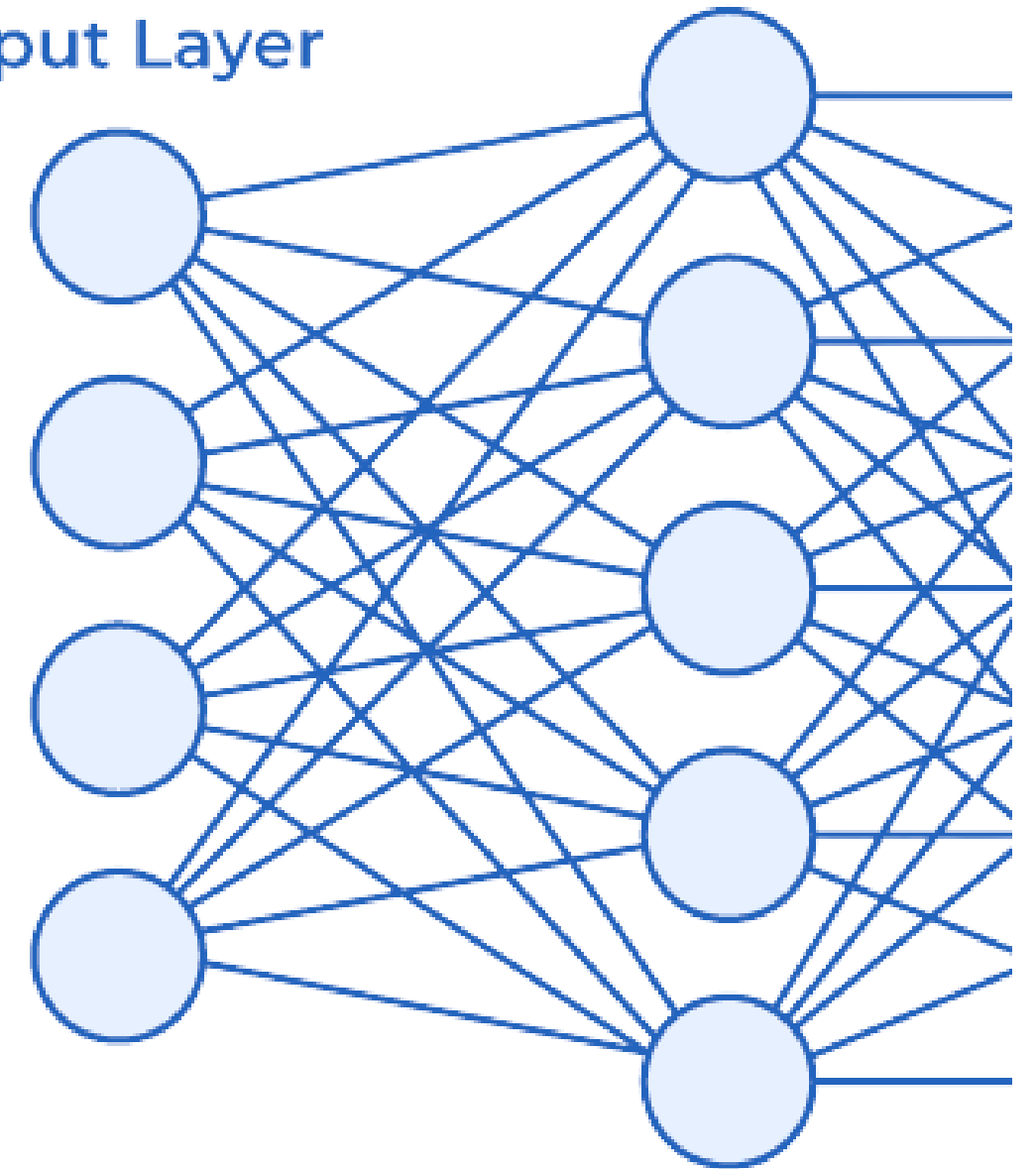
Kelly	worked	at	Google	in	San	Francisco	0	0	0
Kelly	is	a	Software	Engineer	at	Google	in	San	Francisco
Kelly	is	very	talented	0	0	0	0	0	0

Problems in ANN

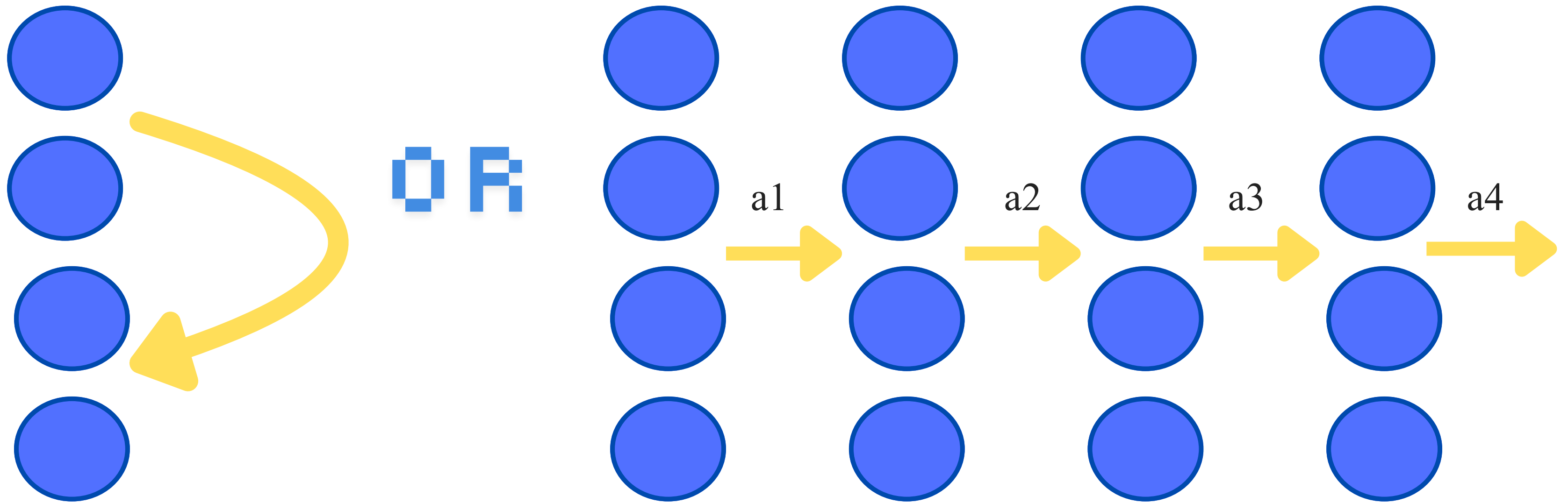
Problem 02: Does not take input sequence into consideration



Input Layer



RNNs



RNNs: Named Entity Recognition

Shayan worked at MERL in Karachi

RNNs: Named Entity Recognition

PERSON



Shayan worked at MERL in Karachi

RNNs: Named Entity Recognition



RNNs: Named Entity Recognition



RNNs: Named Entity Recognition



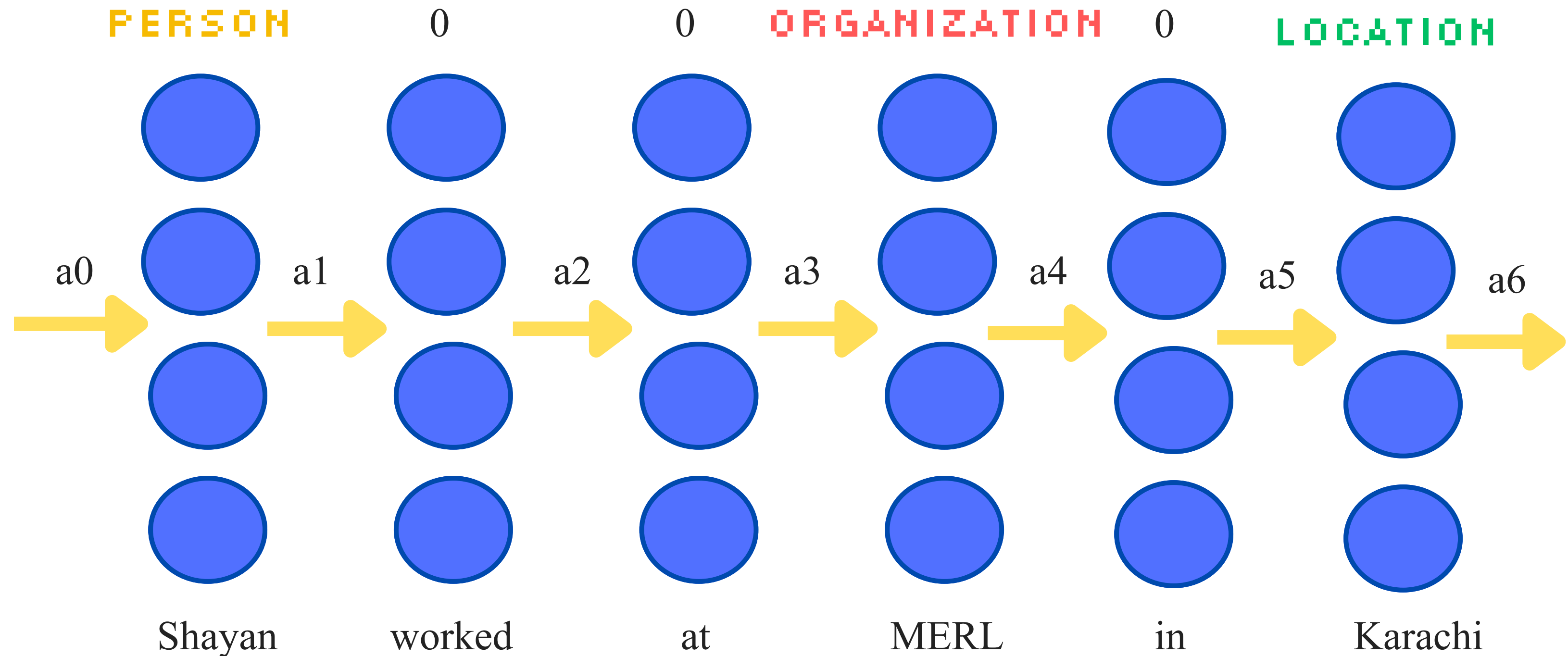
Diagram illustrating Named Entity Recognition (NER) for the sentence "Hey, show me the best places to visit in Karachi during Jan 2022". The entities are categorized as follows:

- LOCATION** (Green): Karachi
- DATE** (Blue): Jan 2022

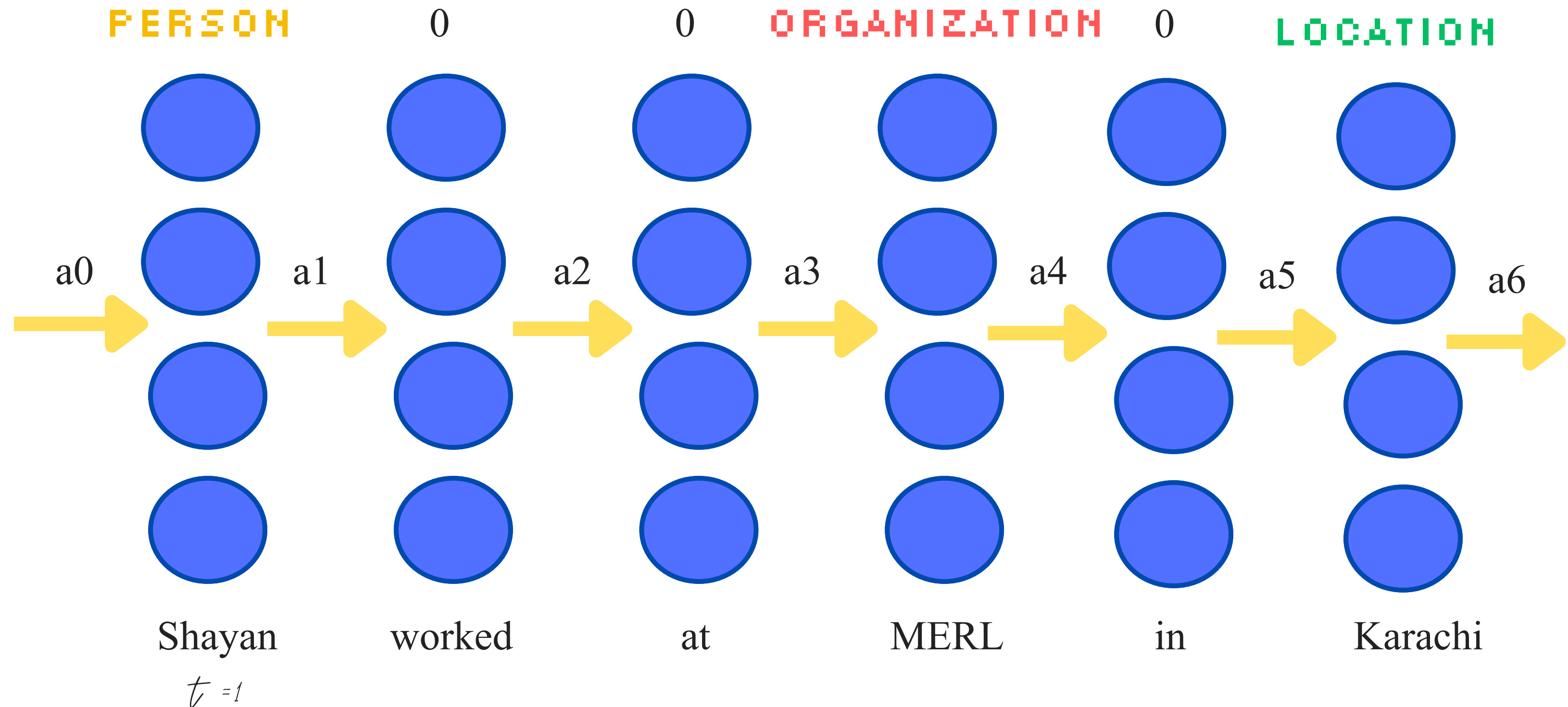
The diagram uses colored lines to connect the entity labels to their corresponding words in the sentence.

Hey, show me the best places to visit in Karachi during Jan 2022

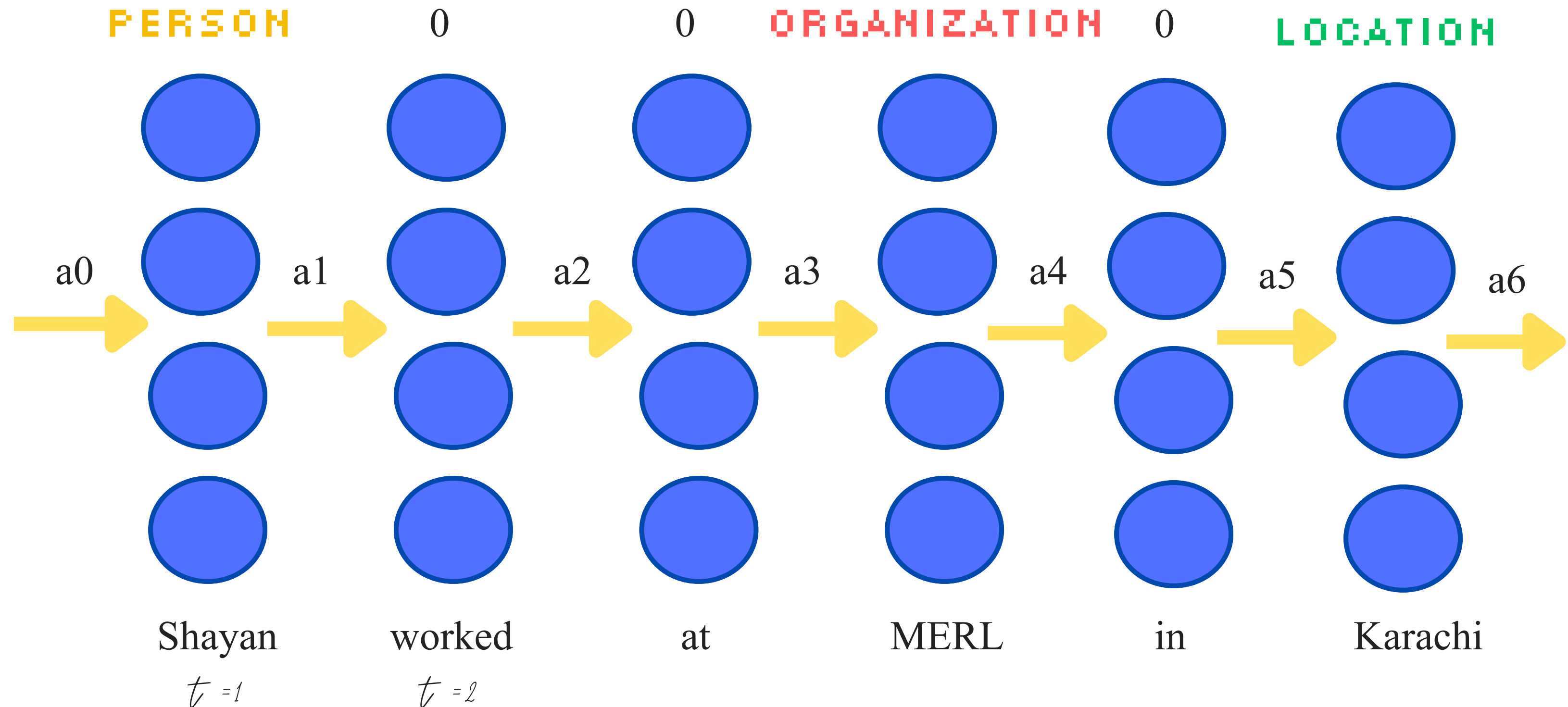
RNNs: Named Entity Recognition



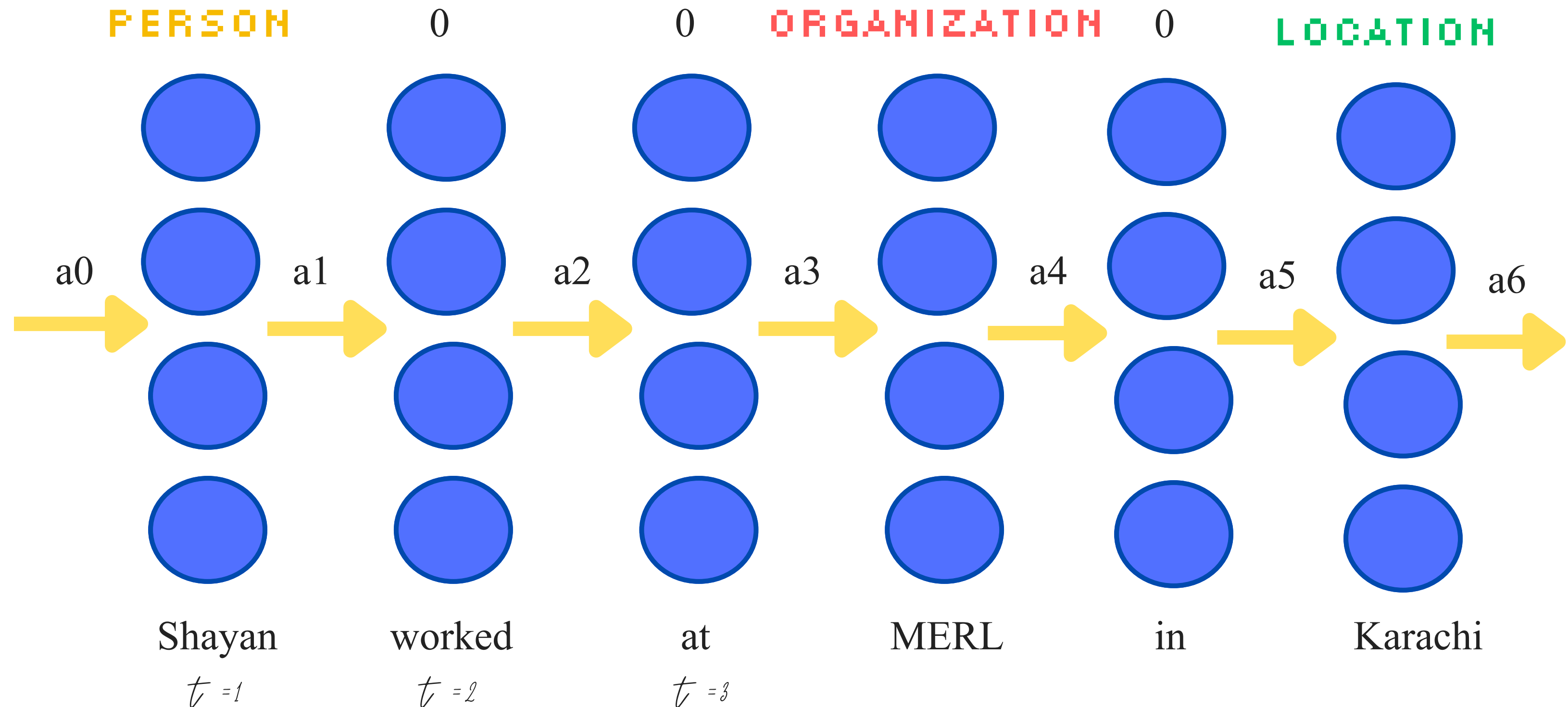
RNNs: Named Entity Recognition



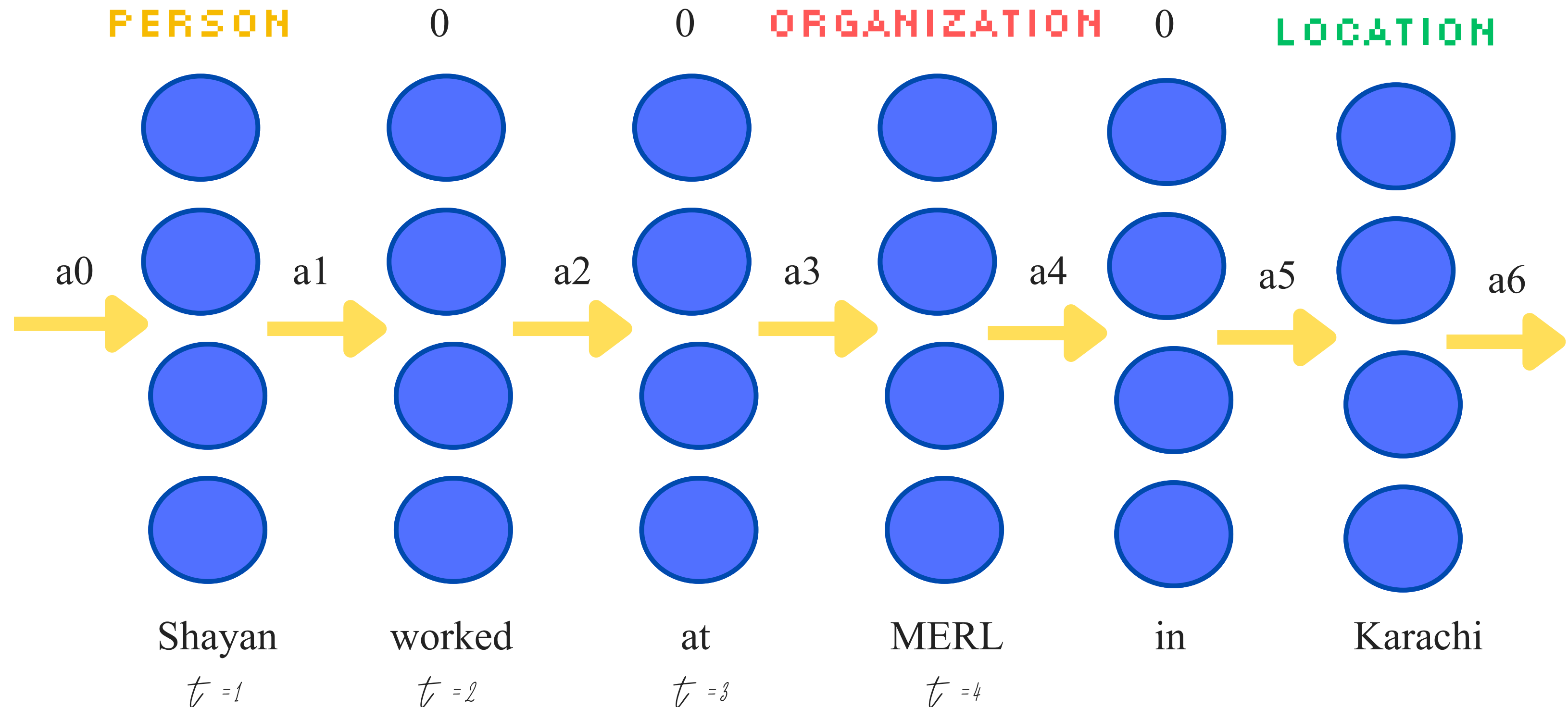
RNNs: Named Entity Recognition



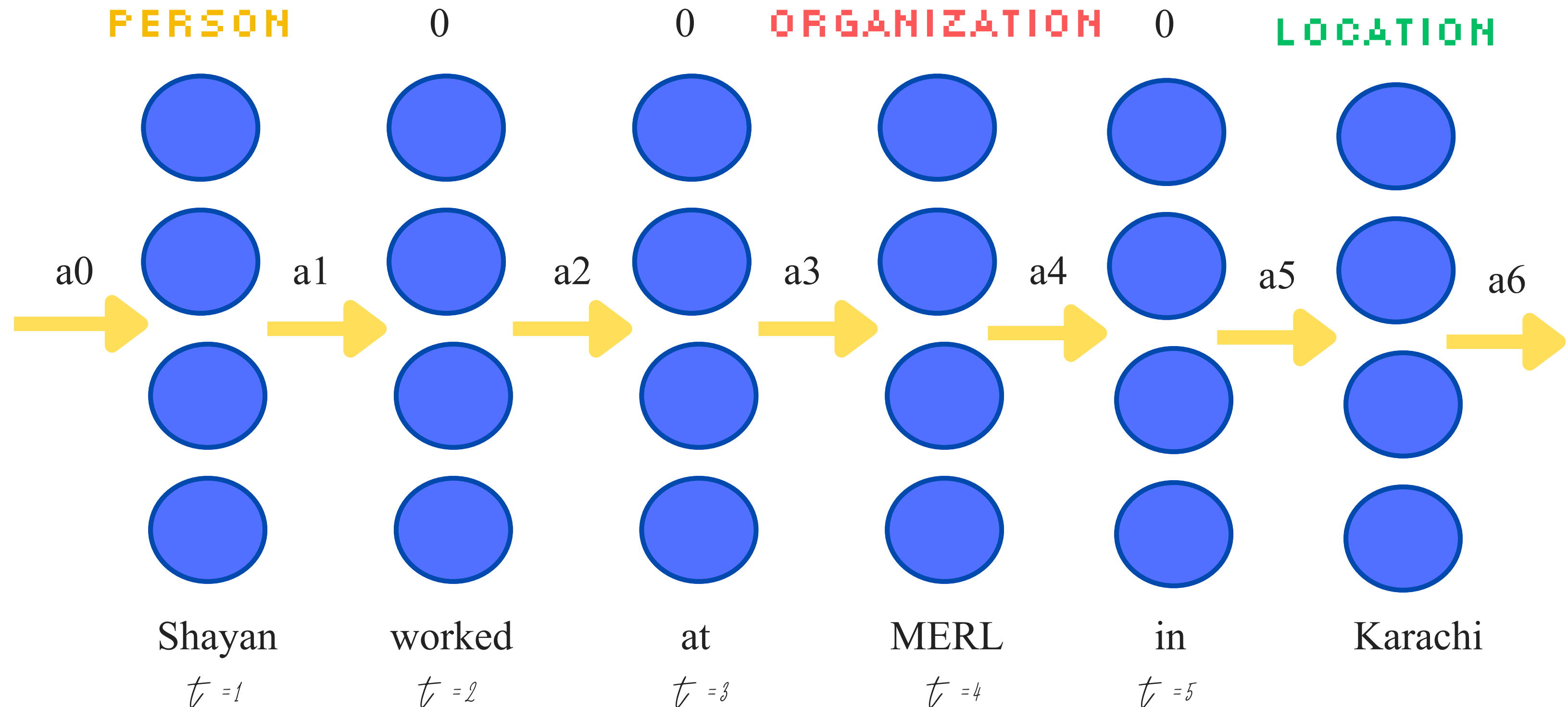
RNNs: Named Entity Recognition



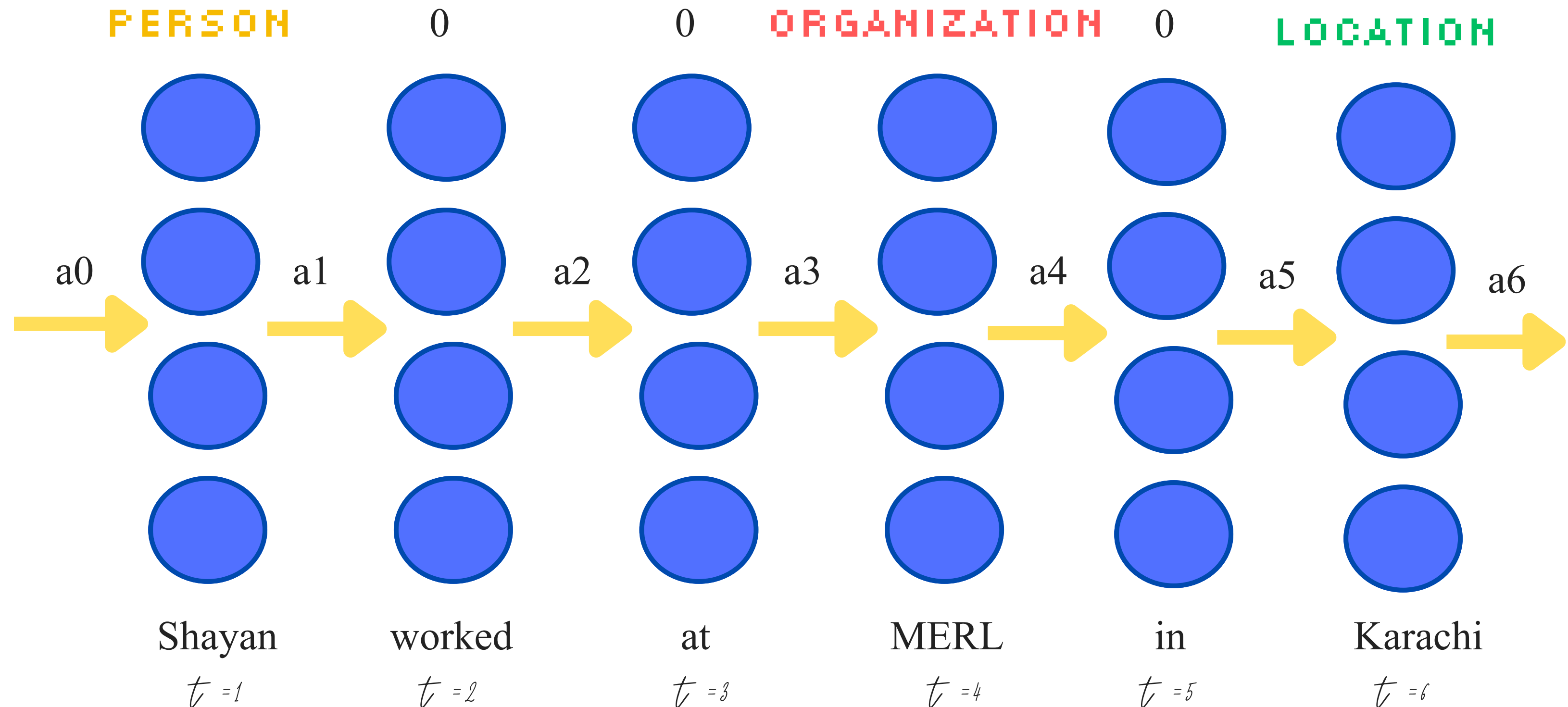
RNNs: Named Entity Recognition



RNNs: Named Entity Recognition



RNNs: Named Entity Recognition



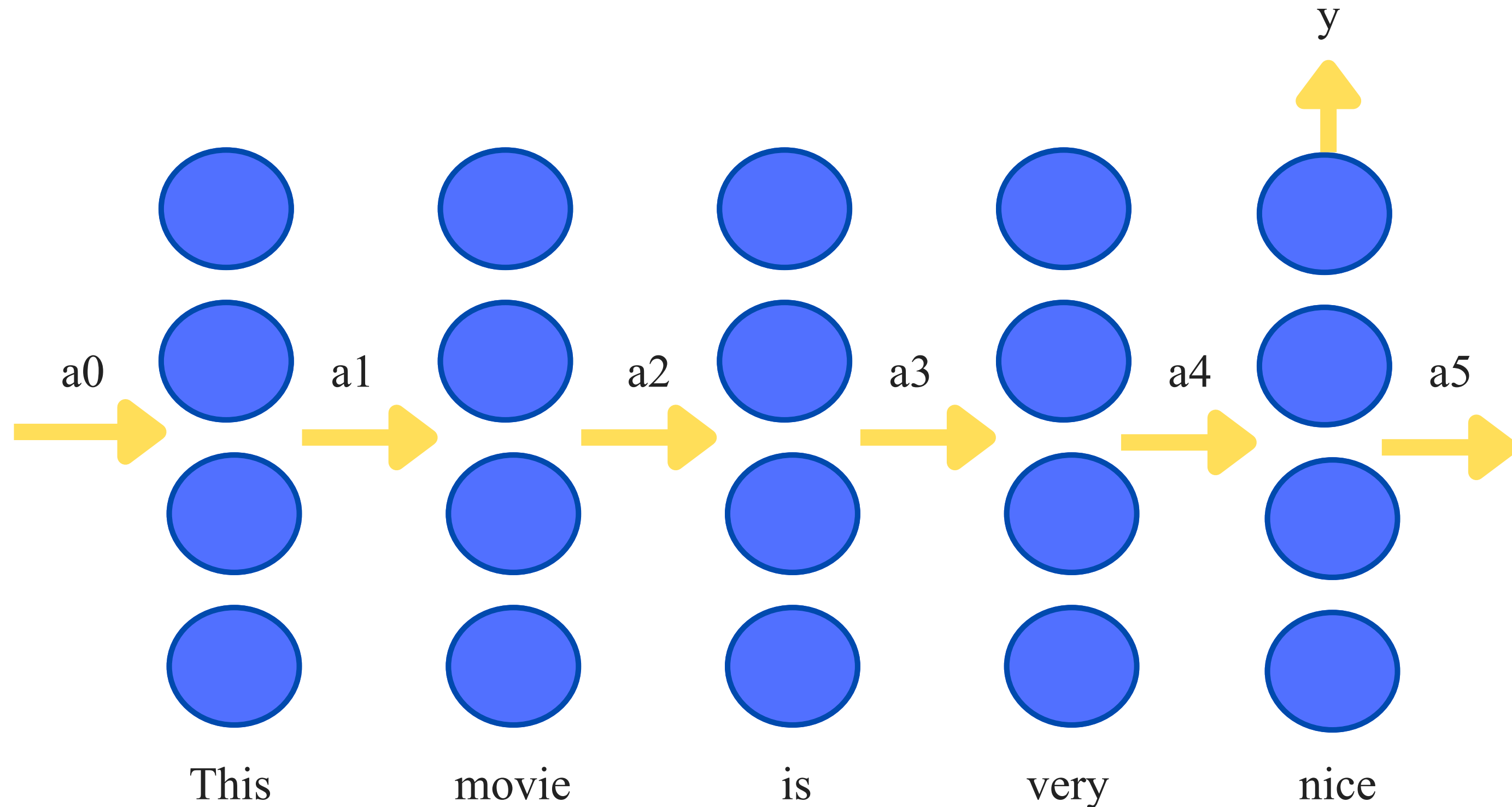
Types of RNNs

This movie is very nice



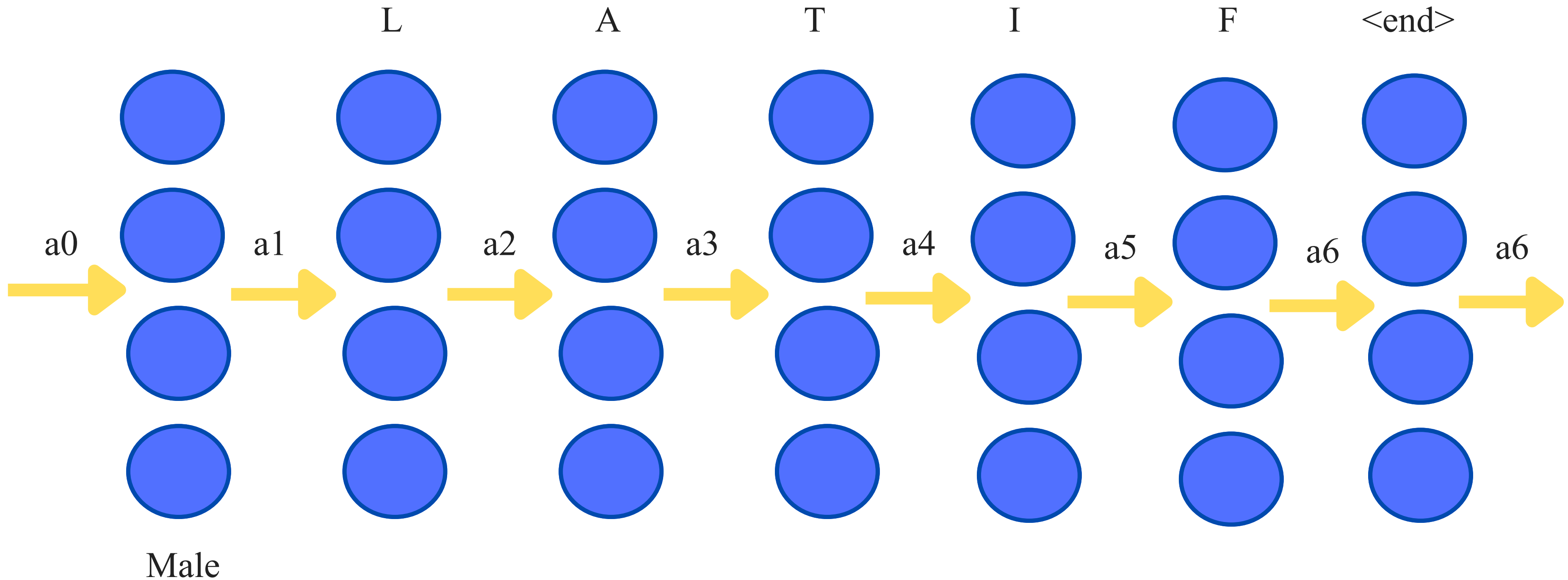
Types of RNNs

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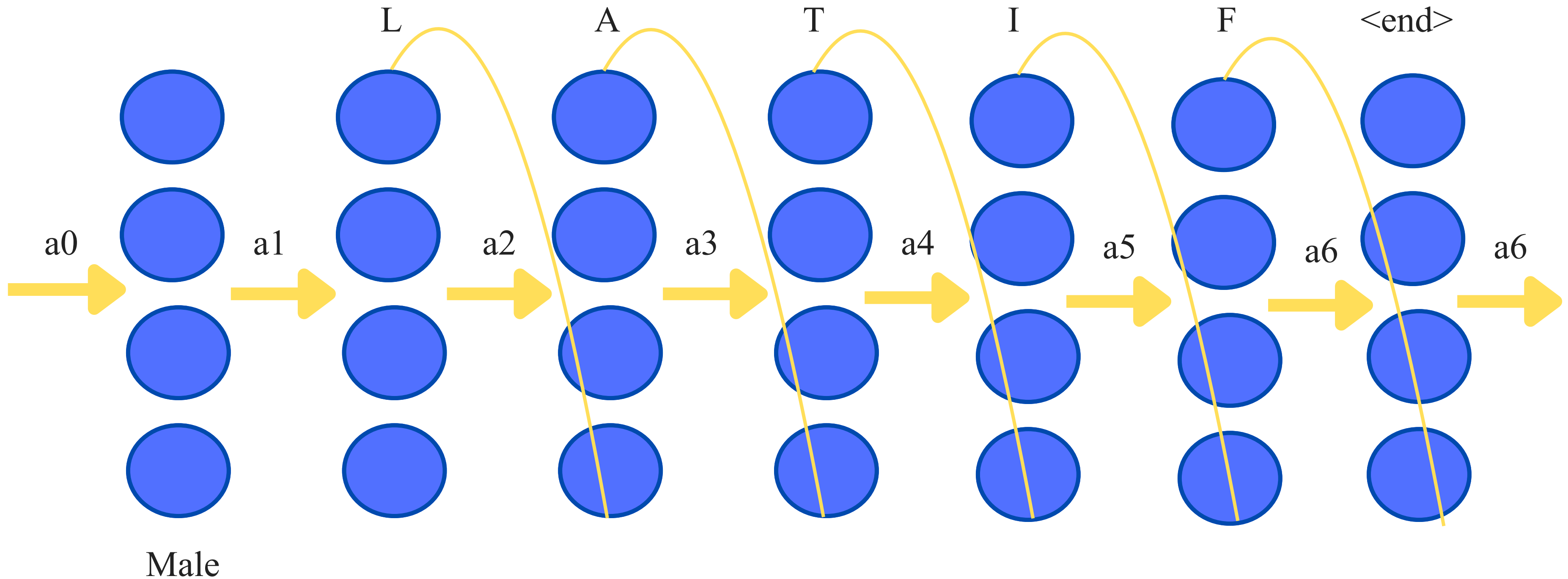
Types of RNNs

Baby Name Generation



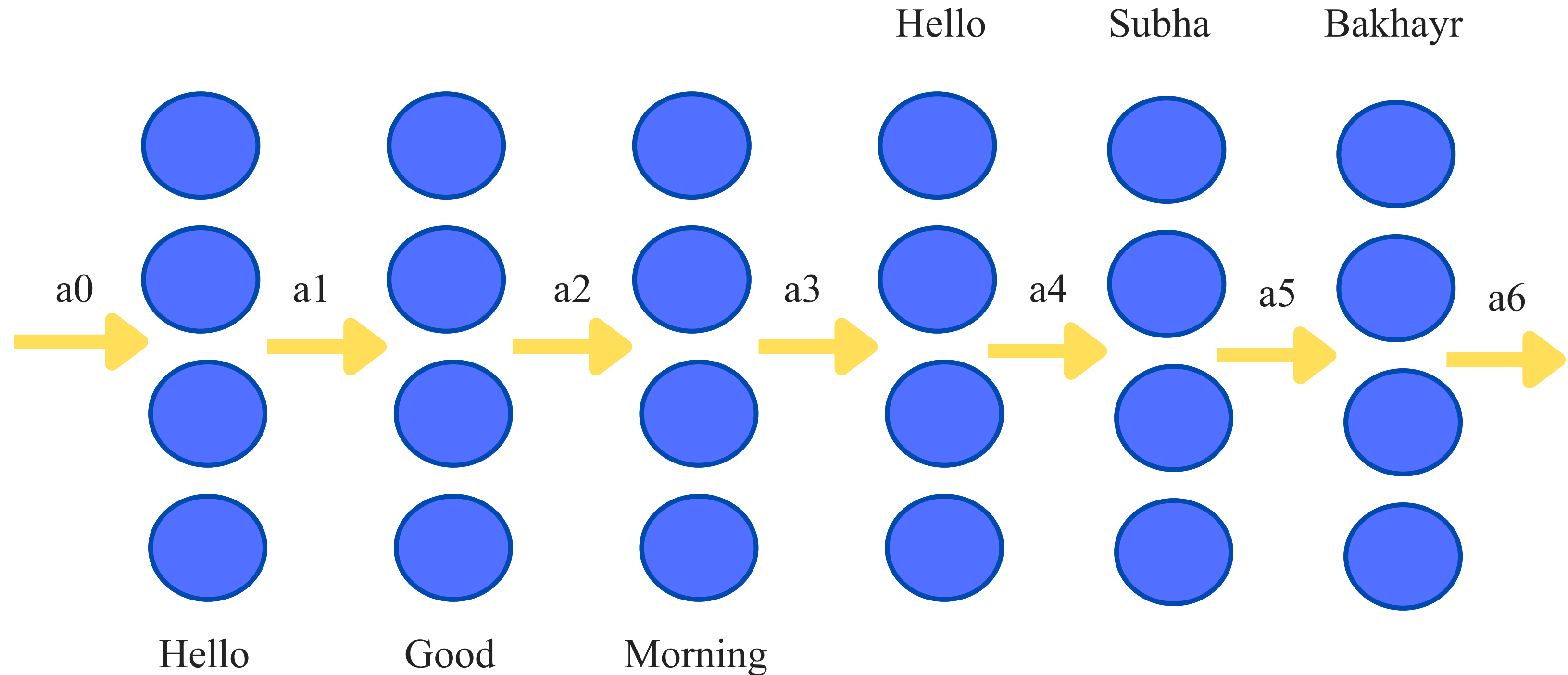
Types of RNNs

Baby Name Generation



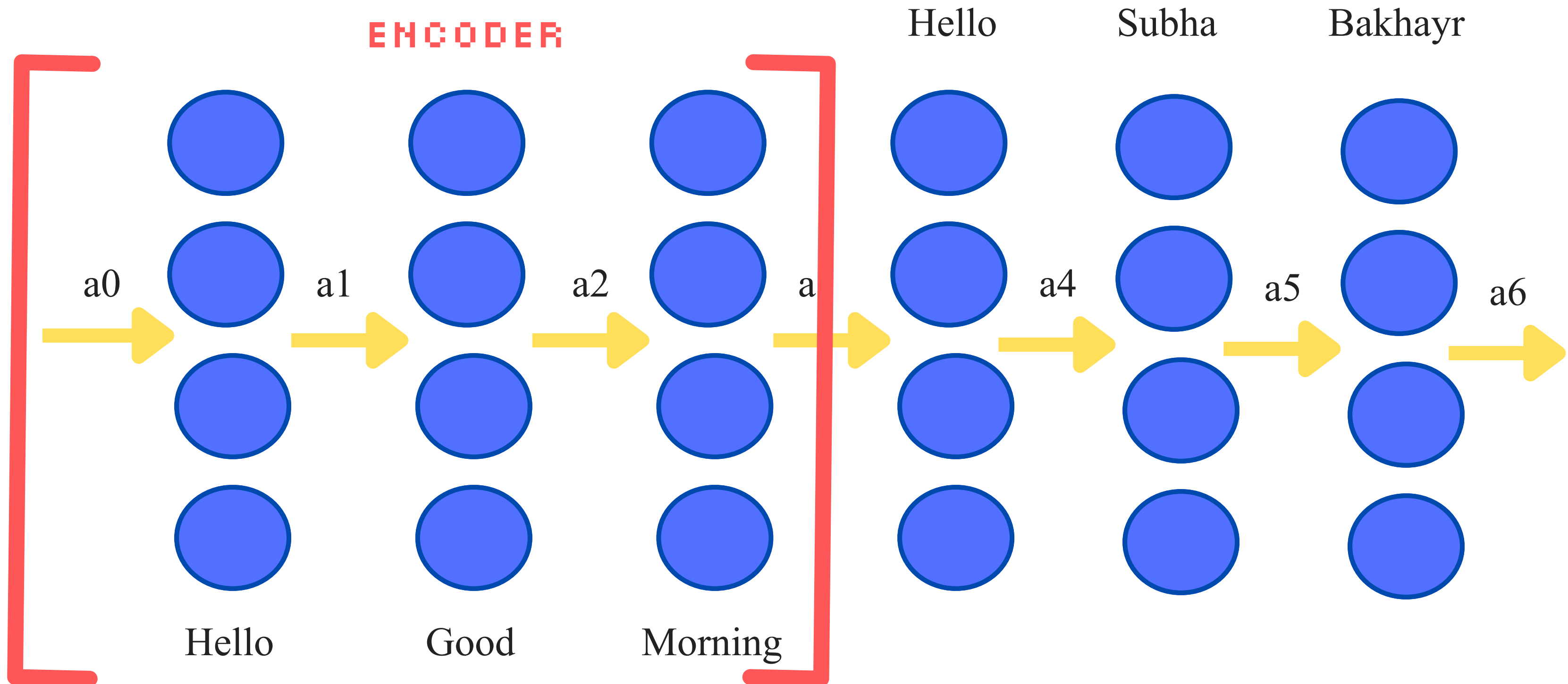
Types of RNNs

Language Translation



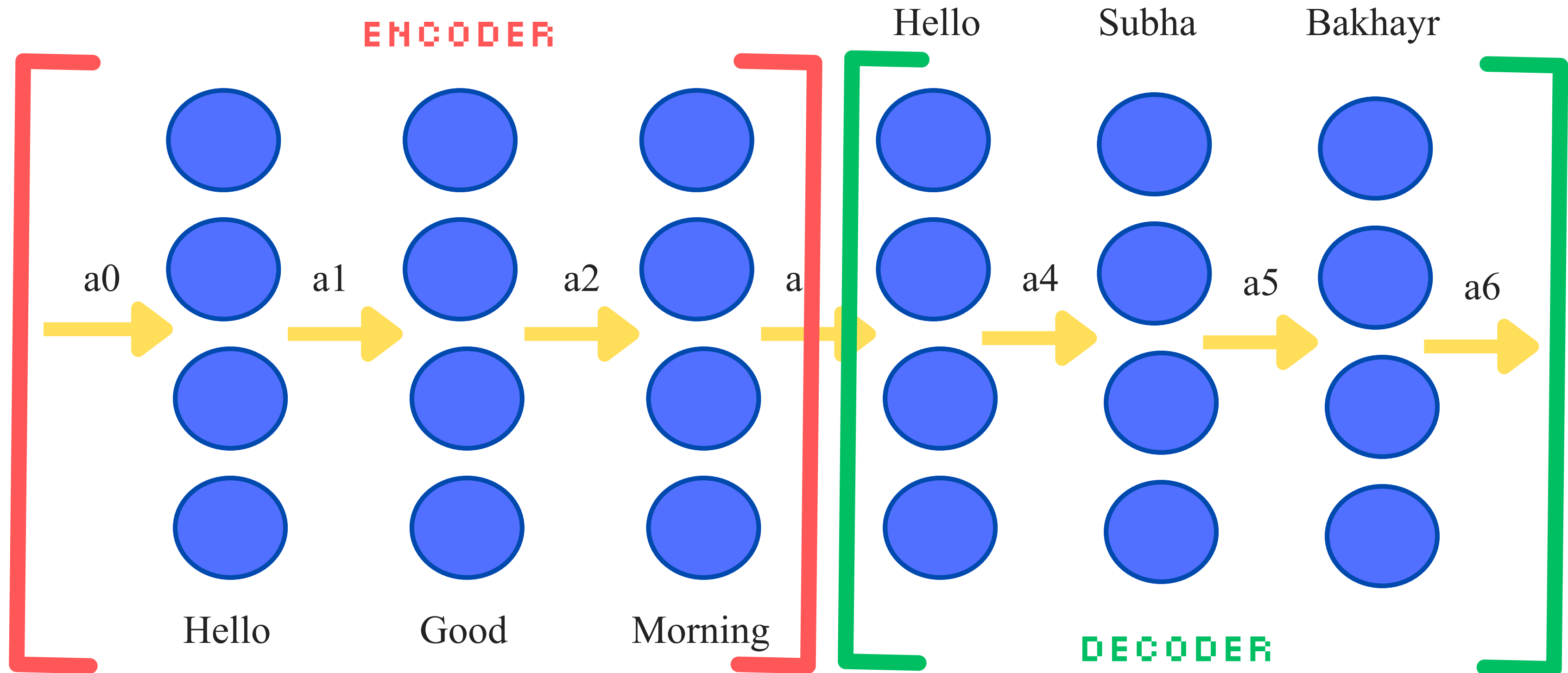
Types of RNNs

Language Translation

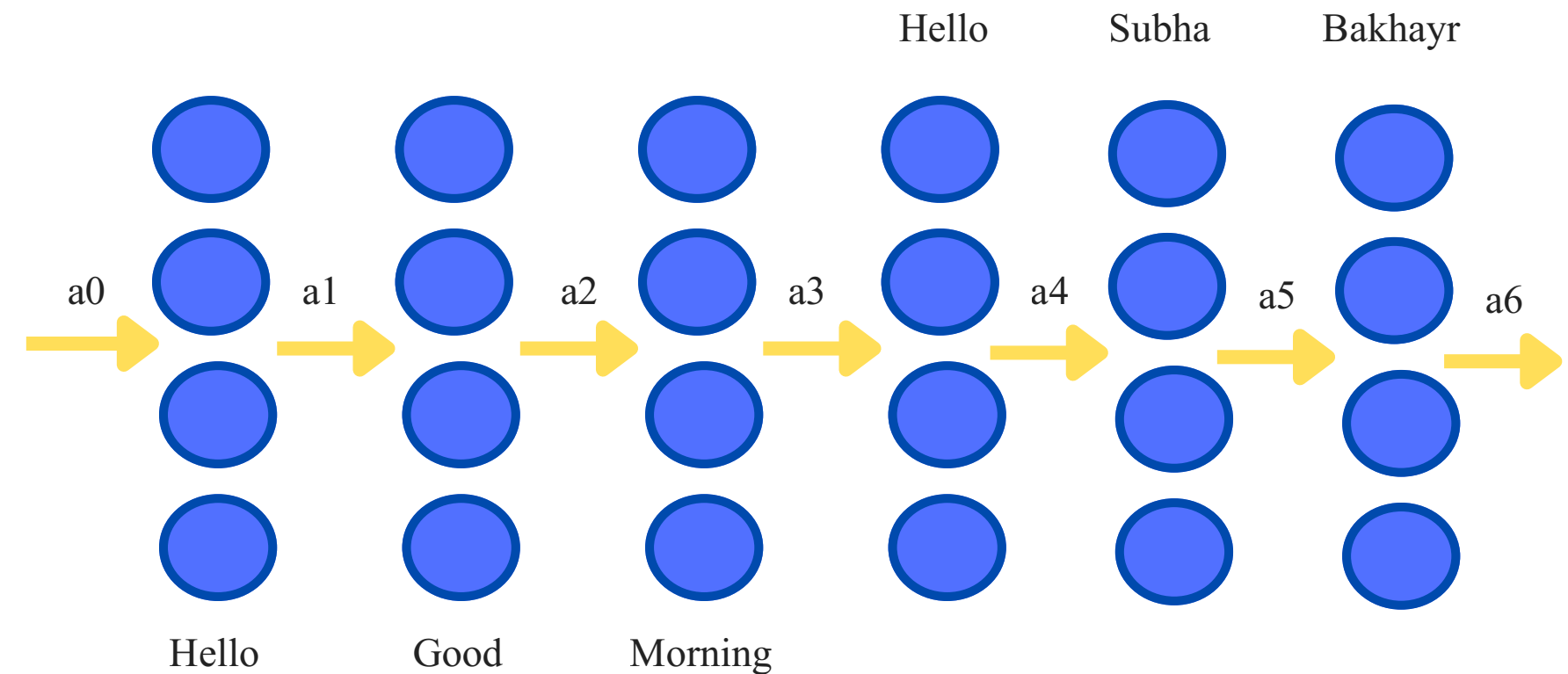
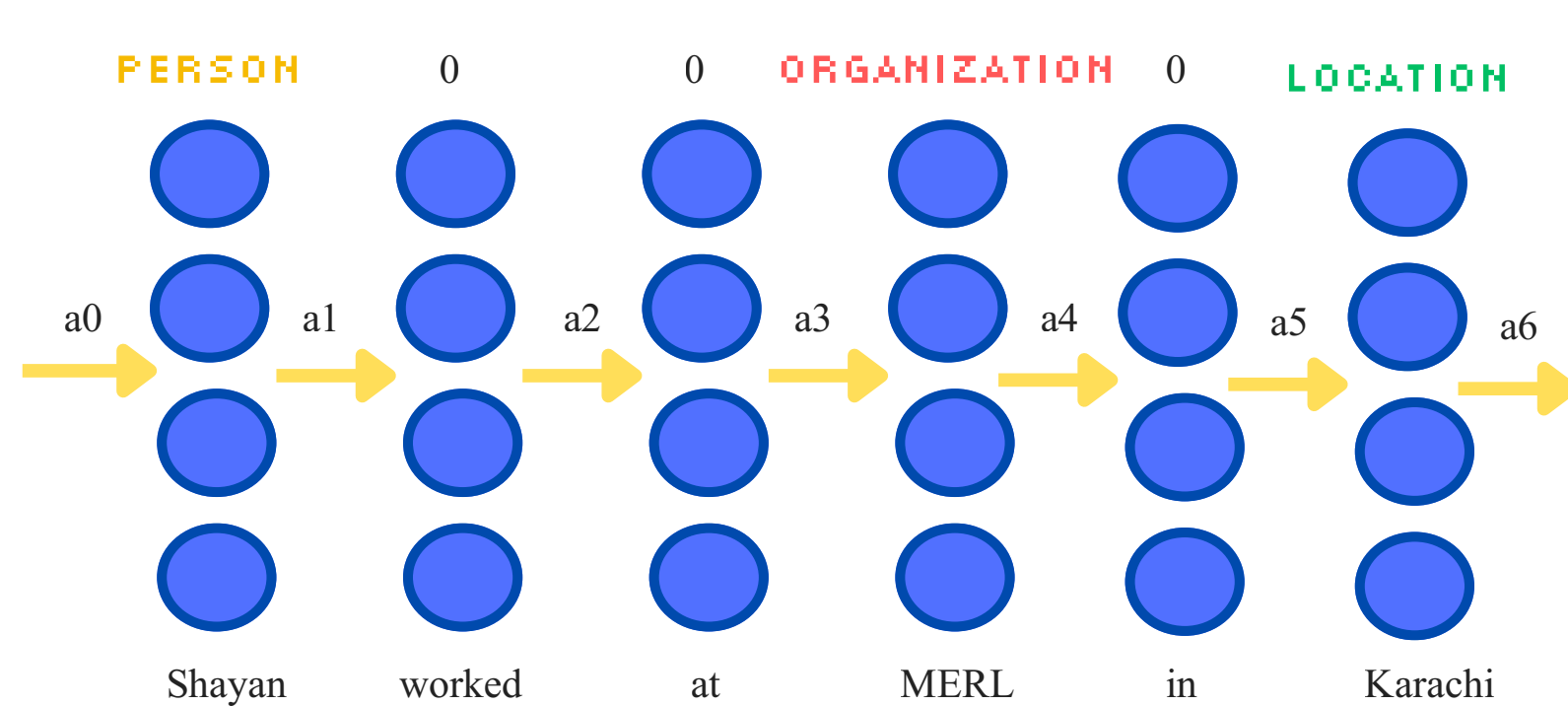
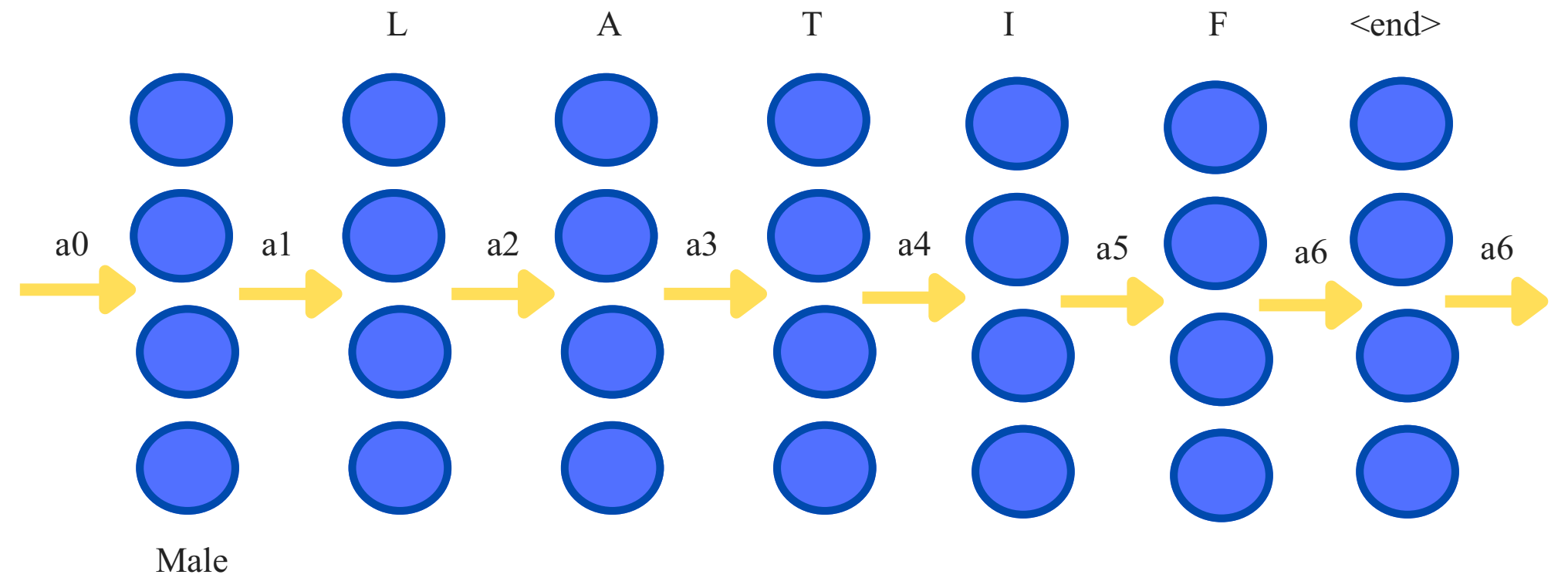
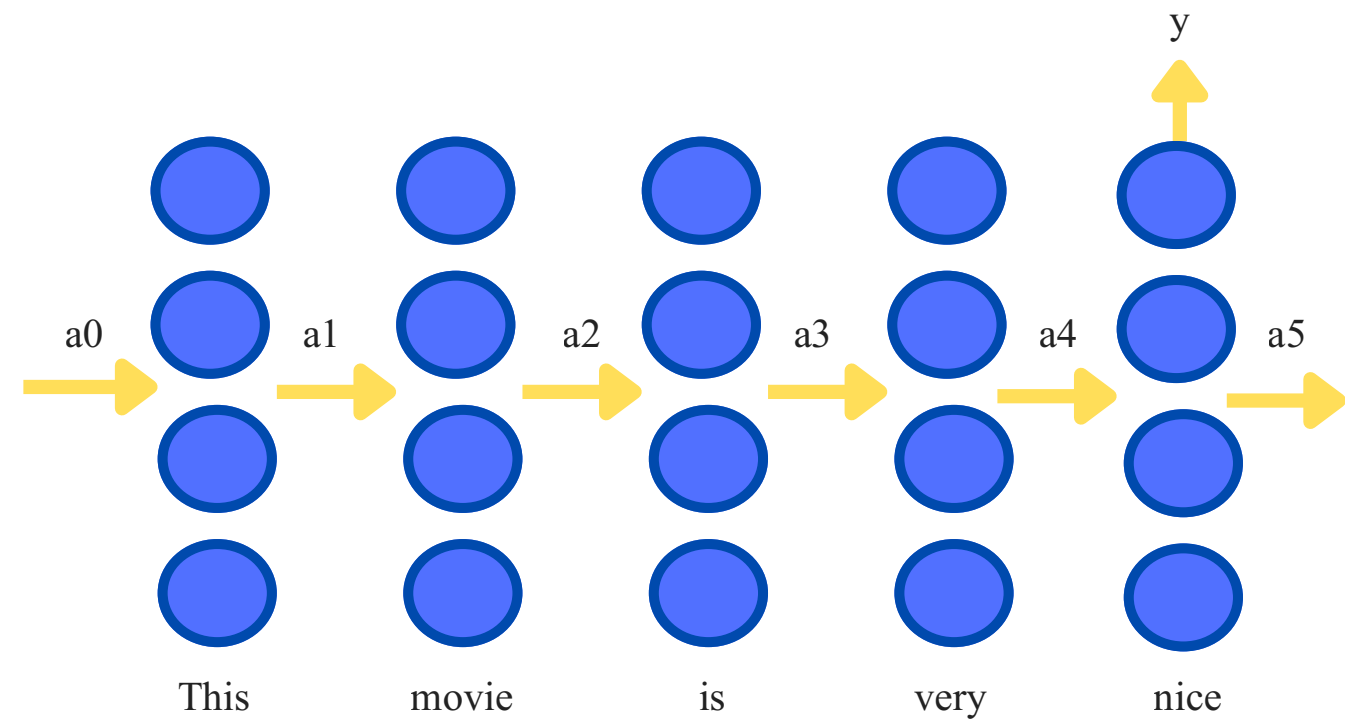


Types of RNNs

Language Translation

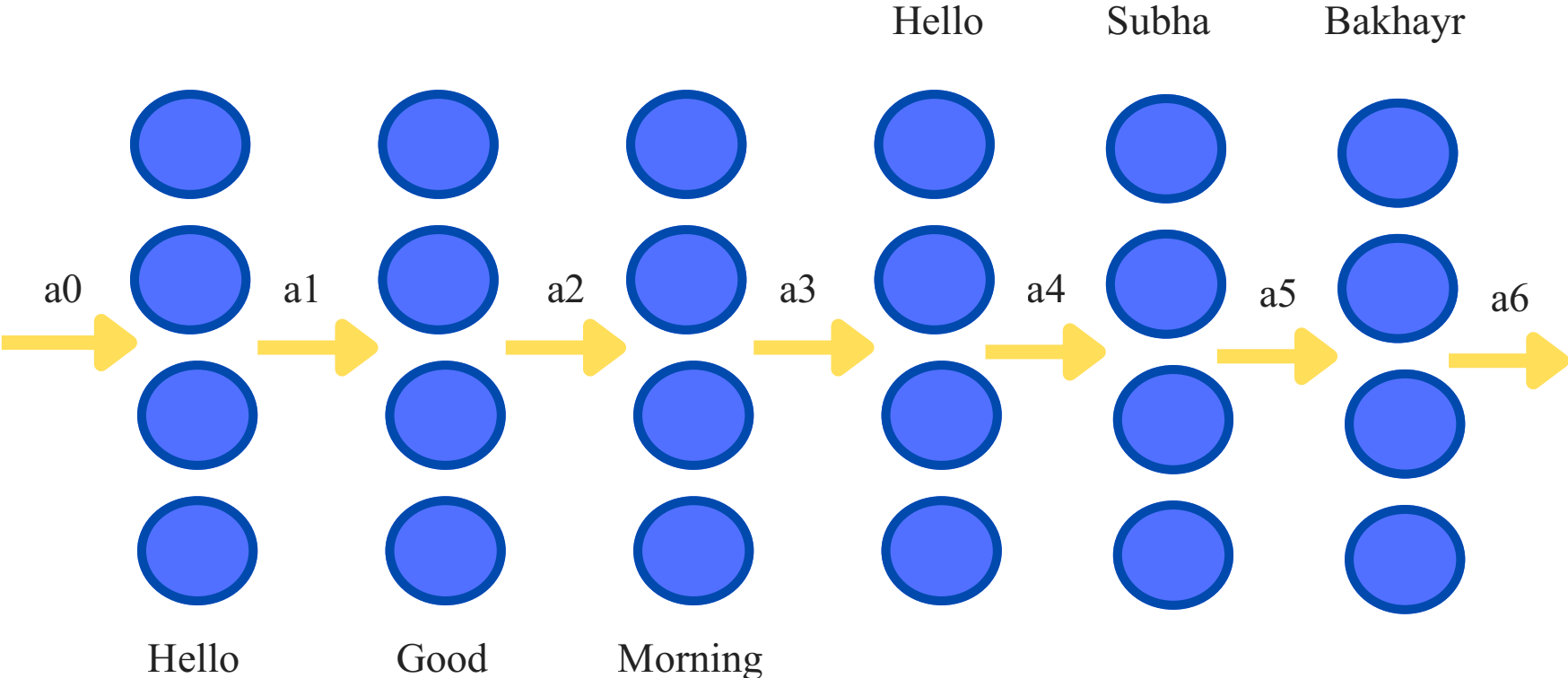
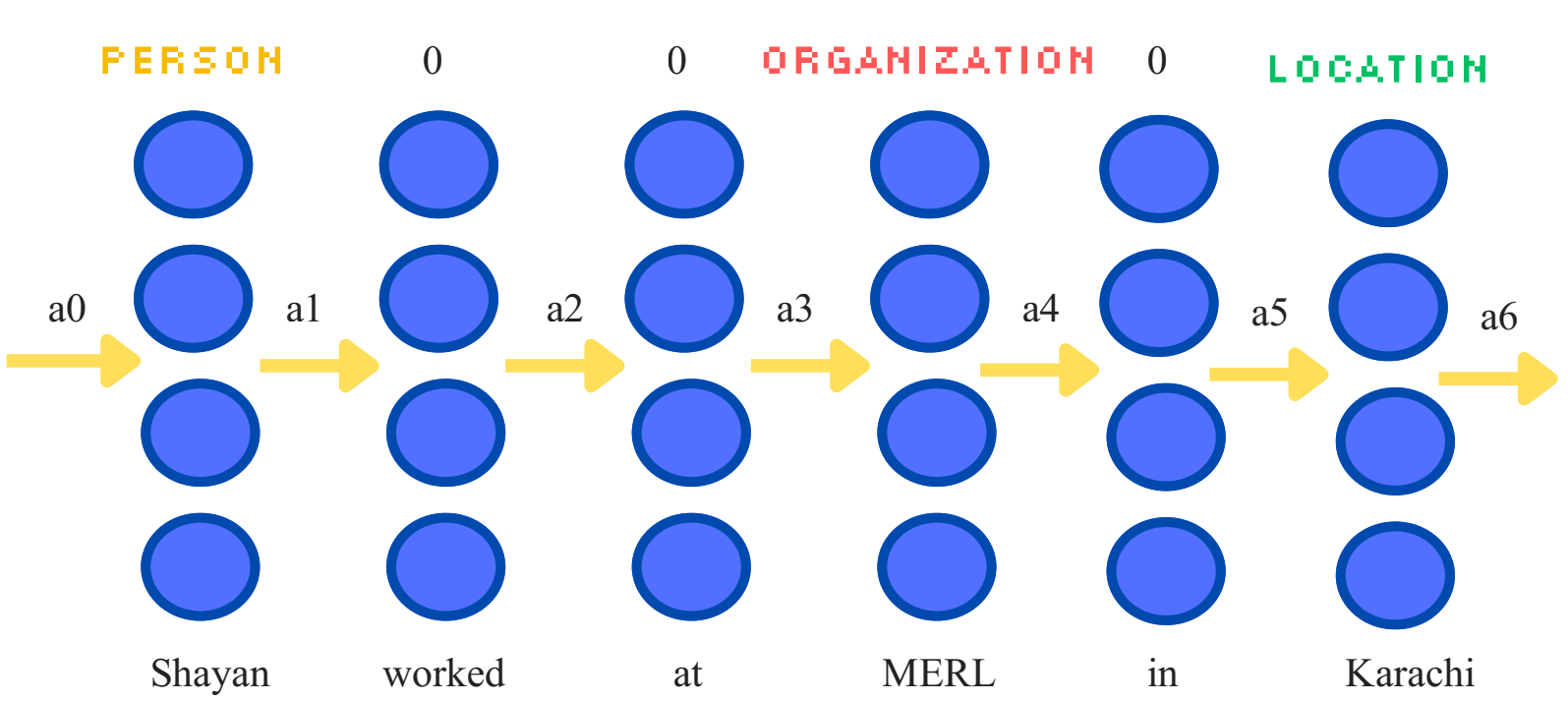
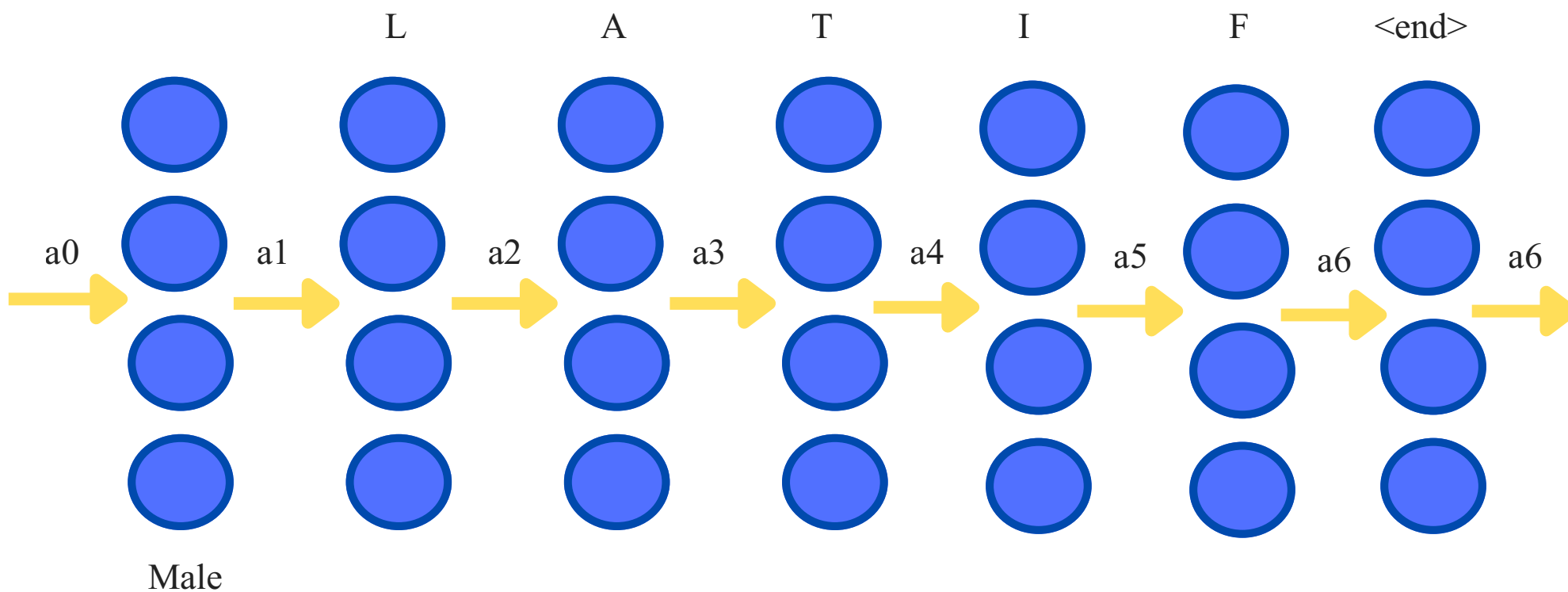
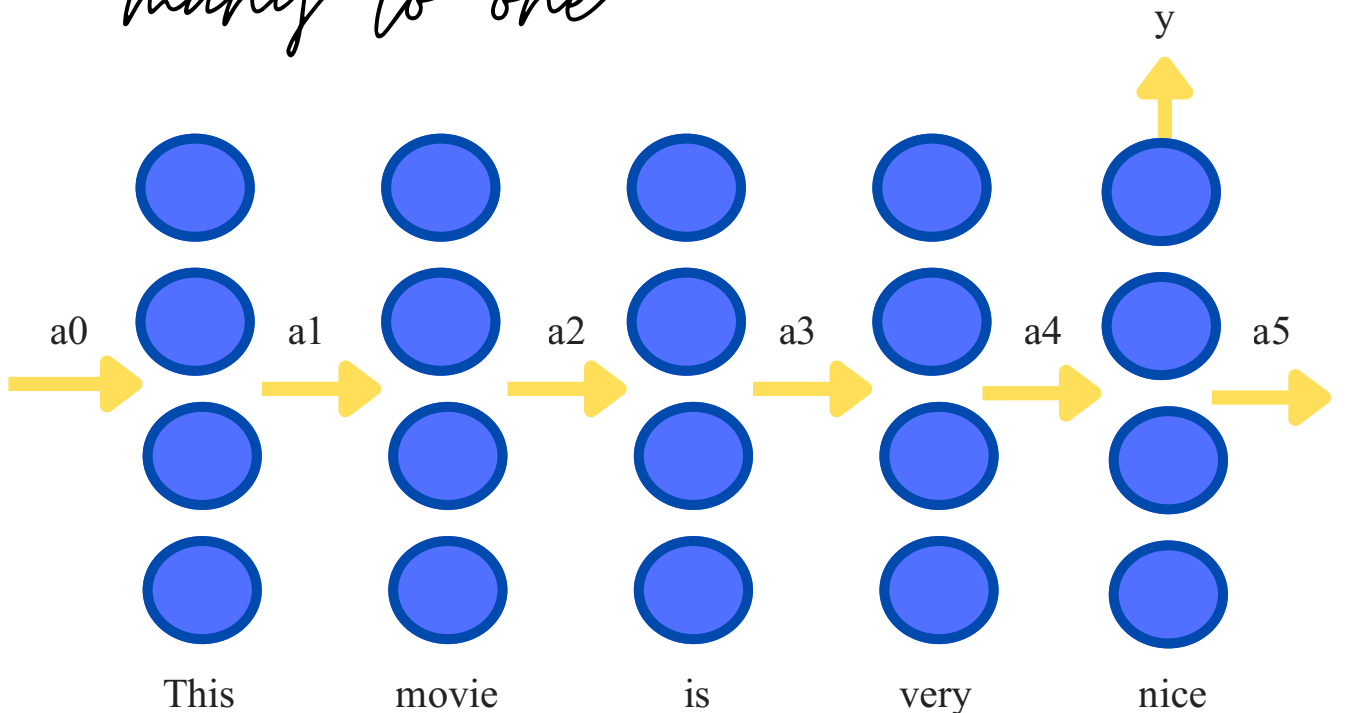


Types of RNNs



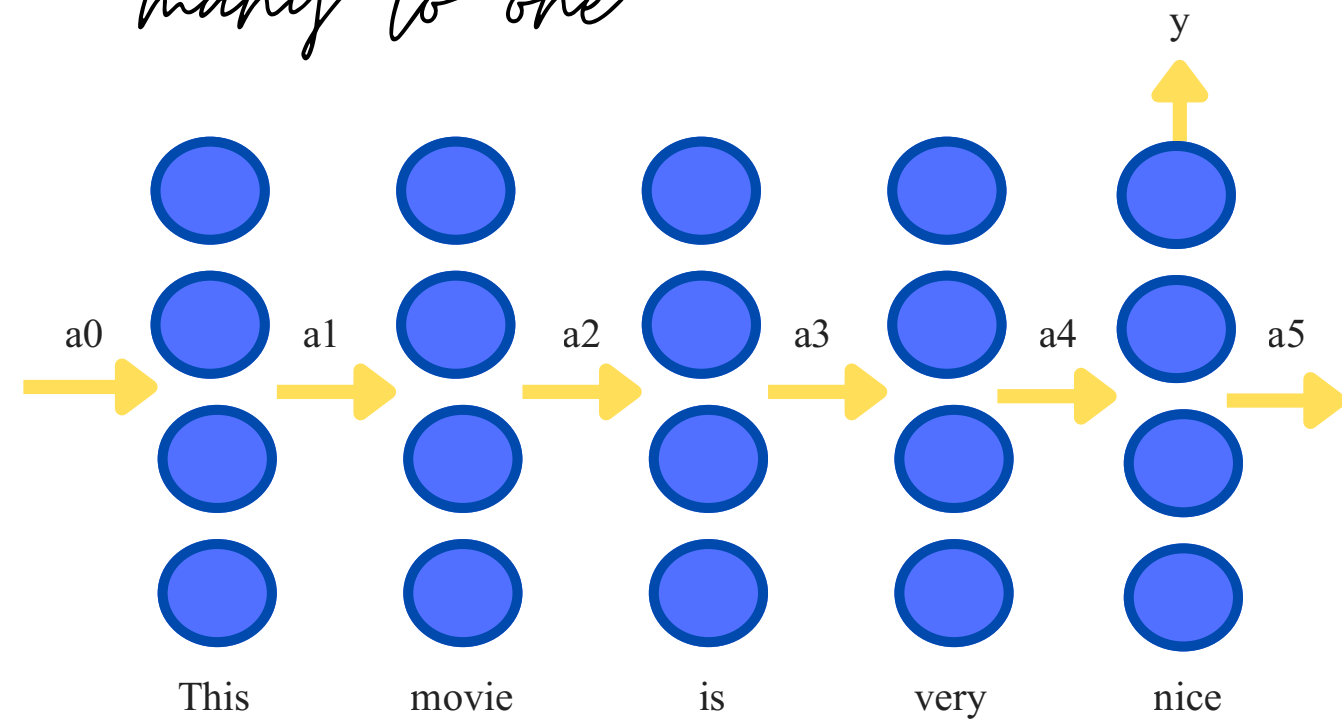
Types of RNNs

many to one

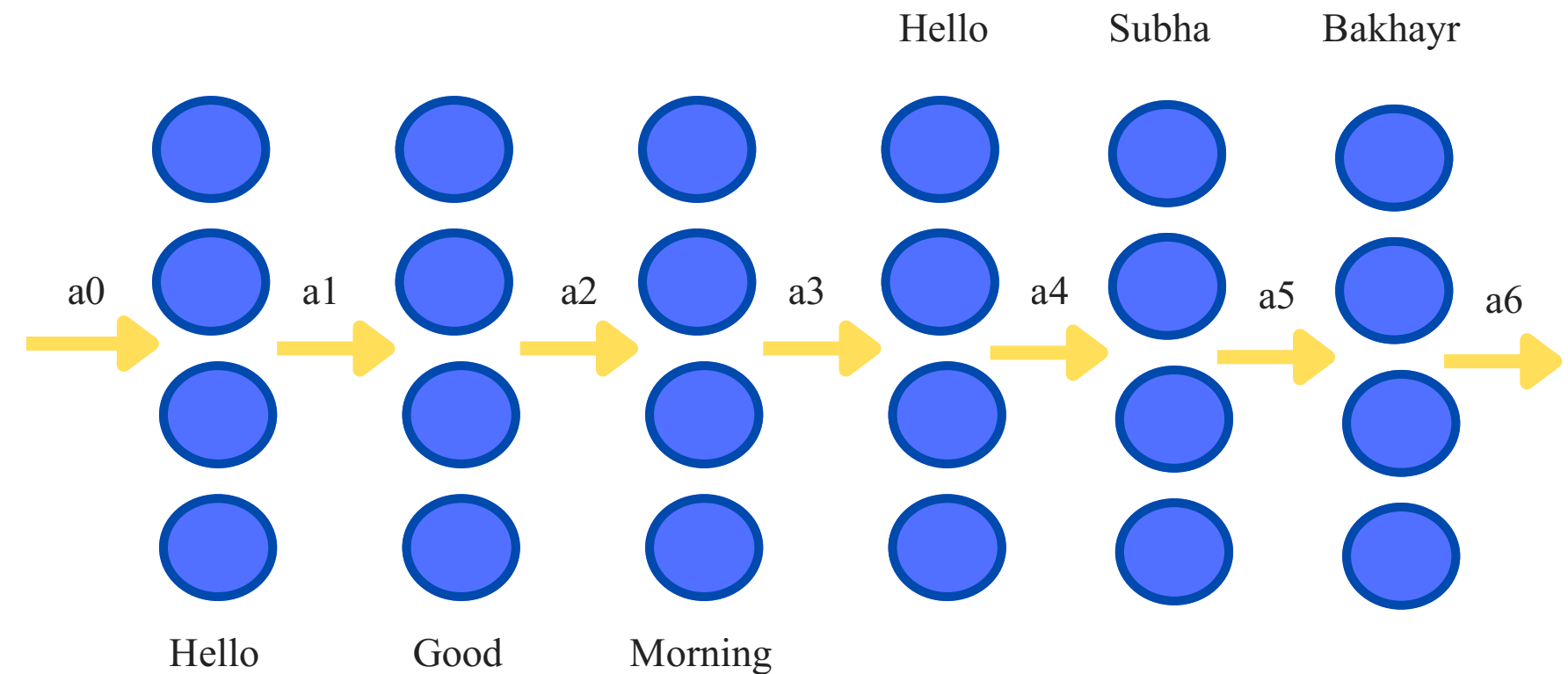
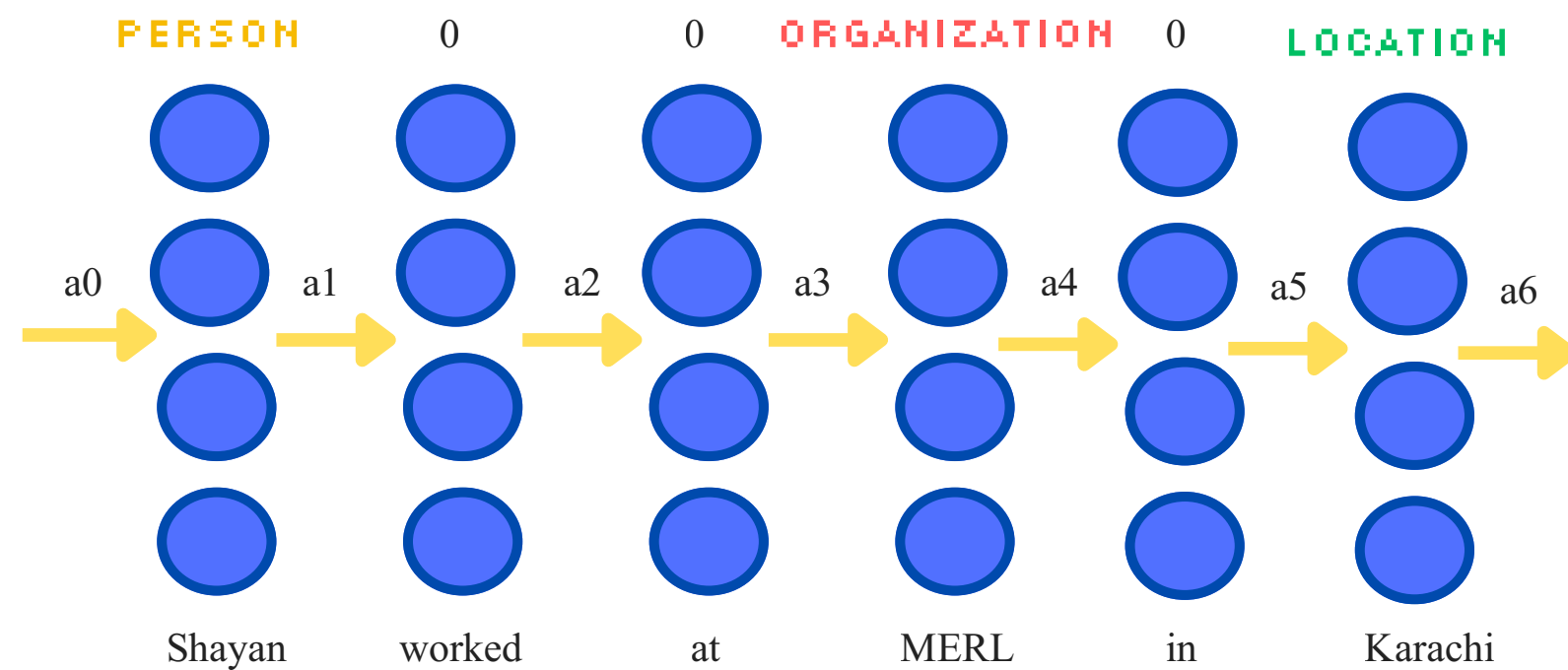
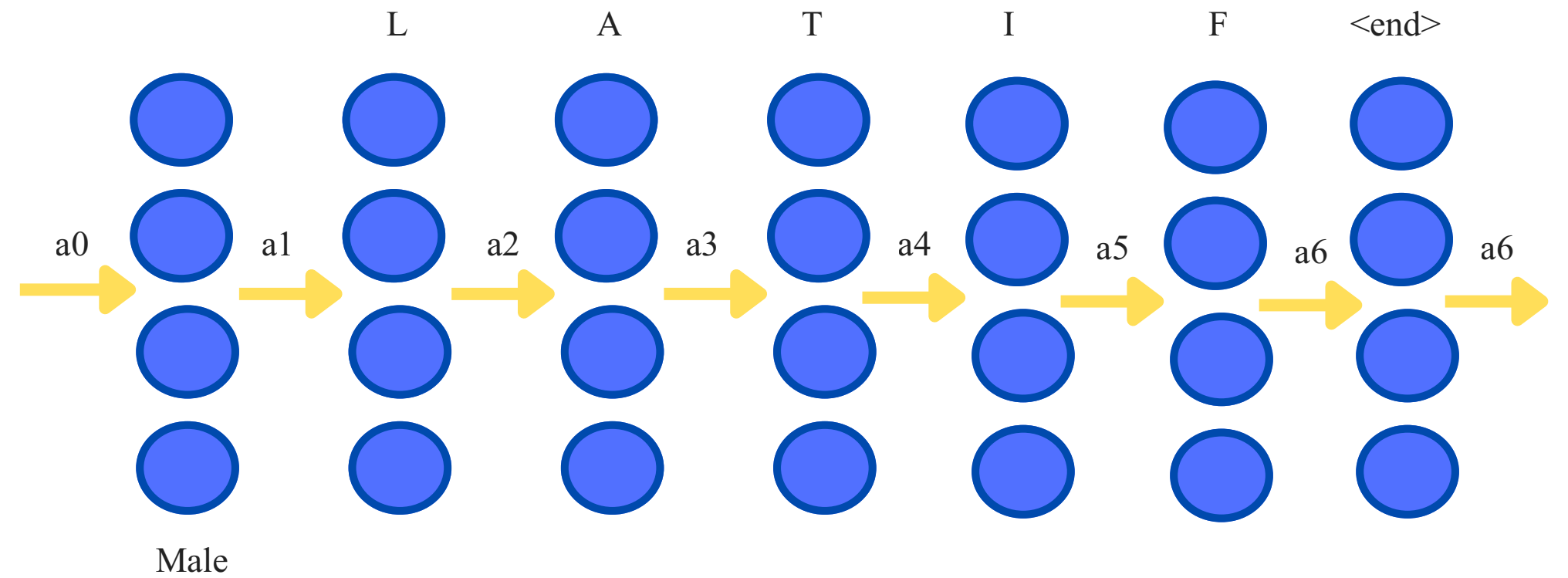


Types of RNNs

many to one

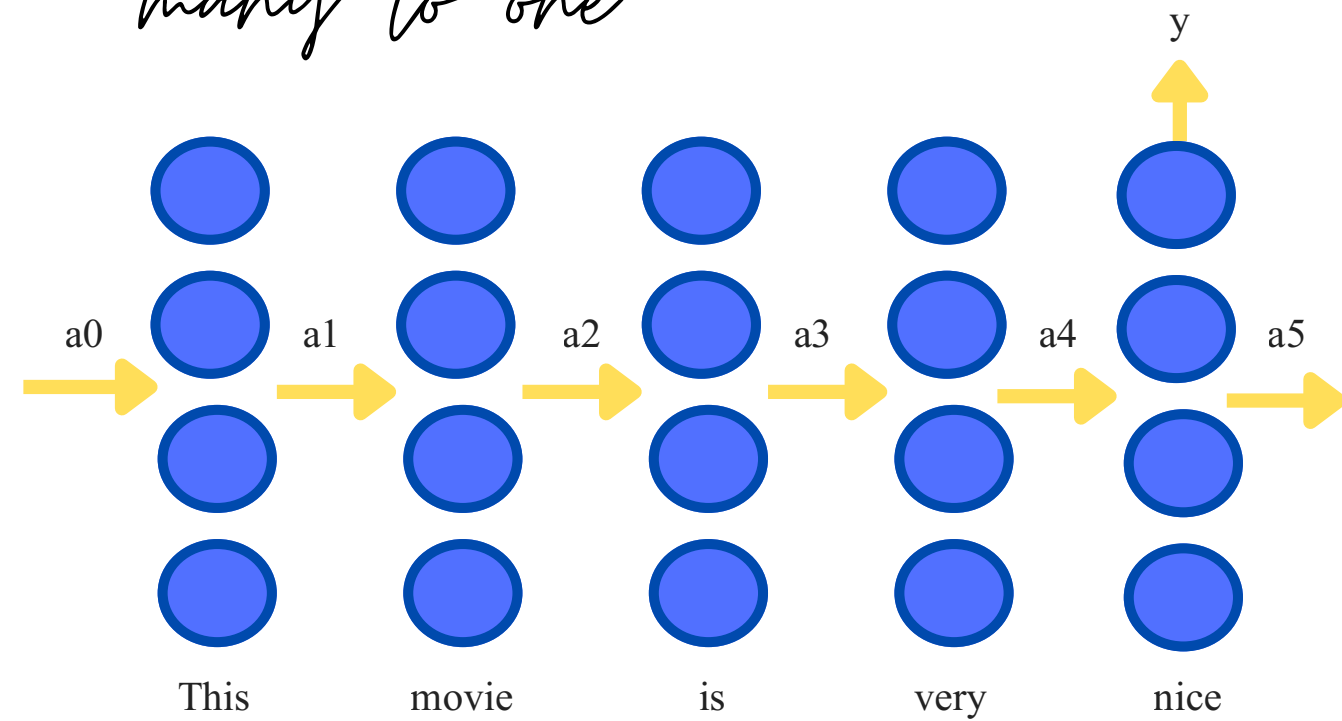


one to many

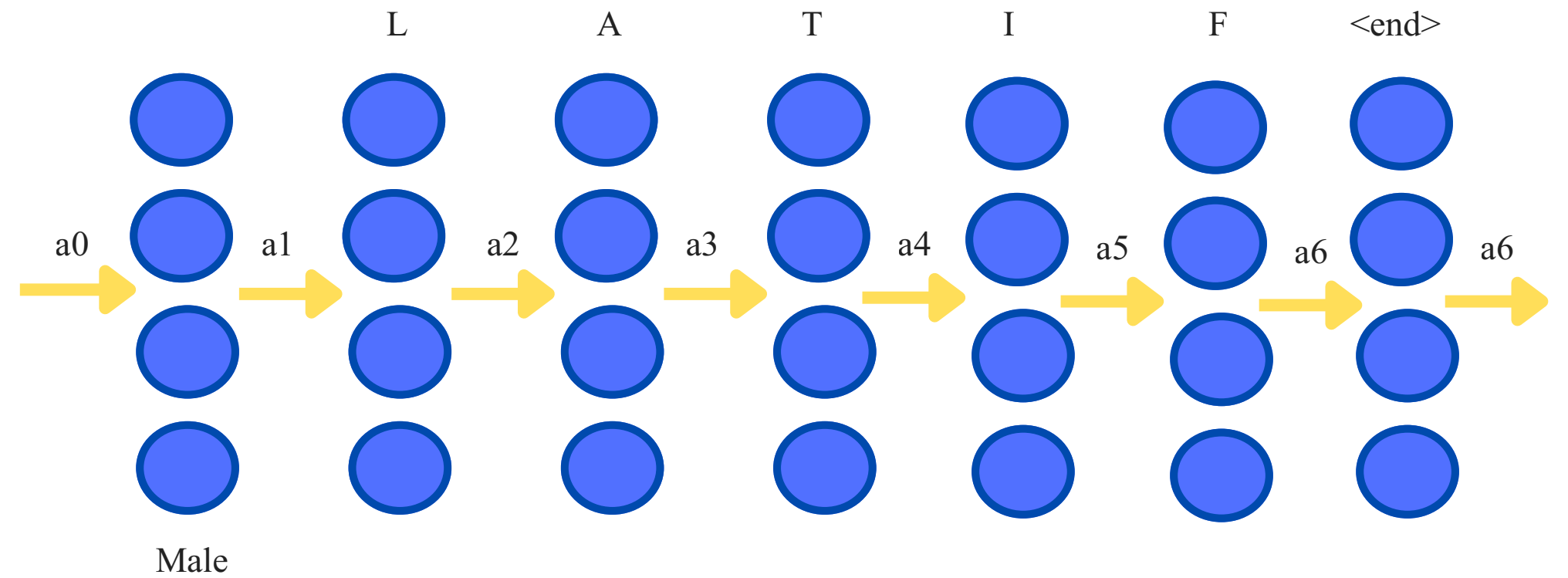


Types of RNNs

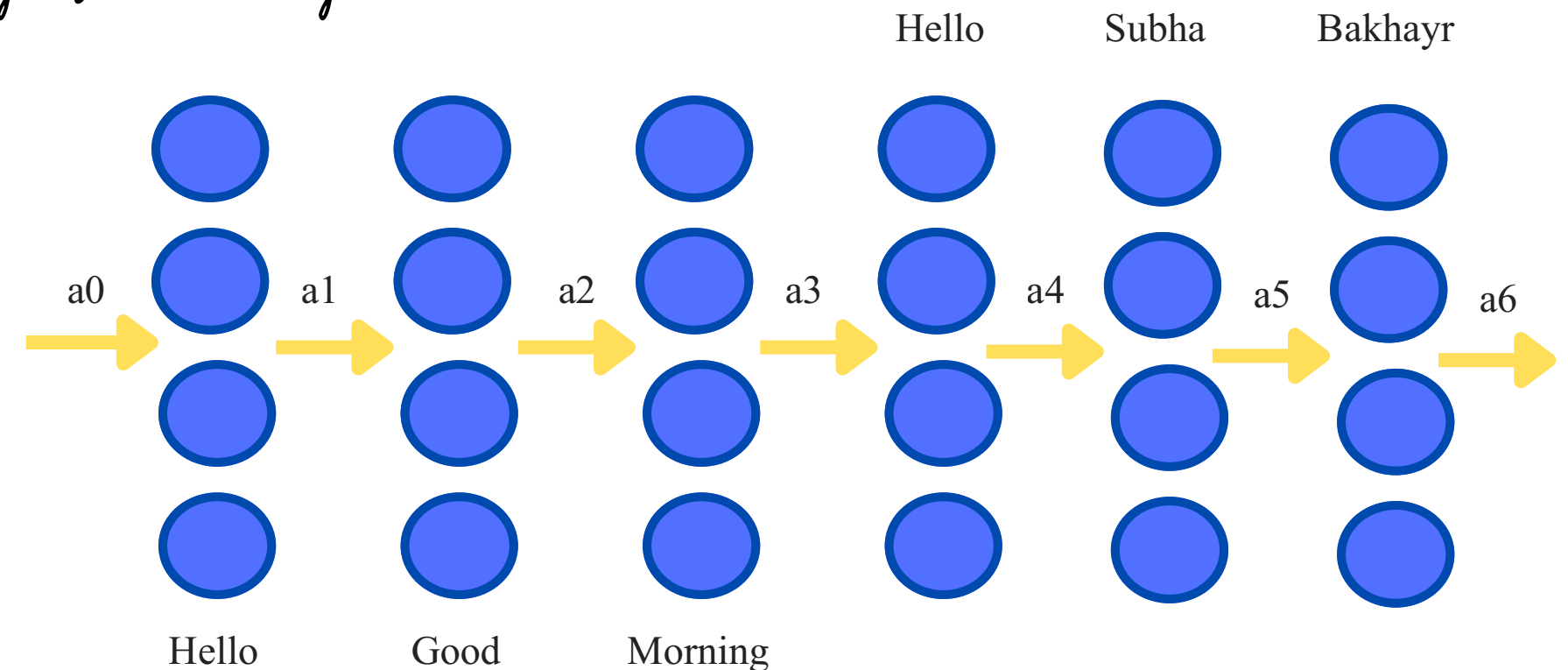
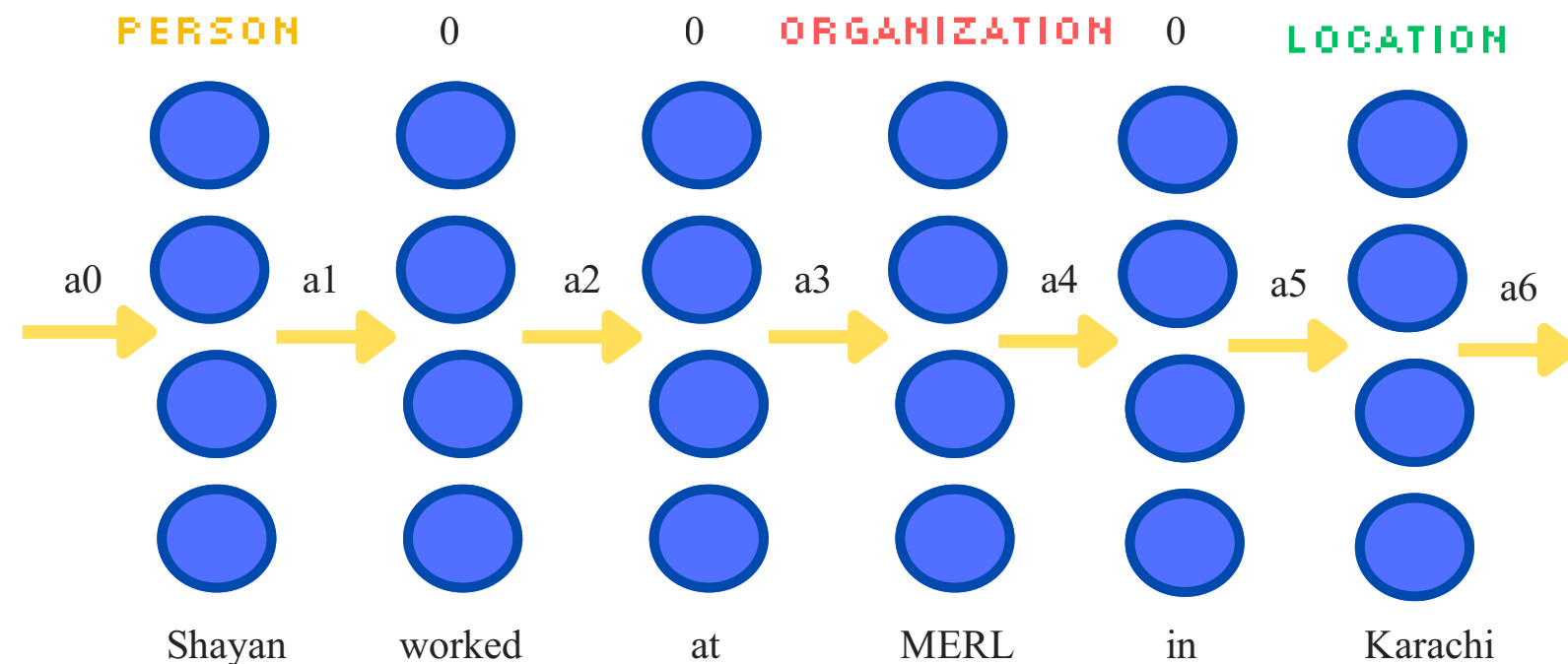
many to one



one to many

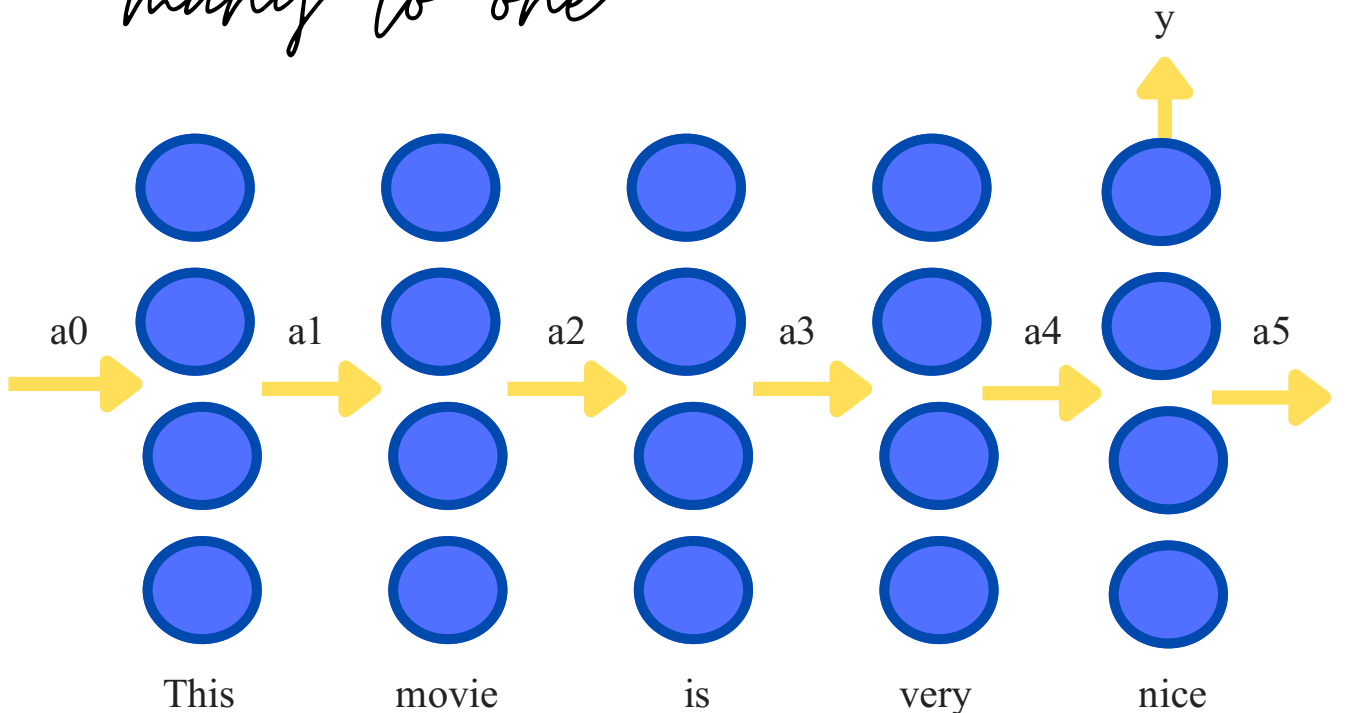


many to many

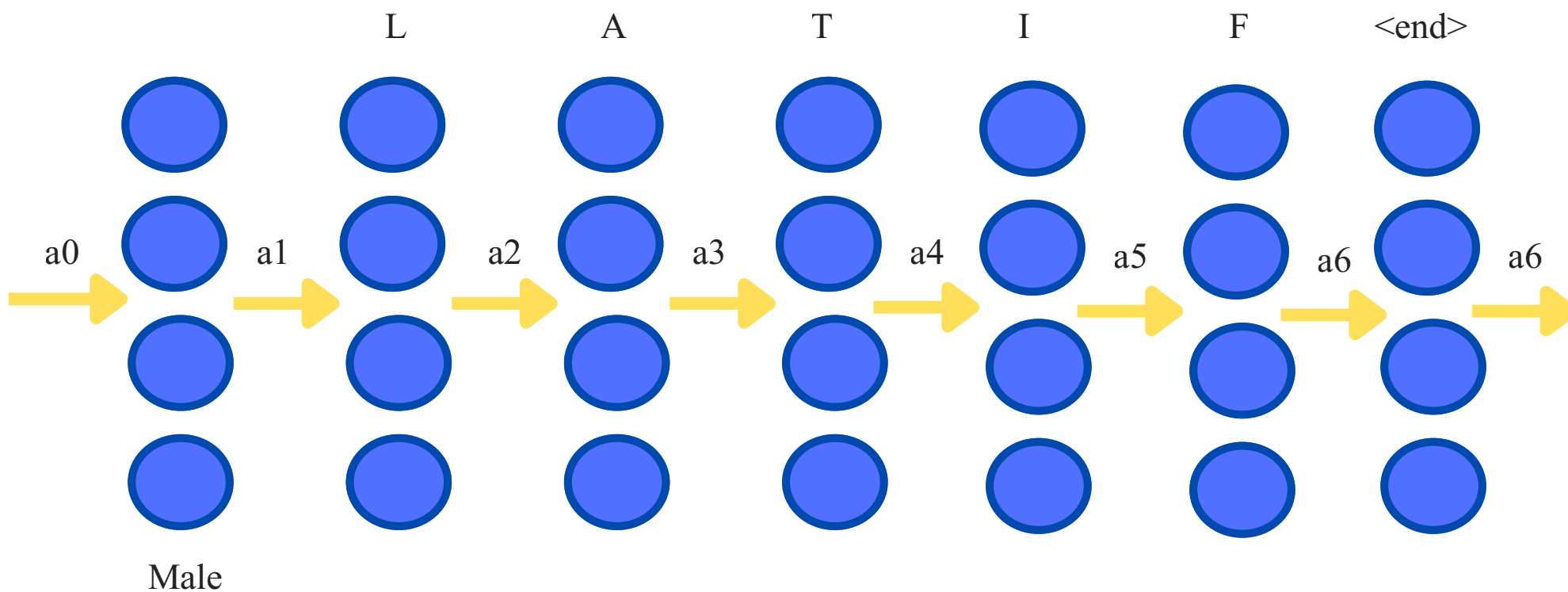


Types of RNNs

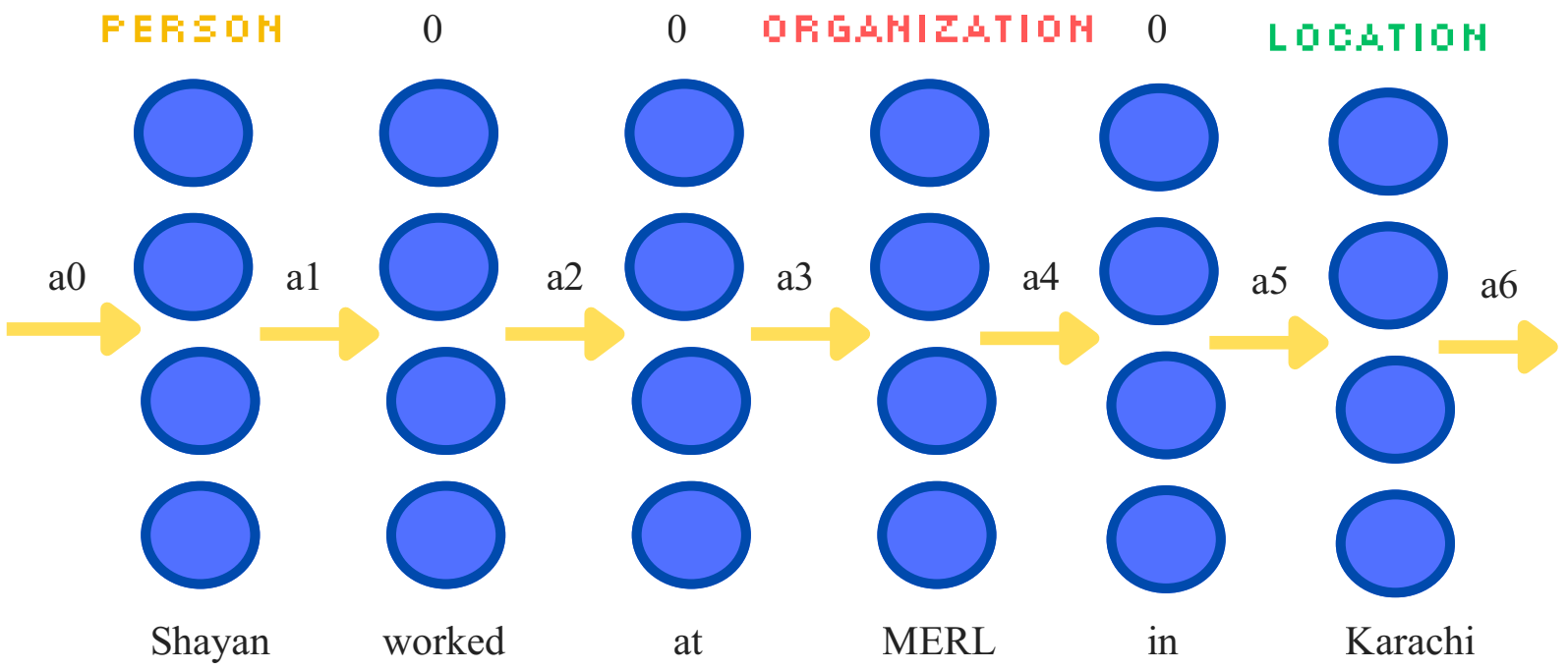
many to one



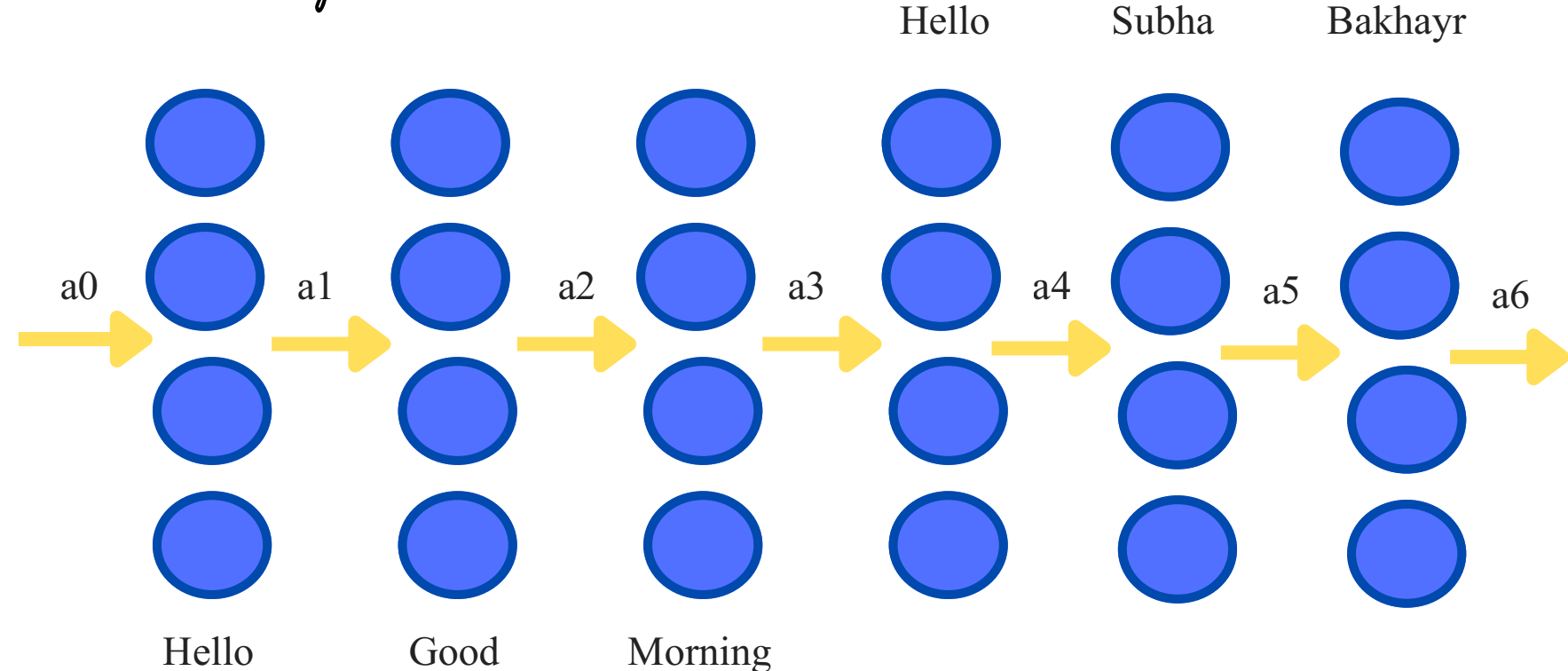
one to many



$t_x = t_y$

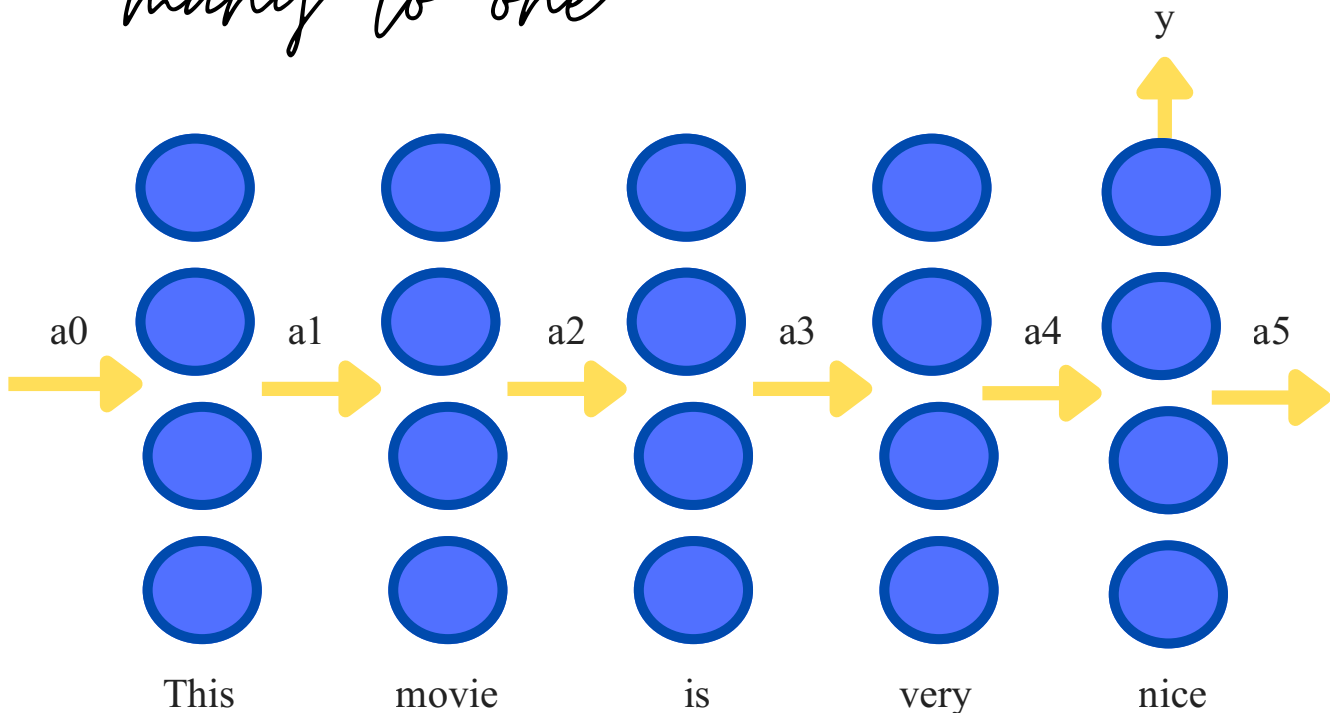


many to many

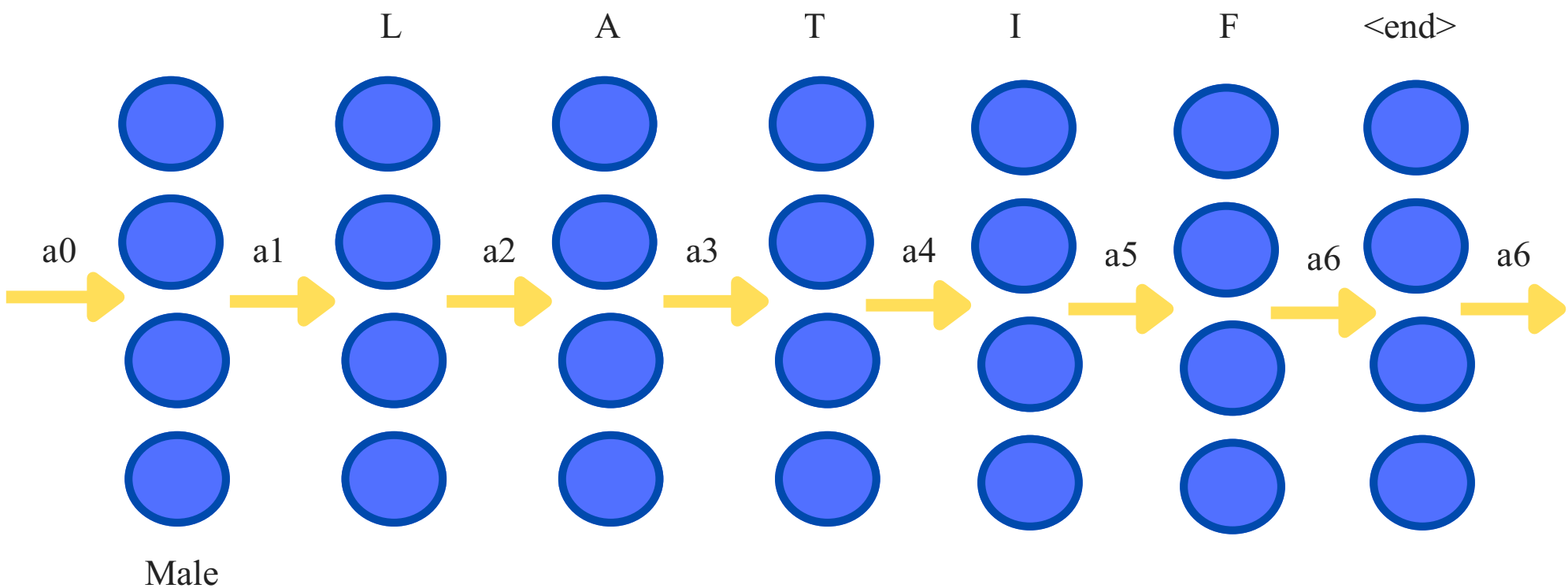


Types of RNNs

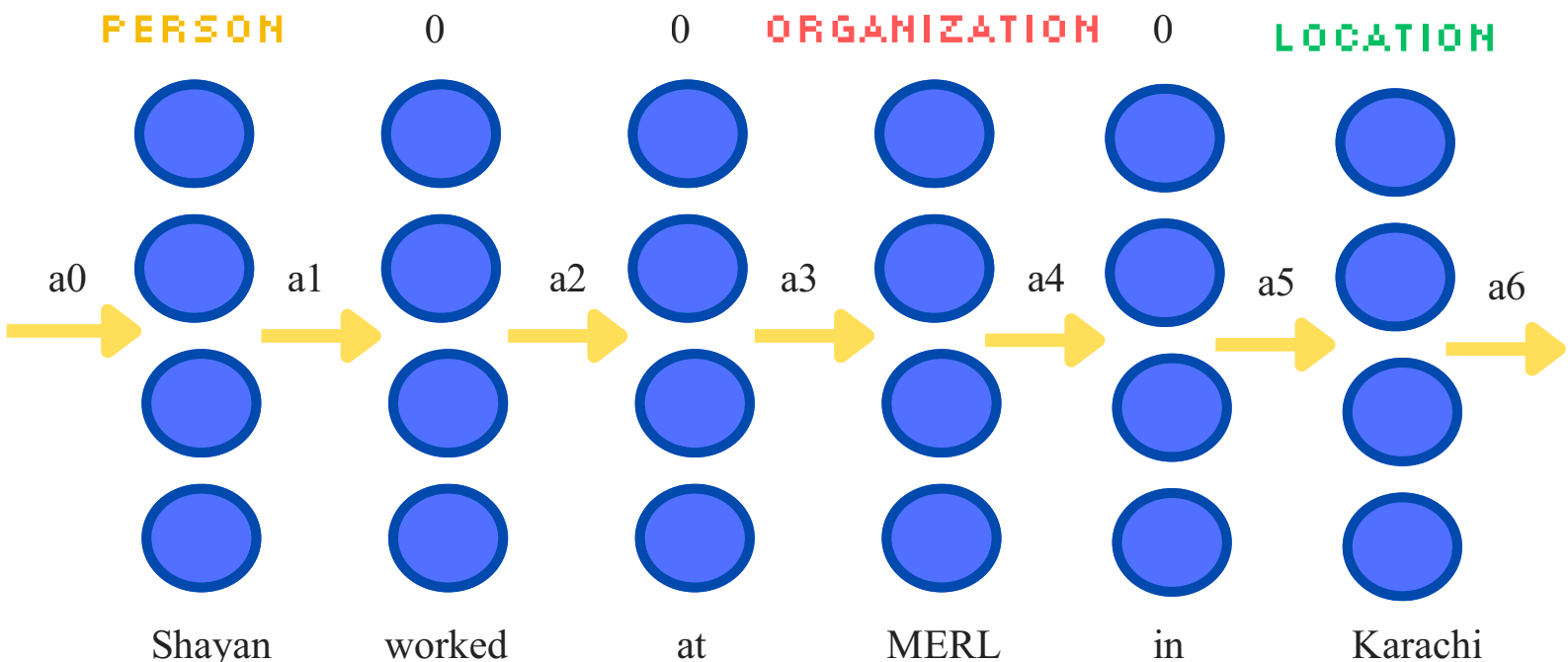
many to one



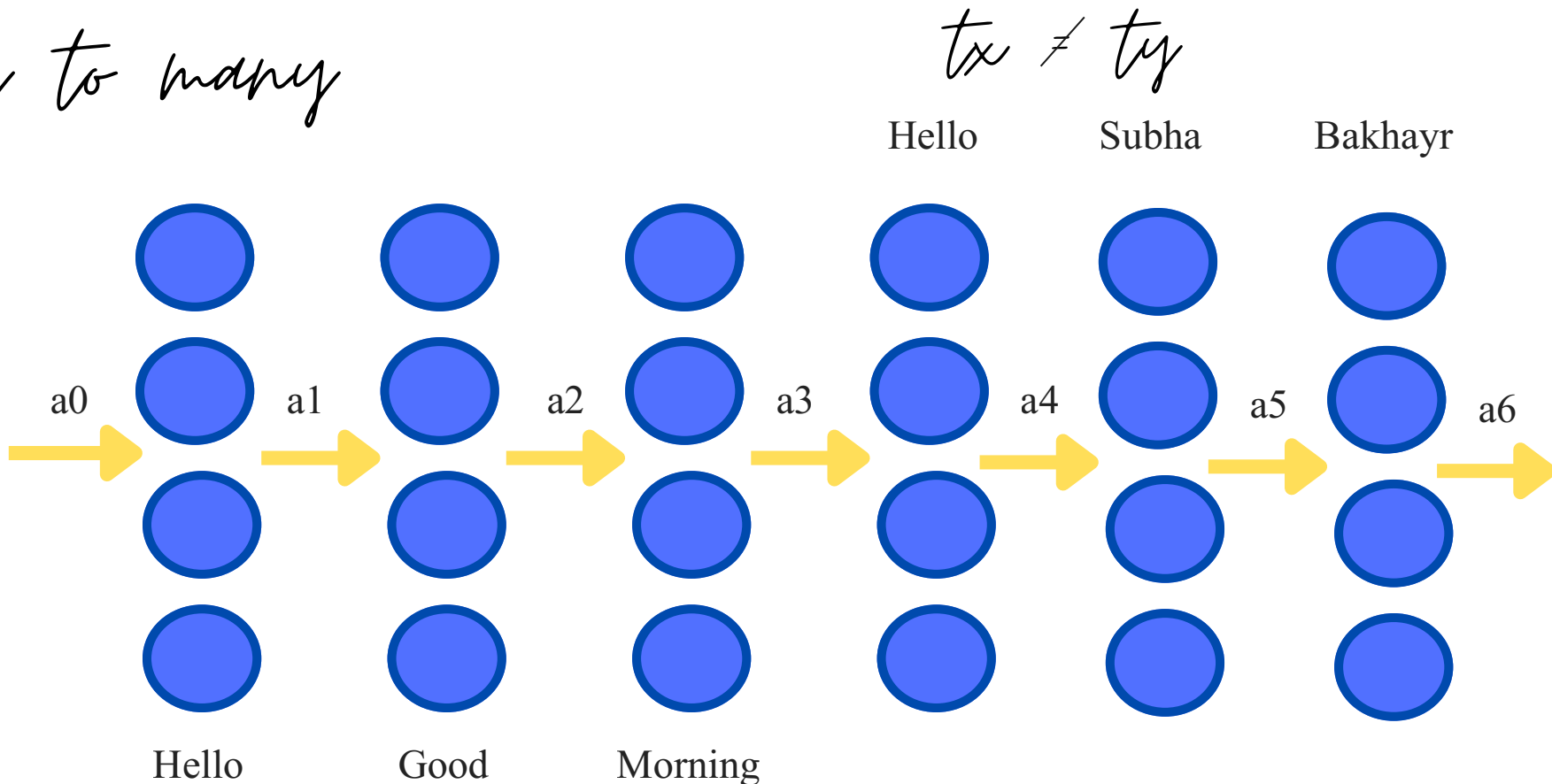
one to many



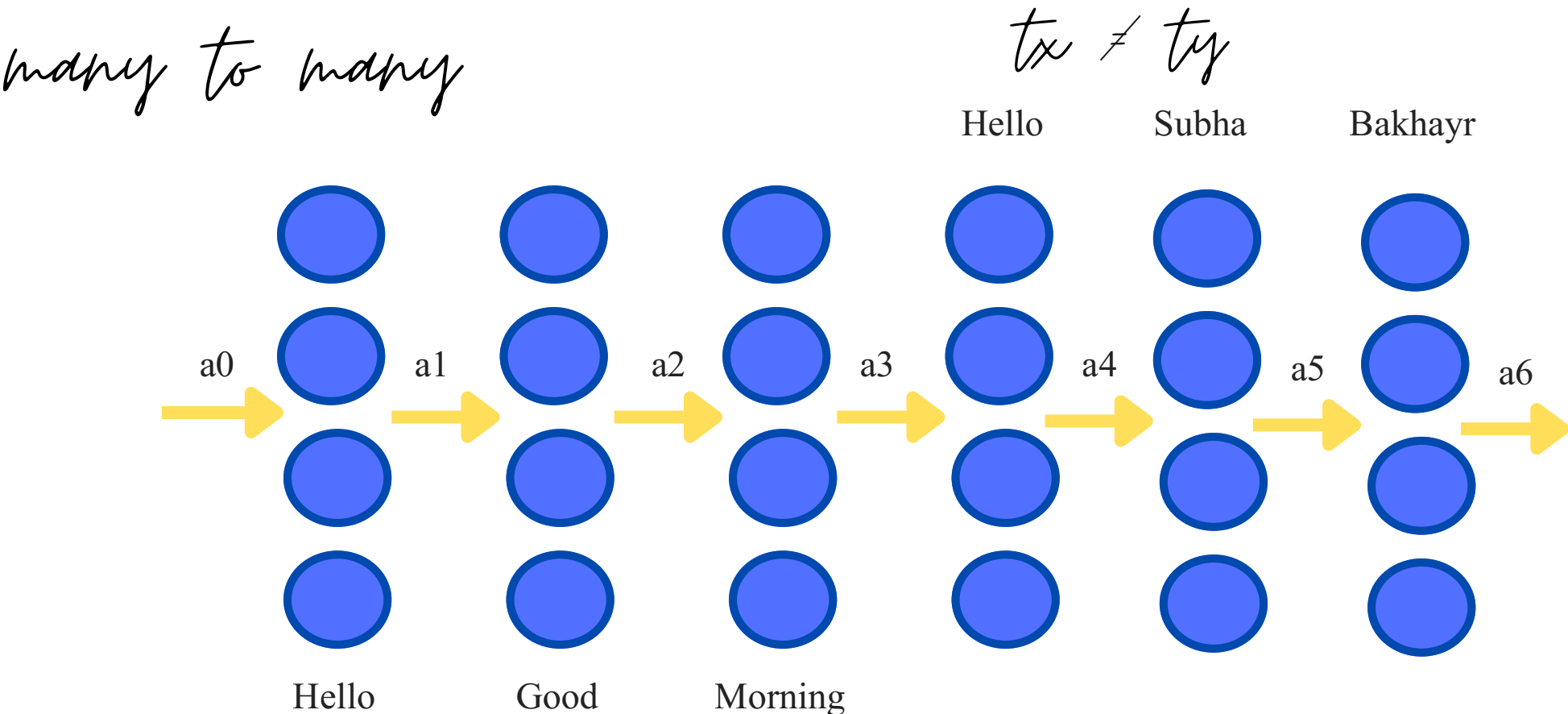
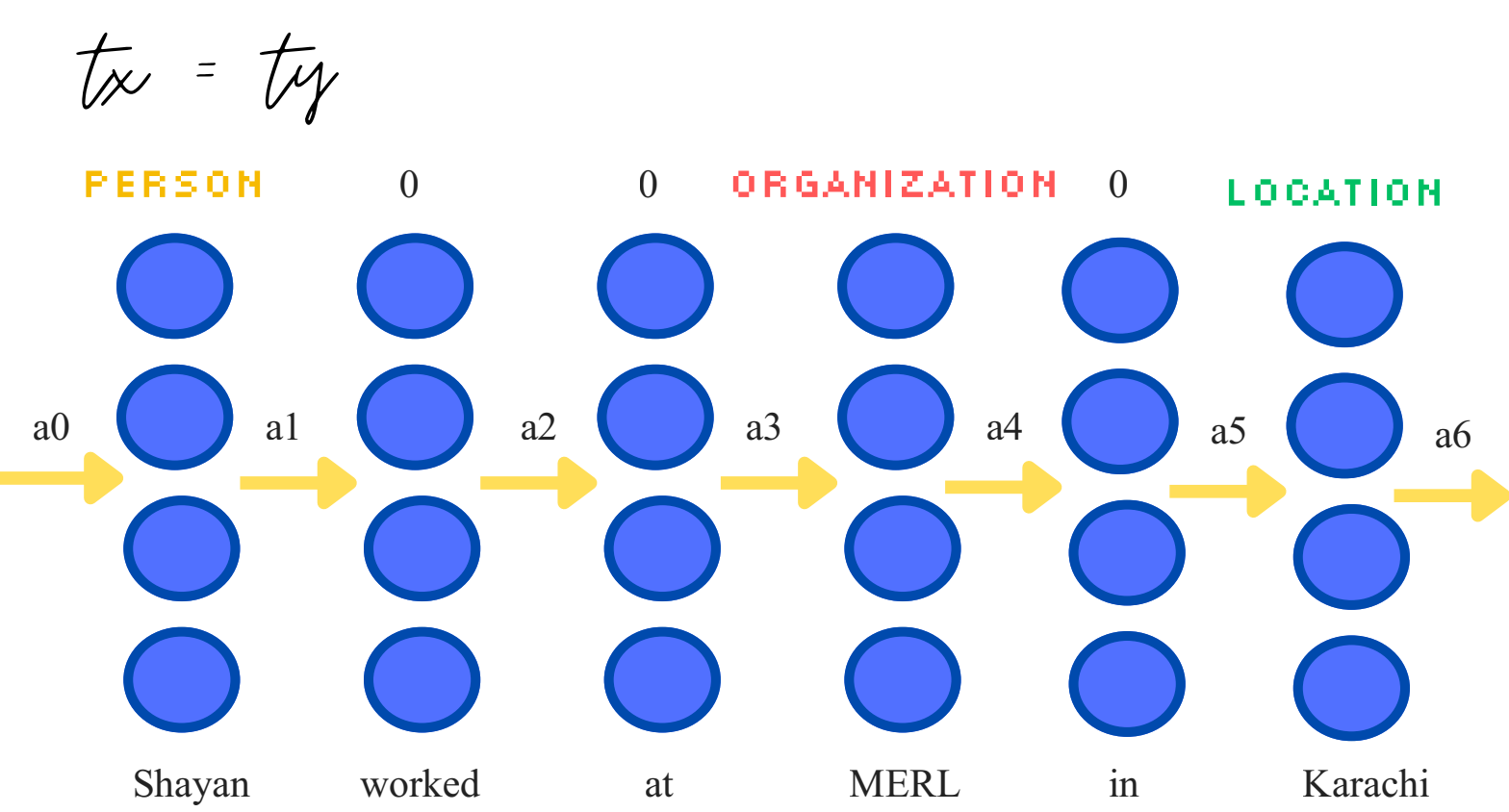
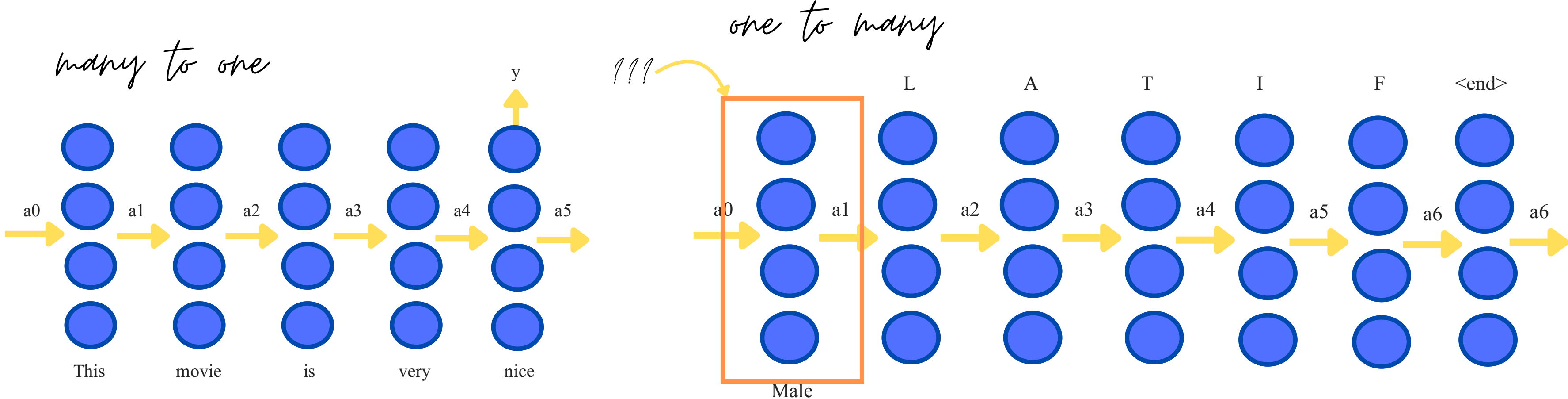
$t_x = t_y$



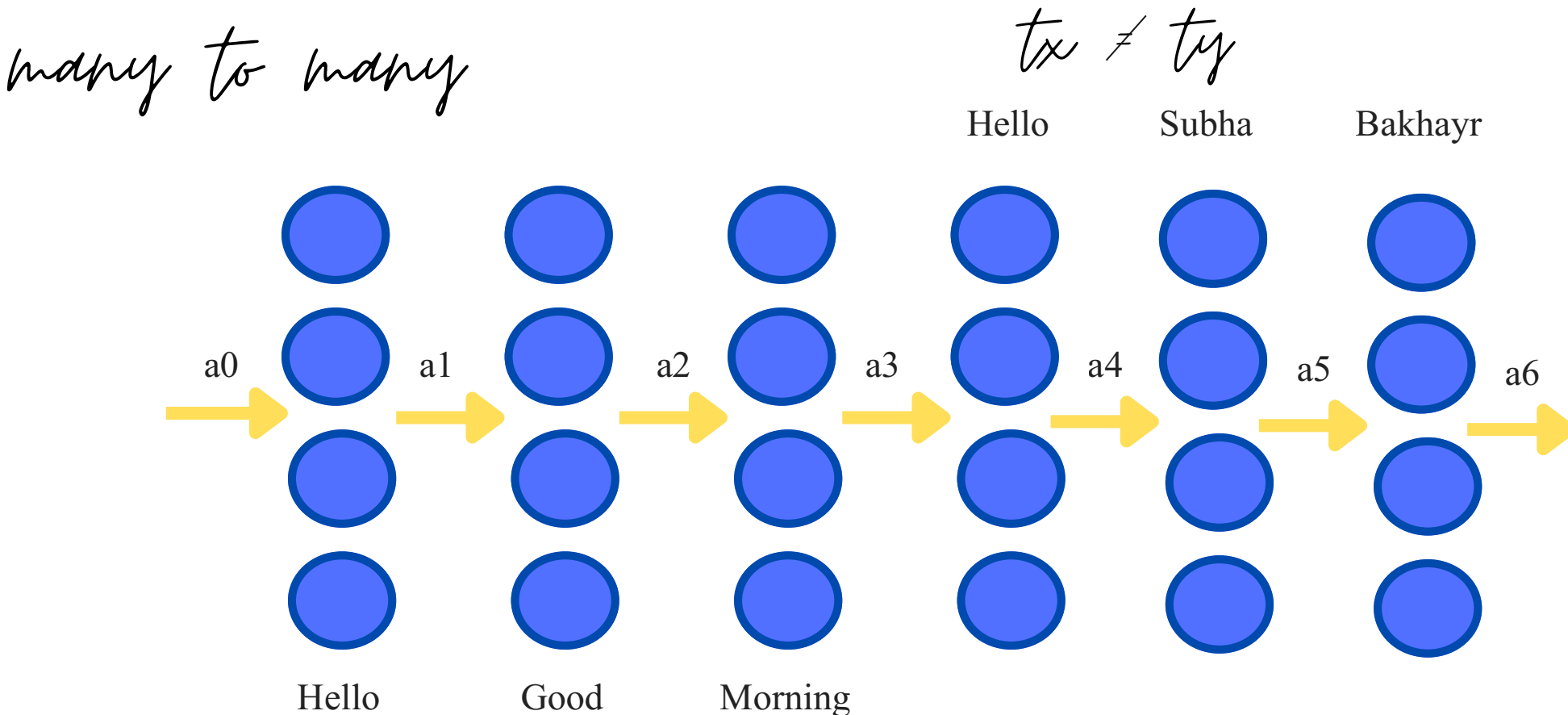
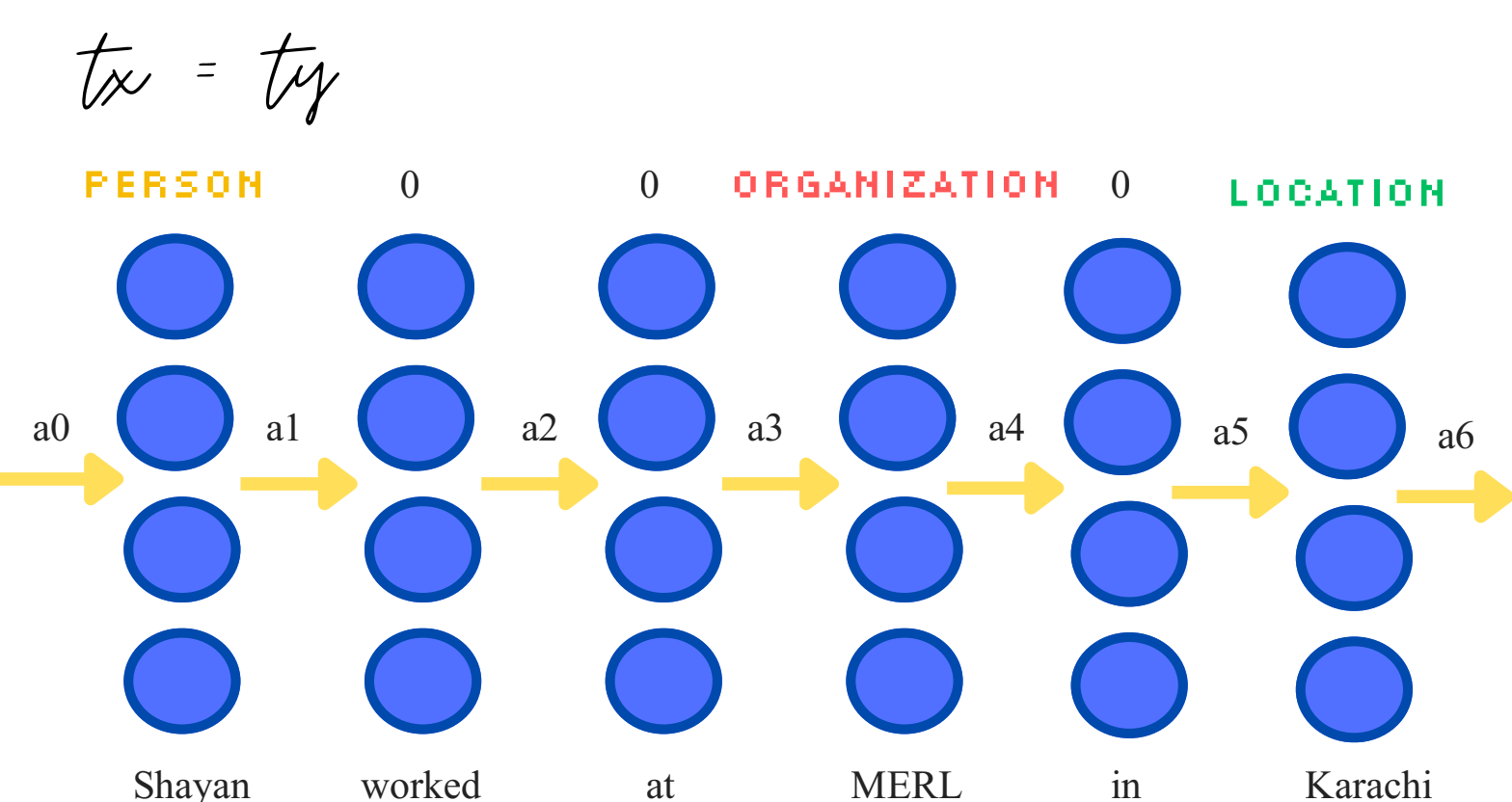
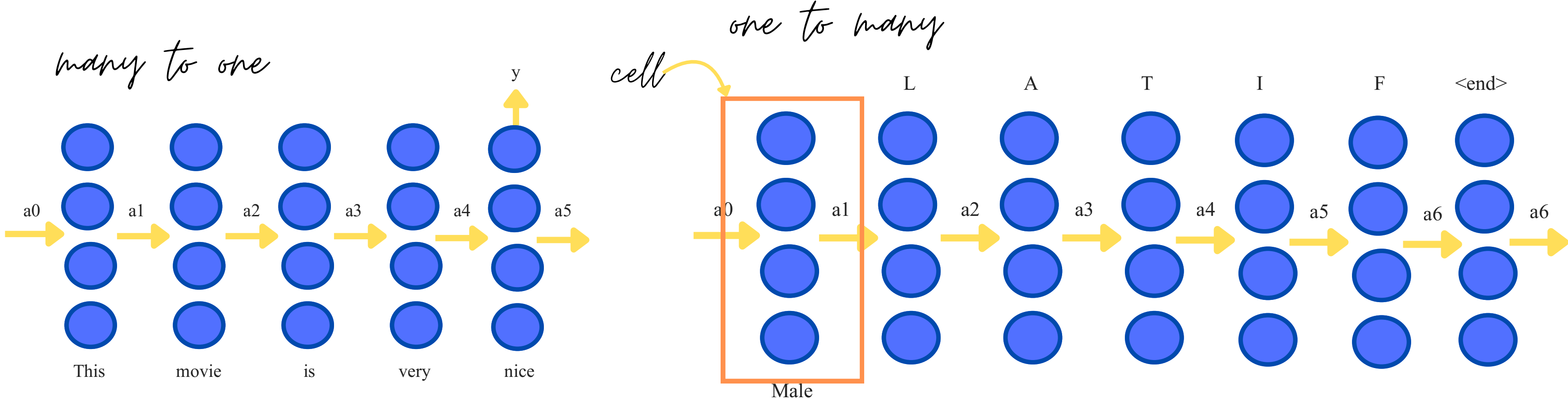
many to many



Types of RNNs



Types of RNNs



RNN Deep Dive

Shayan worked at MERL in Karachi

RNN Deep Dive



VOCABULARY

Shayan worked at MERL in Karachi

RNN Deep Dive

a	1
ali	2
.	.
.	.
. <i>Shayan</i>	. 3456
.	.
.	.
.	.
.	.
. <i>worked</i>	. 8989
.	.
Zebra	9999
<unk>	10,000

VOCABULARY

Shayan worked at MERL in Karachi

RNN Deep Dive

a	1
ali	2
.	.
.	.
. <i>Shayan</i>	. 3456
.	.
.	.
.	.
.	.
. <i>worked</i>	. 8989
.	.
Zebra	9999
<unk>	10,000

VOCABULARY

3456
Shayan

8989
worked

at

MERL

in

Karachi

RNN Deep Dive

a	1	0
ali	2	0
.	.	.
.	.	.
. <i>Shayan</i>	. 3456	1
.	.	.
.	.	.
.	.	.
.	.	.
. <i>worked</i>	. 8989	.
.	.	.
Zebra	9999	0
<unk>	10,000	0

VOCABULARY

3456
Shayan

8989
worked

at

MERL

in

Karachi

RNN Deep Dive

a	1	0	0
ali	2	0	0
.	.	.	.
.	.	.	.
. <i>Shayan</i>	. 3456	1	0
.	.	.	.
.	.	.	.
.	.	.	.
.	.	.	.
. <i>worked</i>	. 8989	.	1
.	.	.	.
Zebra	9999	0	0
<unk>	10,000	0	0

VOCABULARY

3456
Shayan

8989
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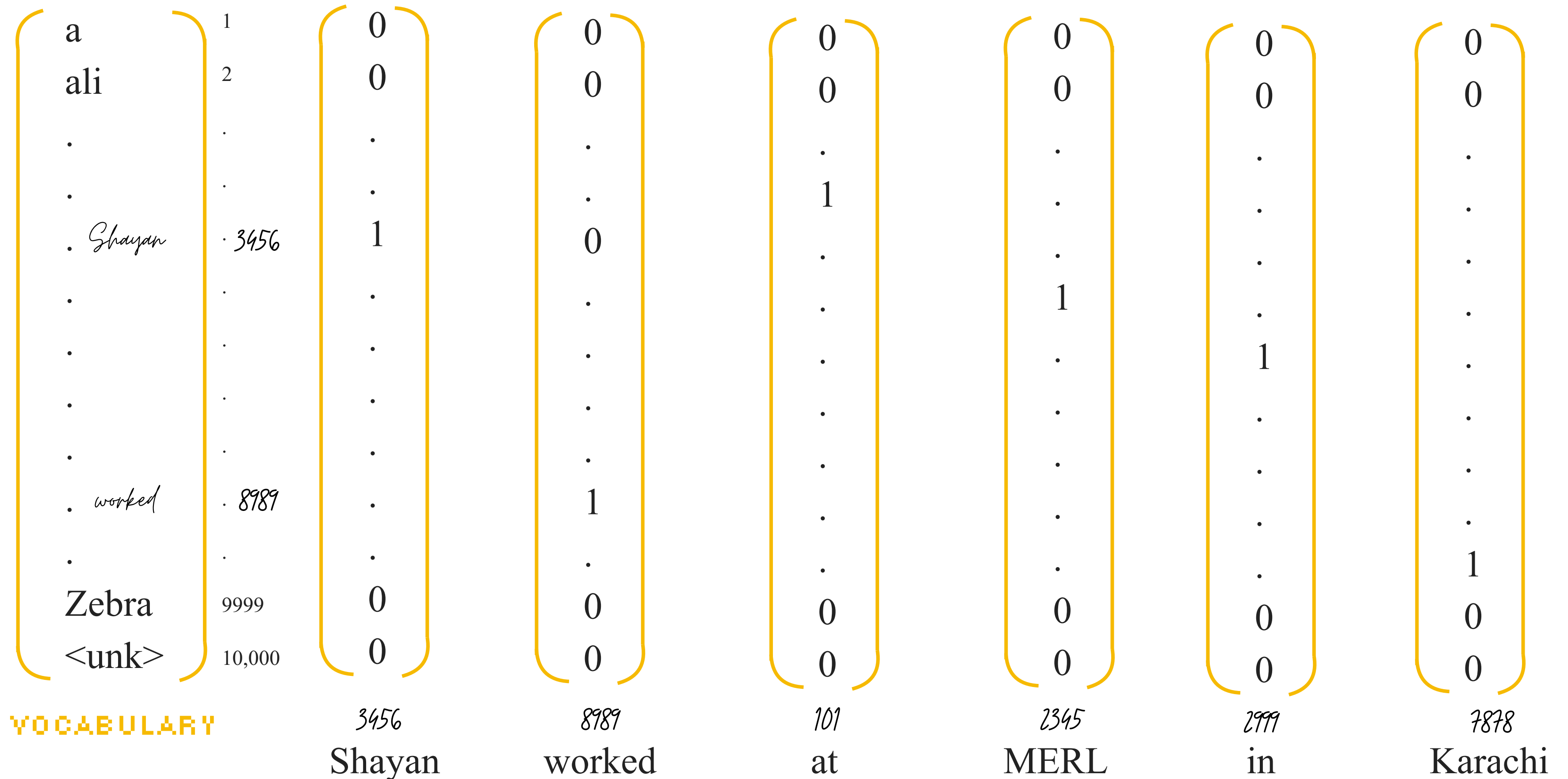
at

MERL

in

Karachi

RNN Deep Dive



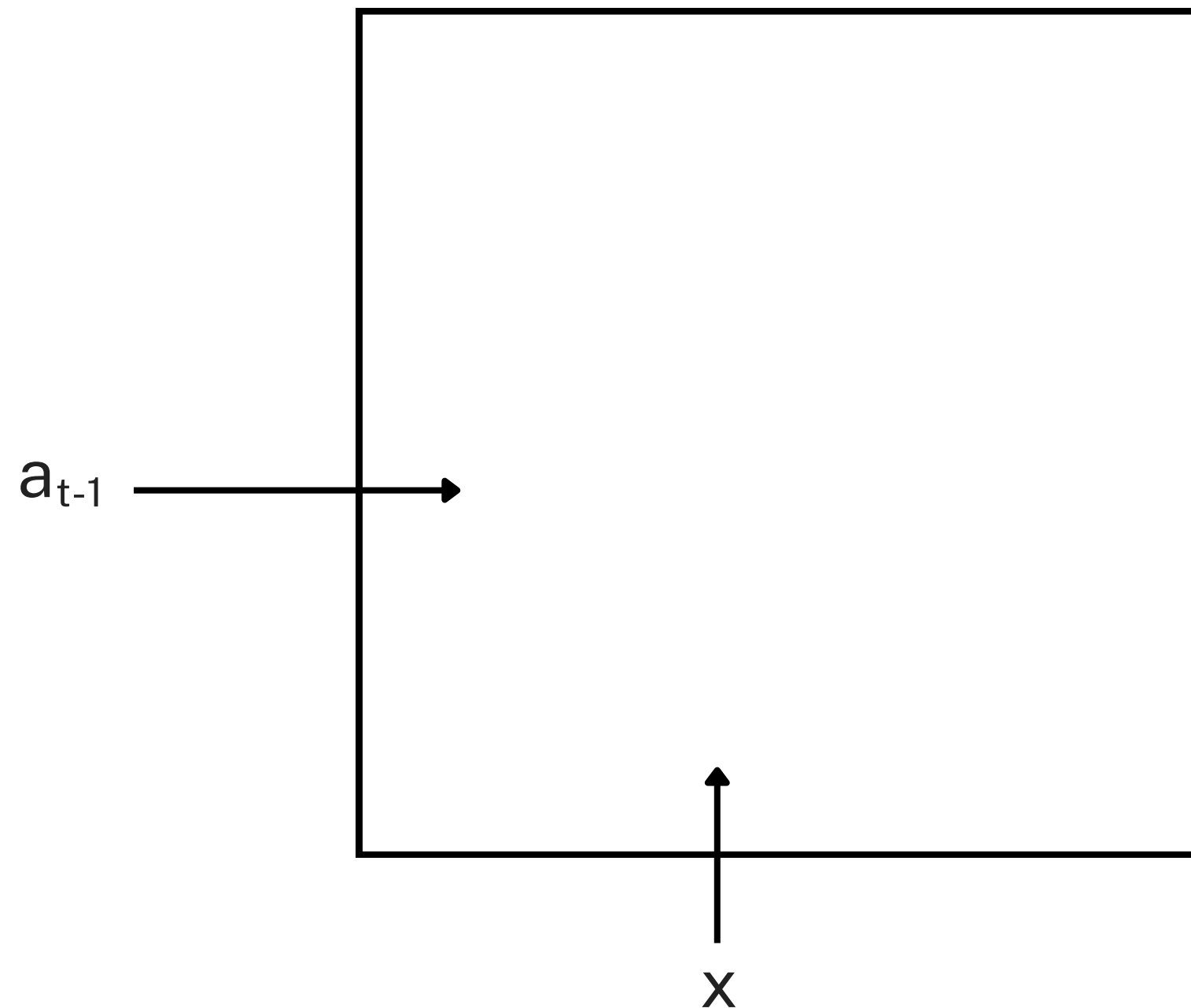
RNN Deep Dive

For ANN

$$a = \text{function}(W * a_{t-1} + B)$$

For RNN

$$a = \text{function}(W_{aa} * a_{t-1} + W_{ax} * X + b_a)$$

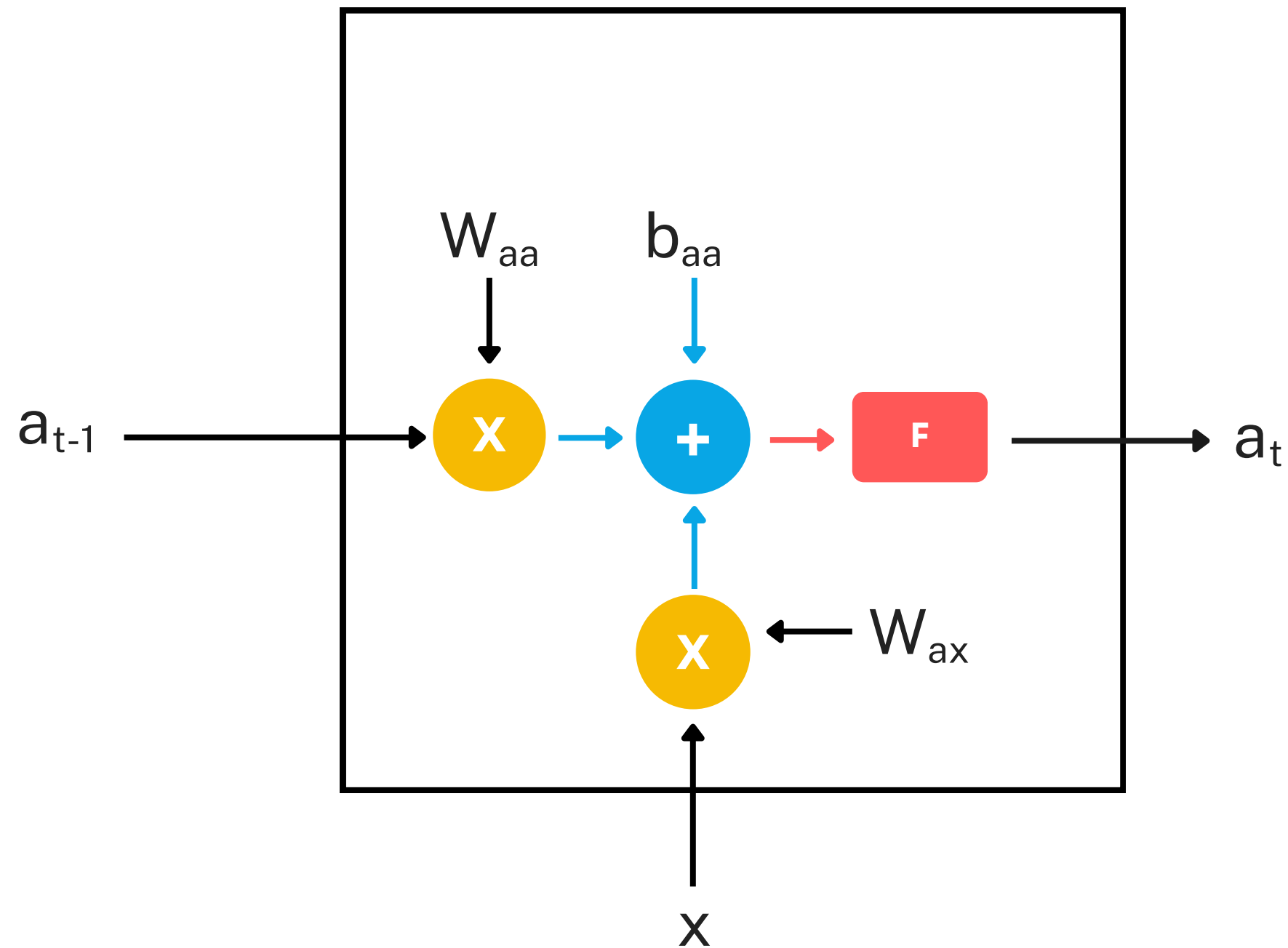


Shayan $\longrightarrow$ $\begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$

RNN Deep Dive

For ANN

$$a = \text{function}(W^*a_{t-1} + B)$$



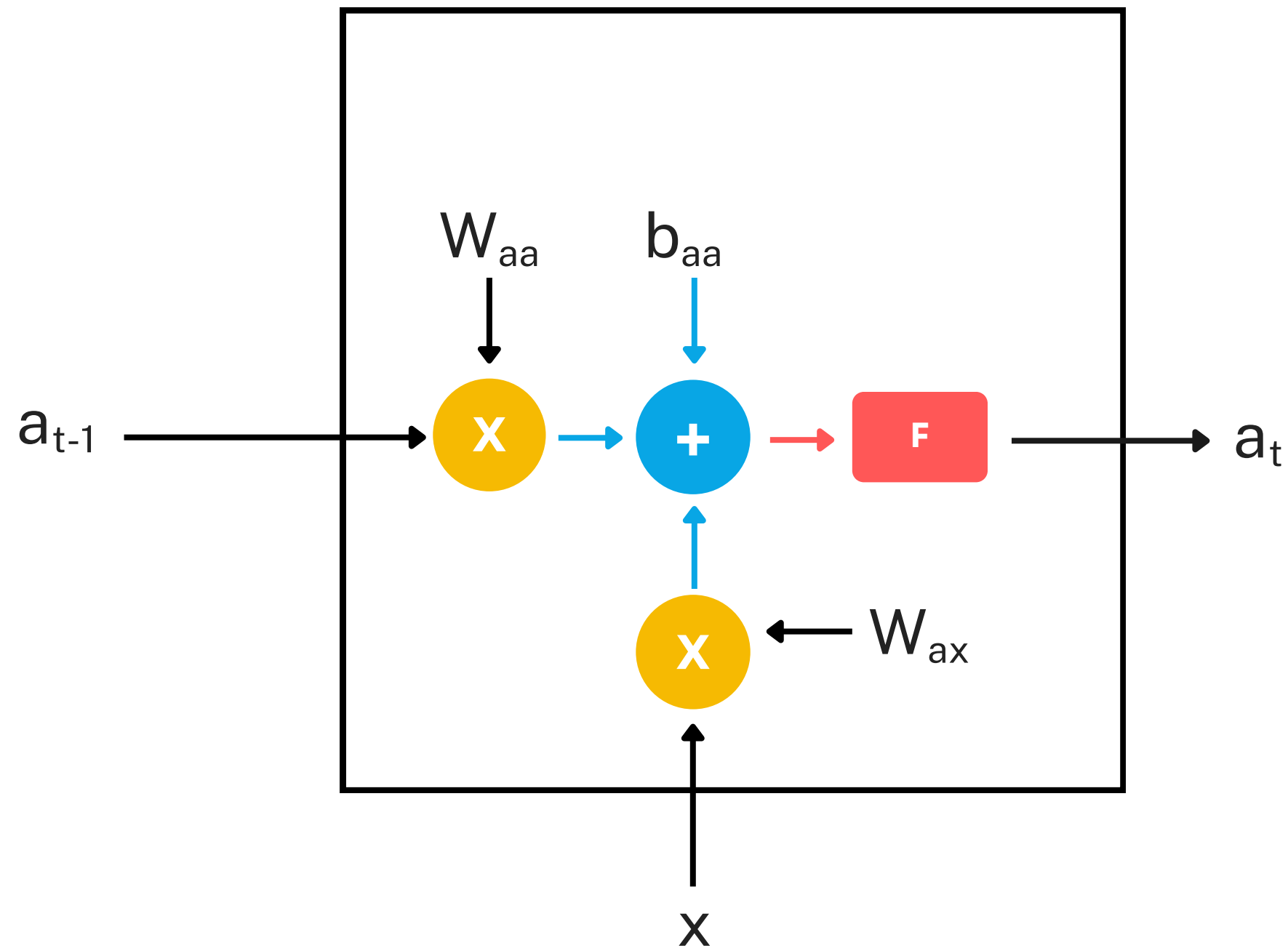
For RNN

$$a = \text{function}(W_{aa}^*a_{t-1} + W_{ax}^*X + b_a)$$

Shayan $\longrightarrow$ $\begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$

RNN Deep Dive

y



Shayan $\longrightarrow \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$

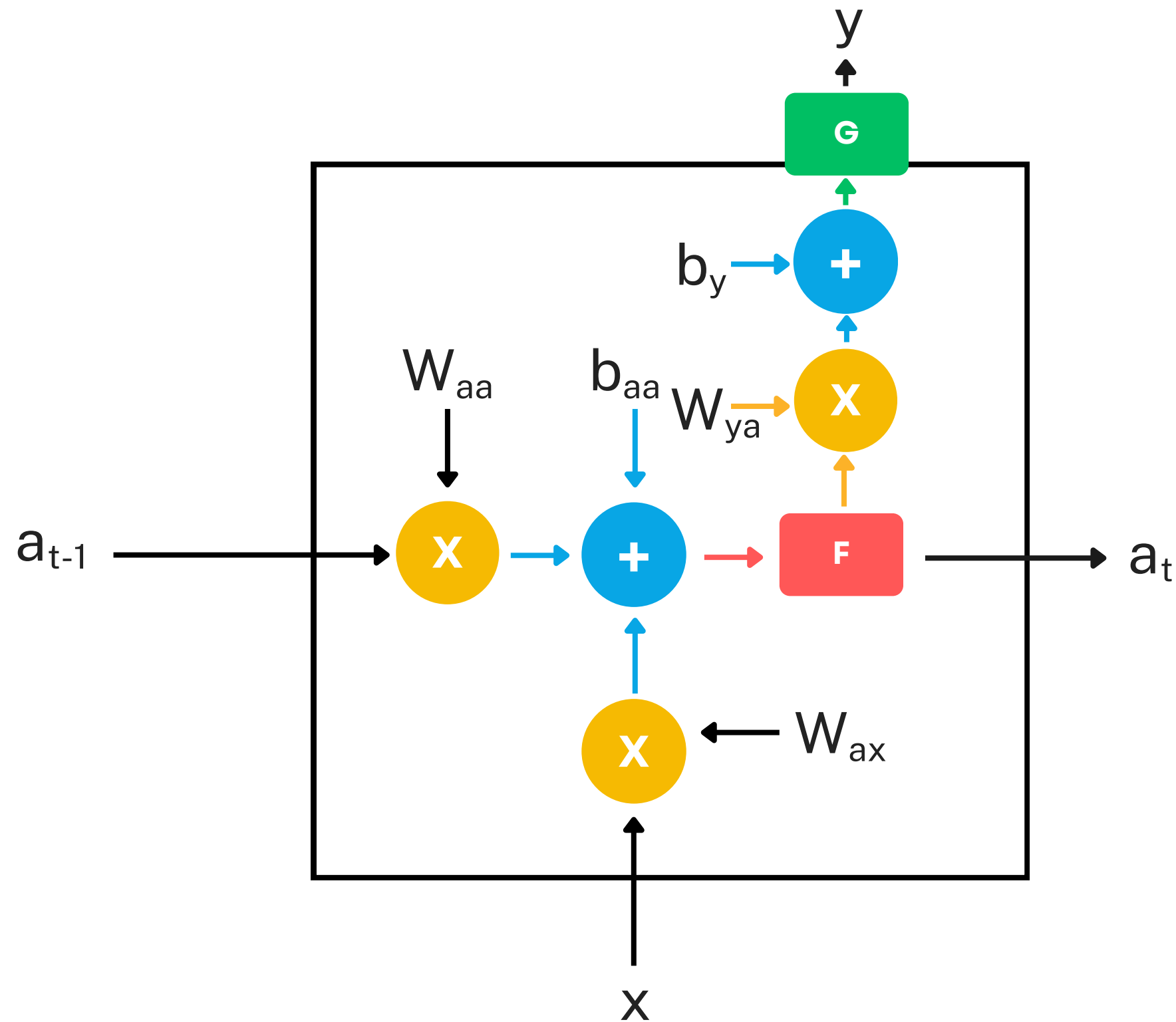
For ANN

$$a = \text{function}(W^*a_{t-1} + B)$$

For RNN

$$a = \text{function}(W_{aa}^*a_{t-1} + W_{ax}^*X + b_a)$$

RNN Deep Dive



Shayan $\longrightarrow \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$

For ANN

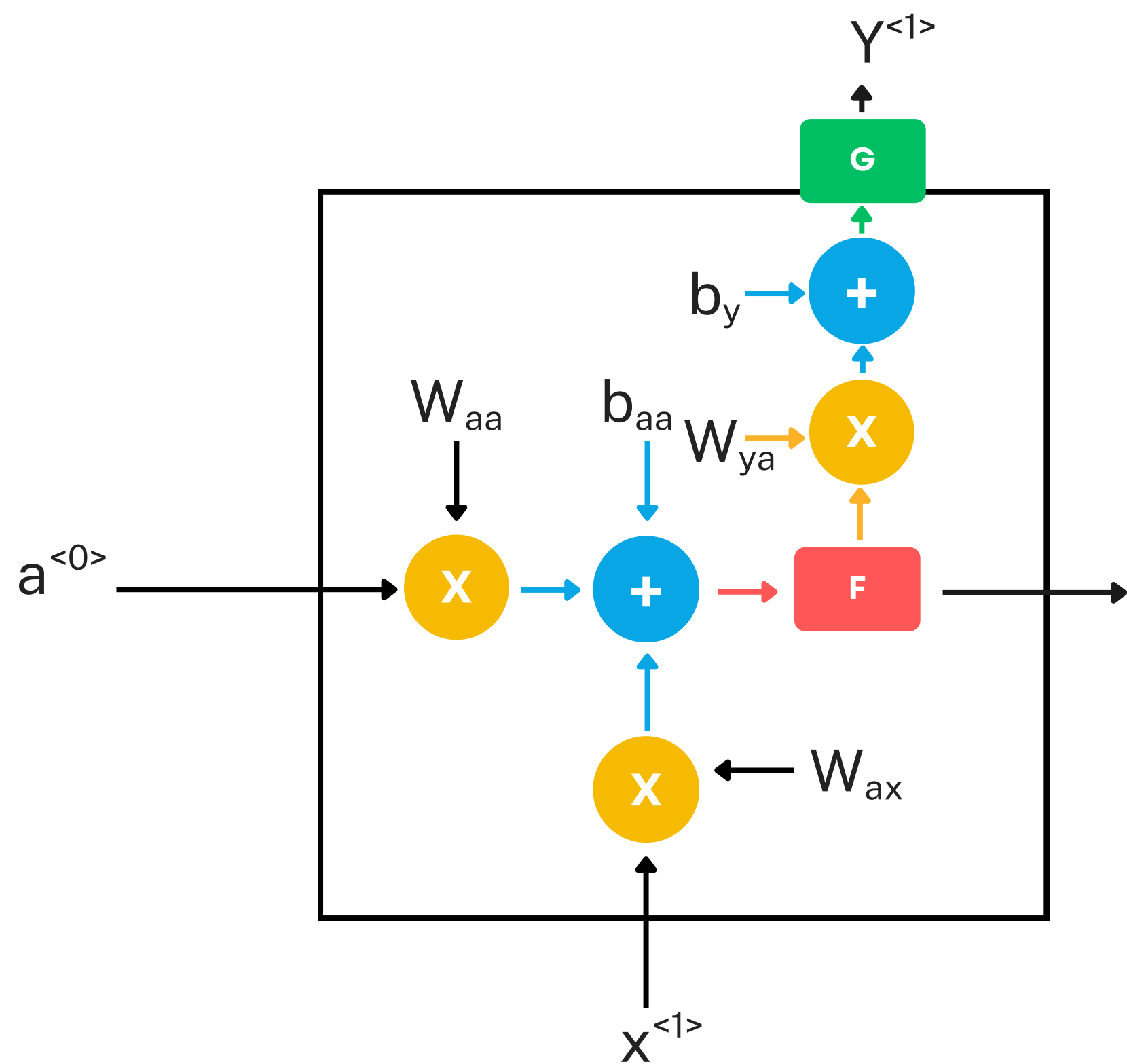
$$a = \text{function}(W * a_{t-1} + B)$$

For RNN

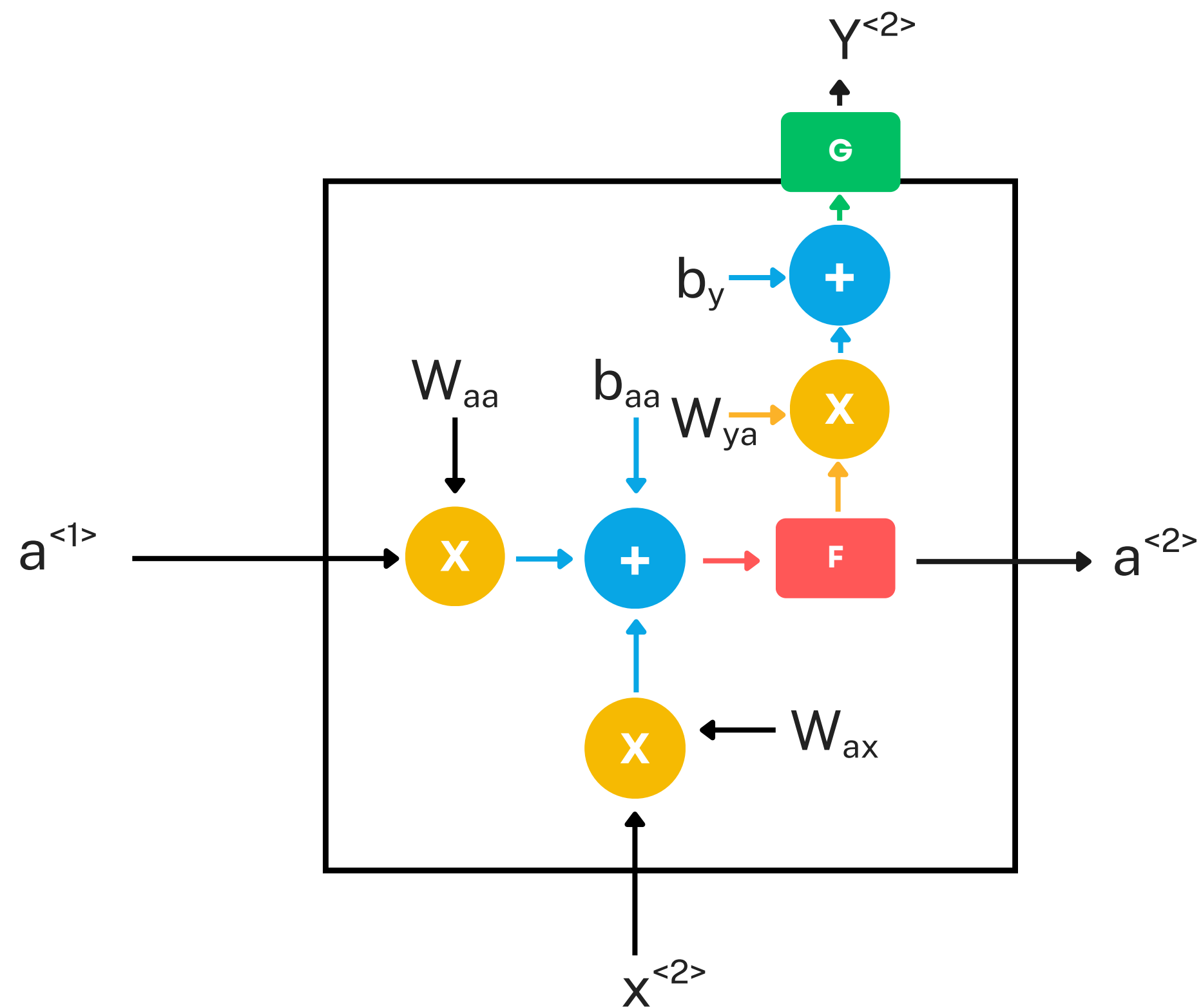
$$a = \text{function}(W_{aa} * a_{t-1} + W_{ax} * X + b_a)$$

$$y = \text{softmax}(W_{ya} * a_t + b_y)$$

RNN Deep Dive



Shayan



worked

RNN Deep Dive

W_{aa}

W_{ax}

W_{ya}

RNN Deep Dive

W_{aa}

W_{ax}

W_{ya}

RNN Deep Dive

W_{aa}

W_{ax}

W_{ya}

what quantity we are multiplying with?

RNN Deep Dive

W_{aa}
W_{ax}
W_{ya}

A diagram illustrating the weight matrices for an RNN. It shows three matrices stacked vertically: W_{aa}, W_{ax}, and W_{ya}. A pink rectangular box highlights the second column of each matrix, which corresponds to the 'a' input in the subscripts. This visualizes how the hidden state 'a' is updated by the previous hidden state 'a' (W_{aa}), the current input 'x' (W_{ax}), and the initial hidden state 'y' (W_{ya}).

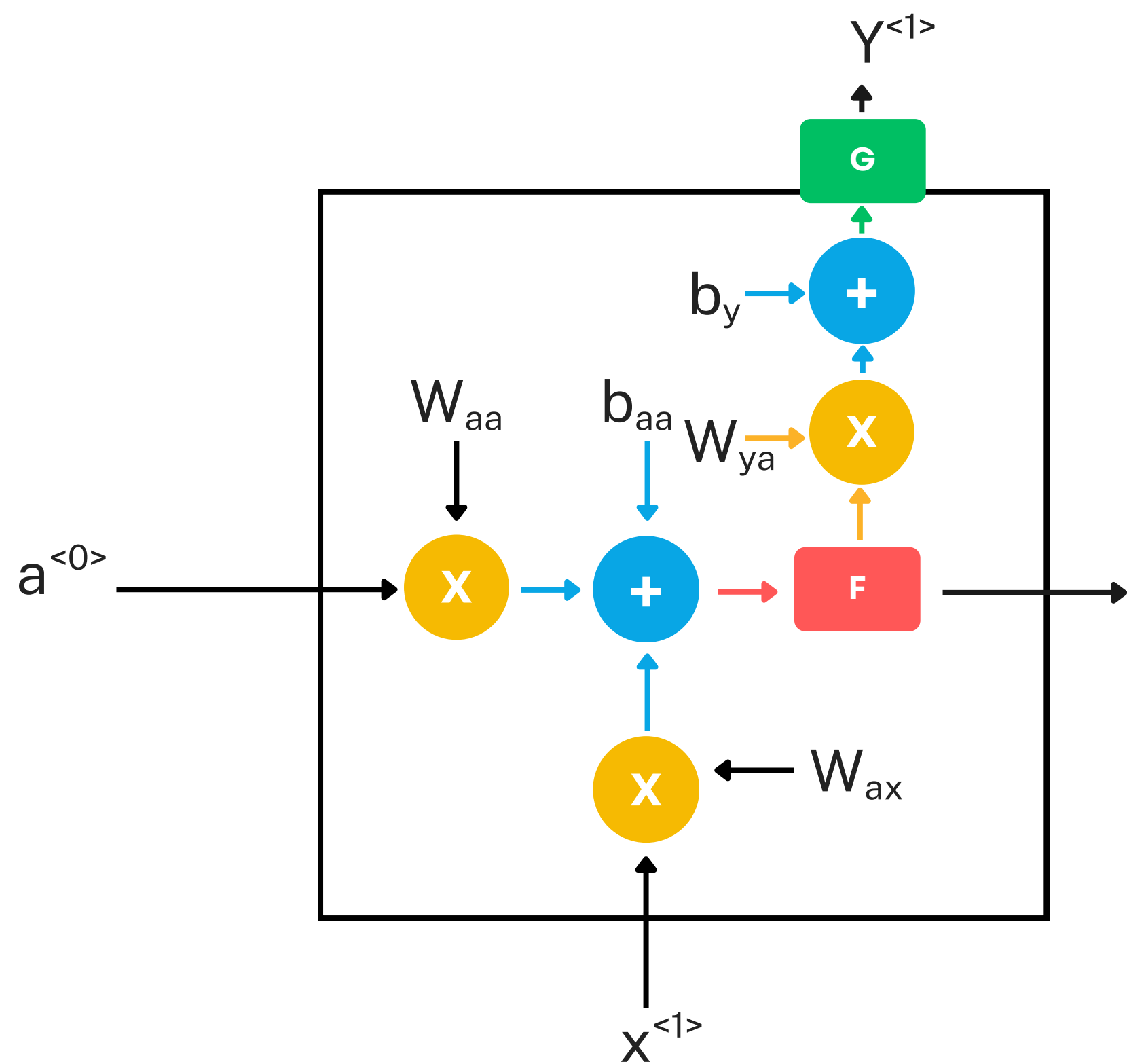
RNN Deep Dive

Waa
Wax
Wya

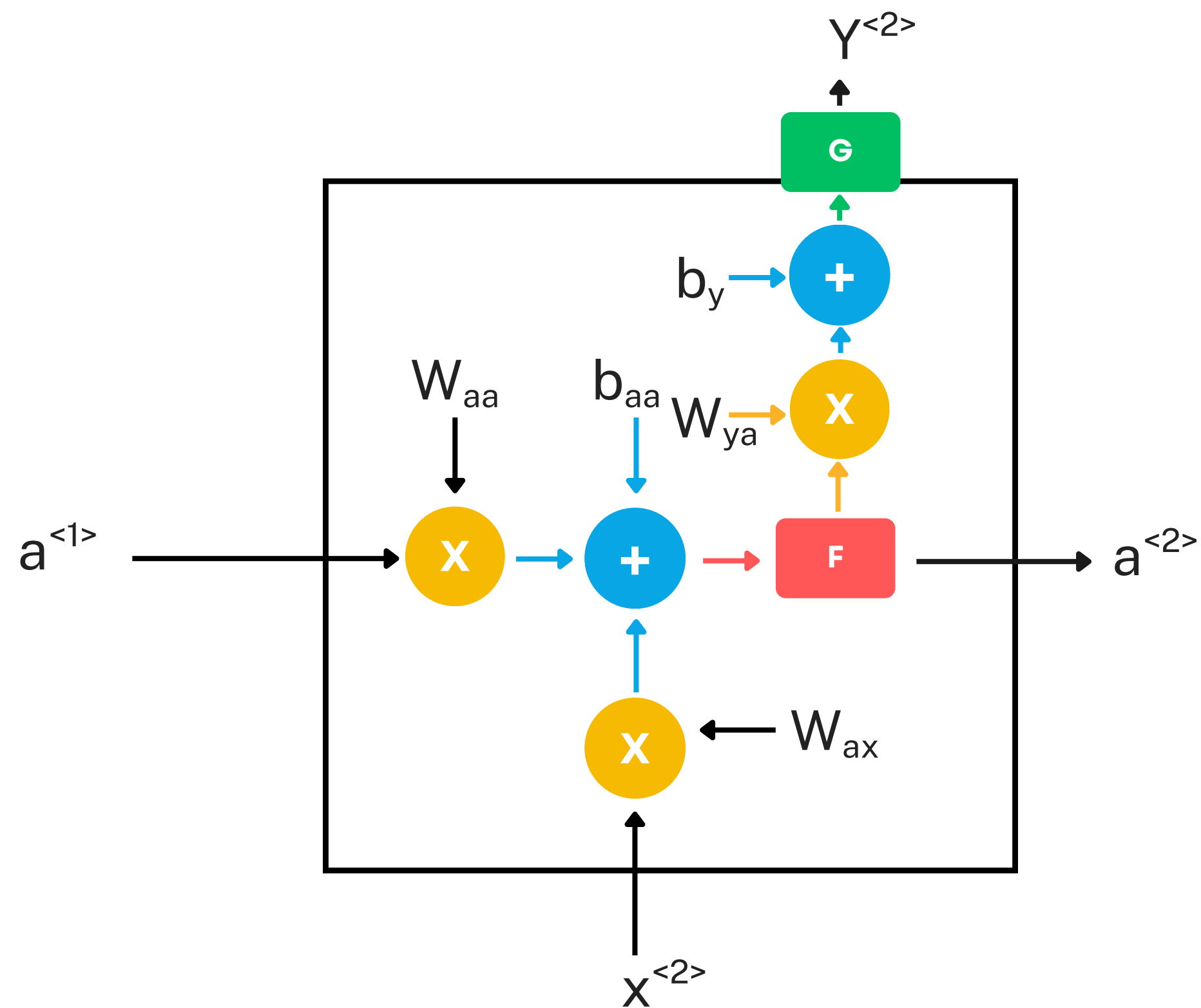


what are we producing?

RNN Deep Dive

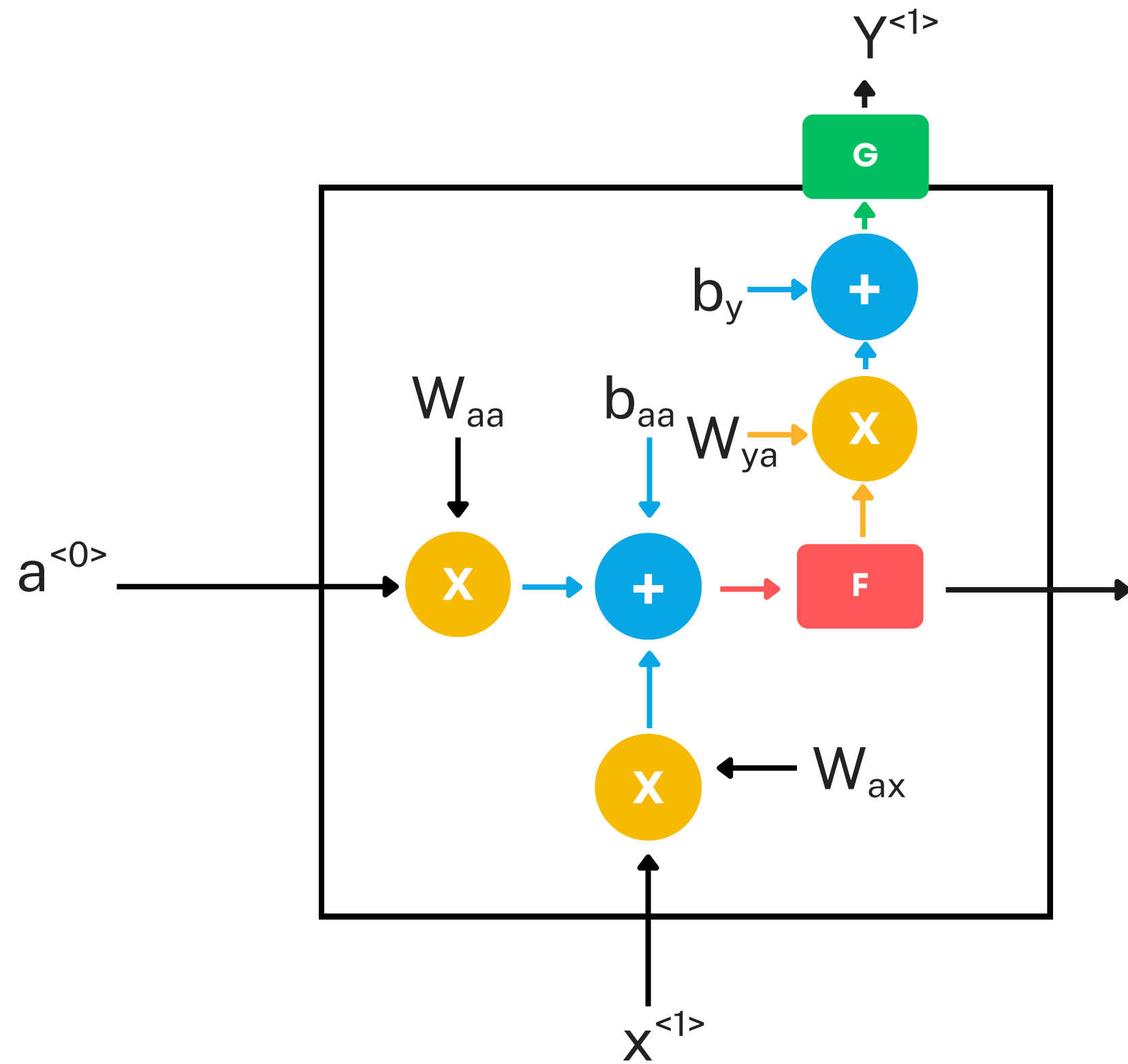


Shayan



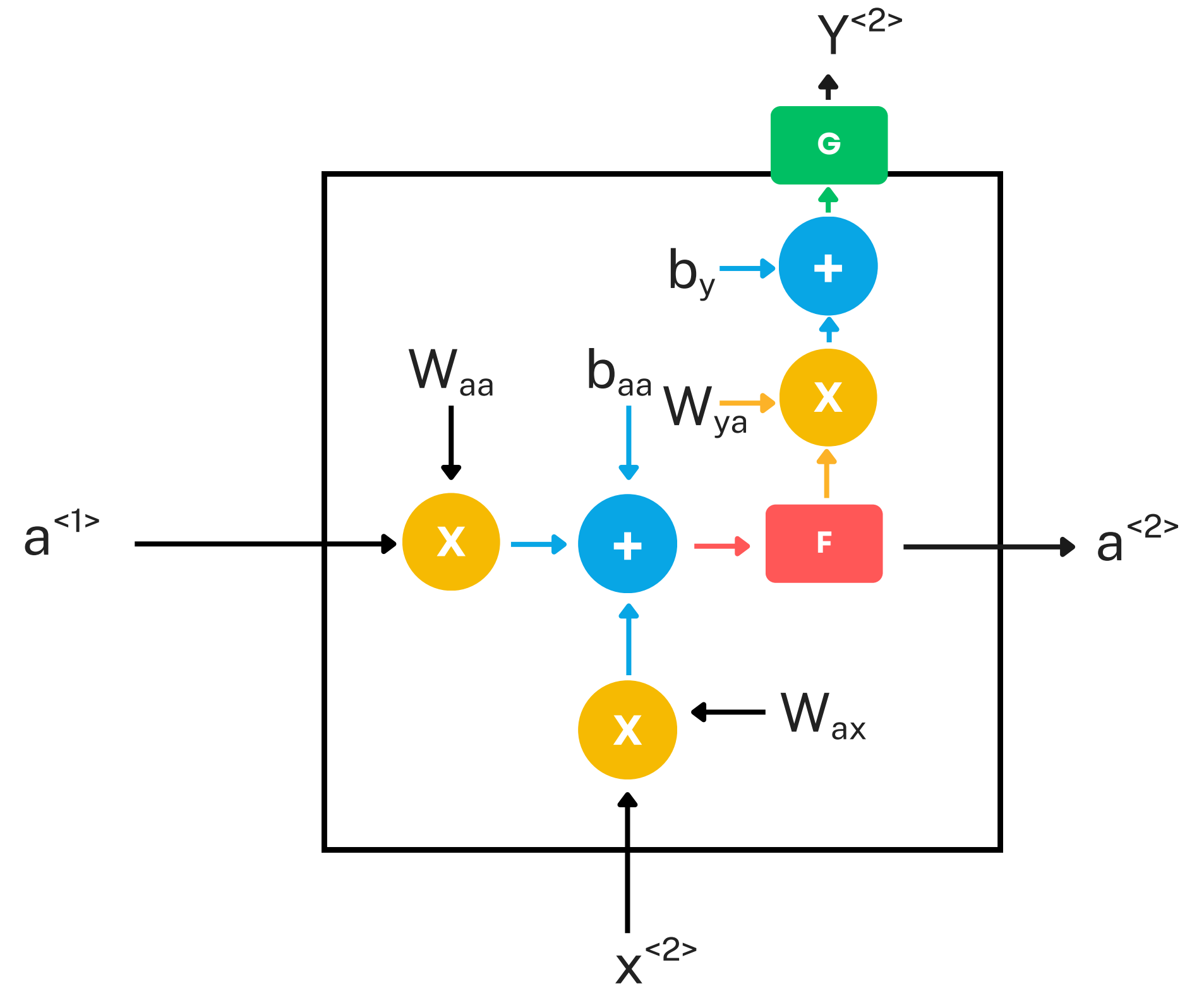
worked

RNN Deep Dive



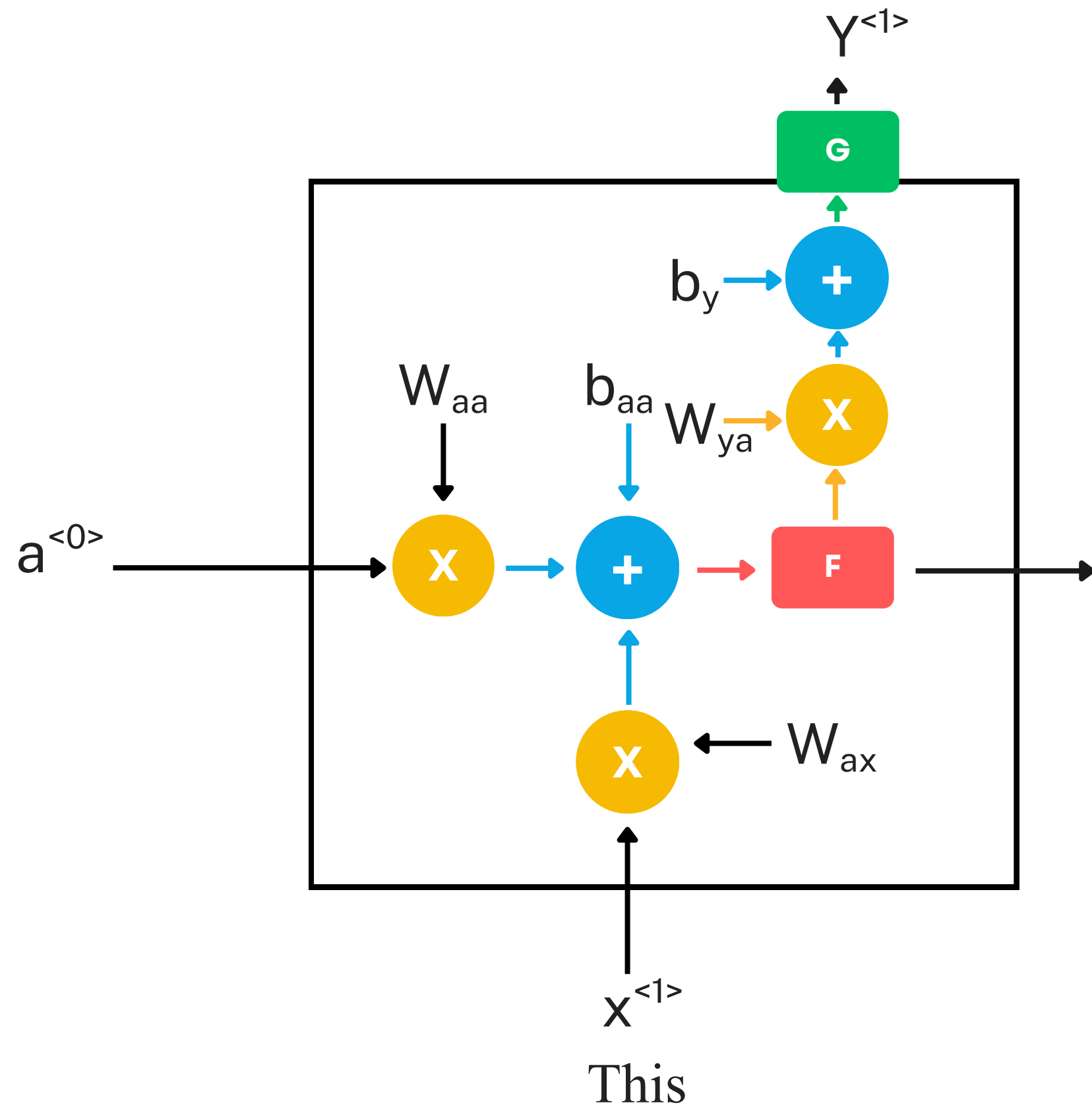
Shayan

many to many

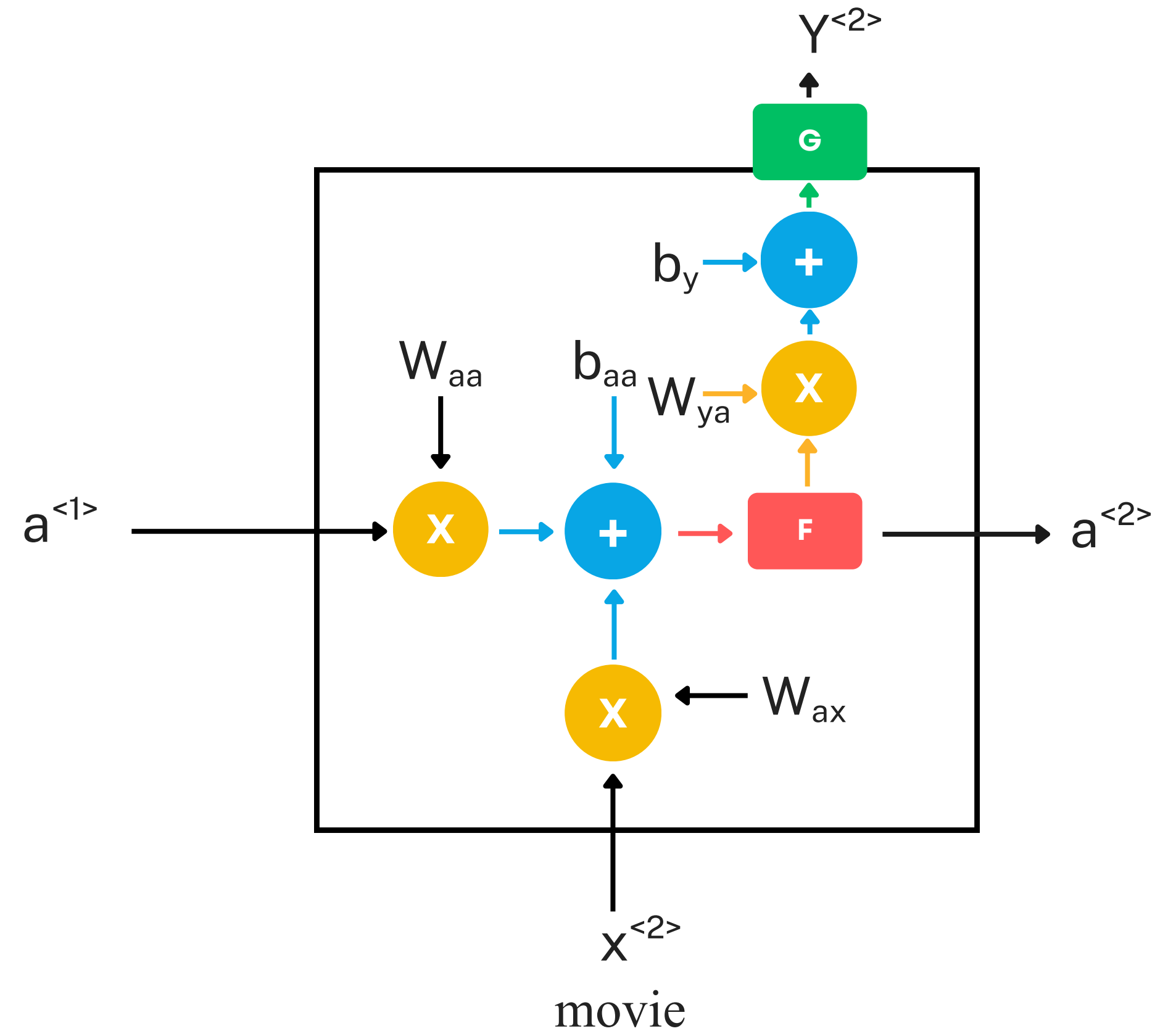


worked

RNN Deep Dive

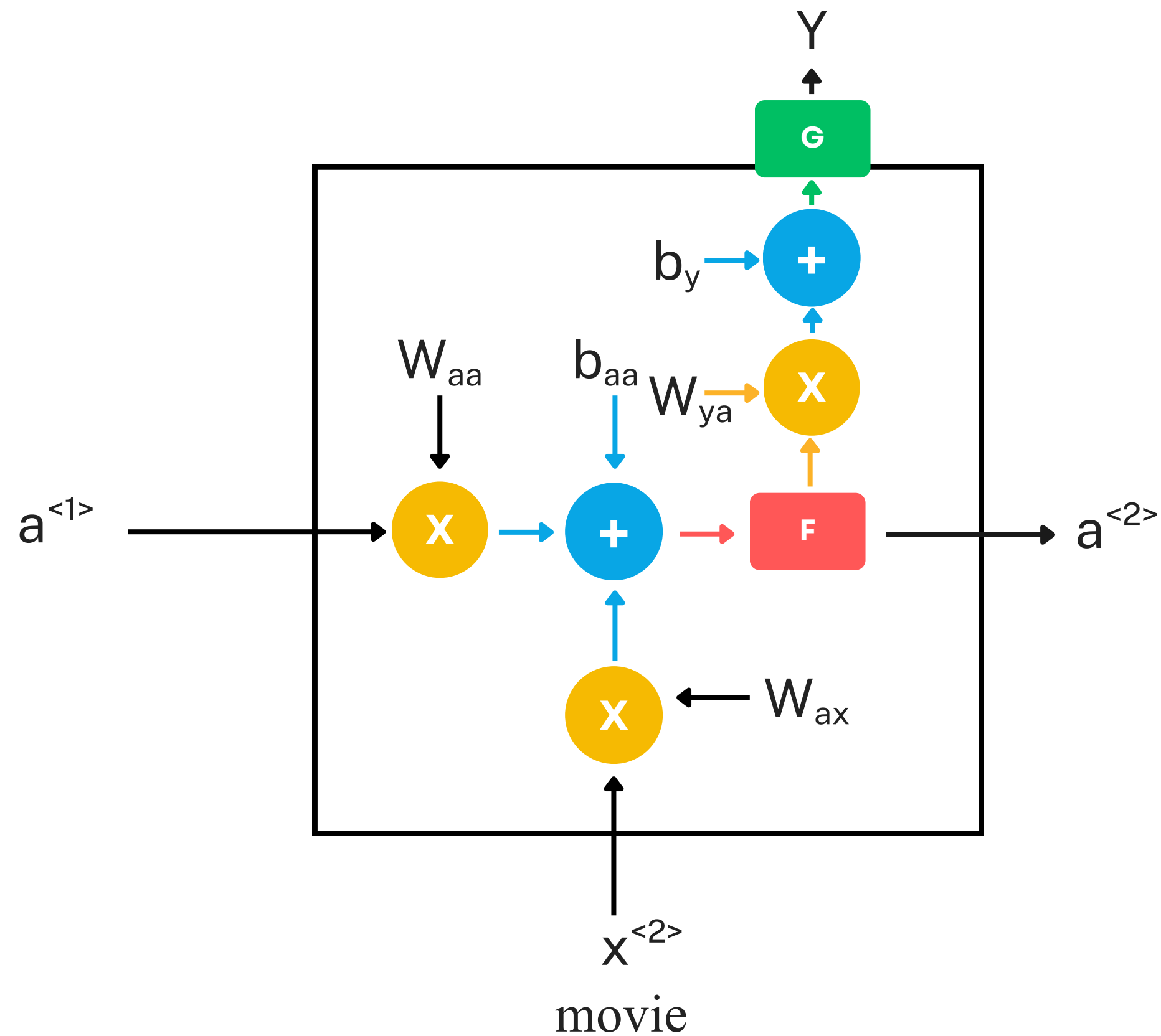
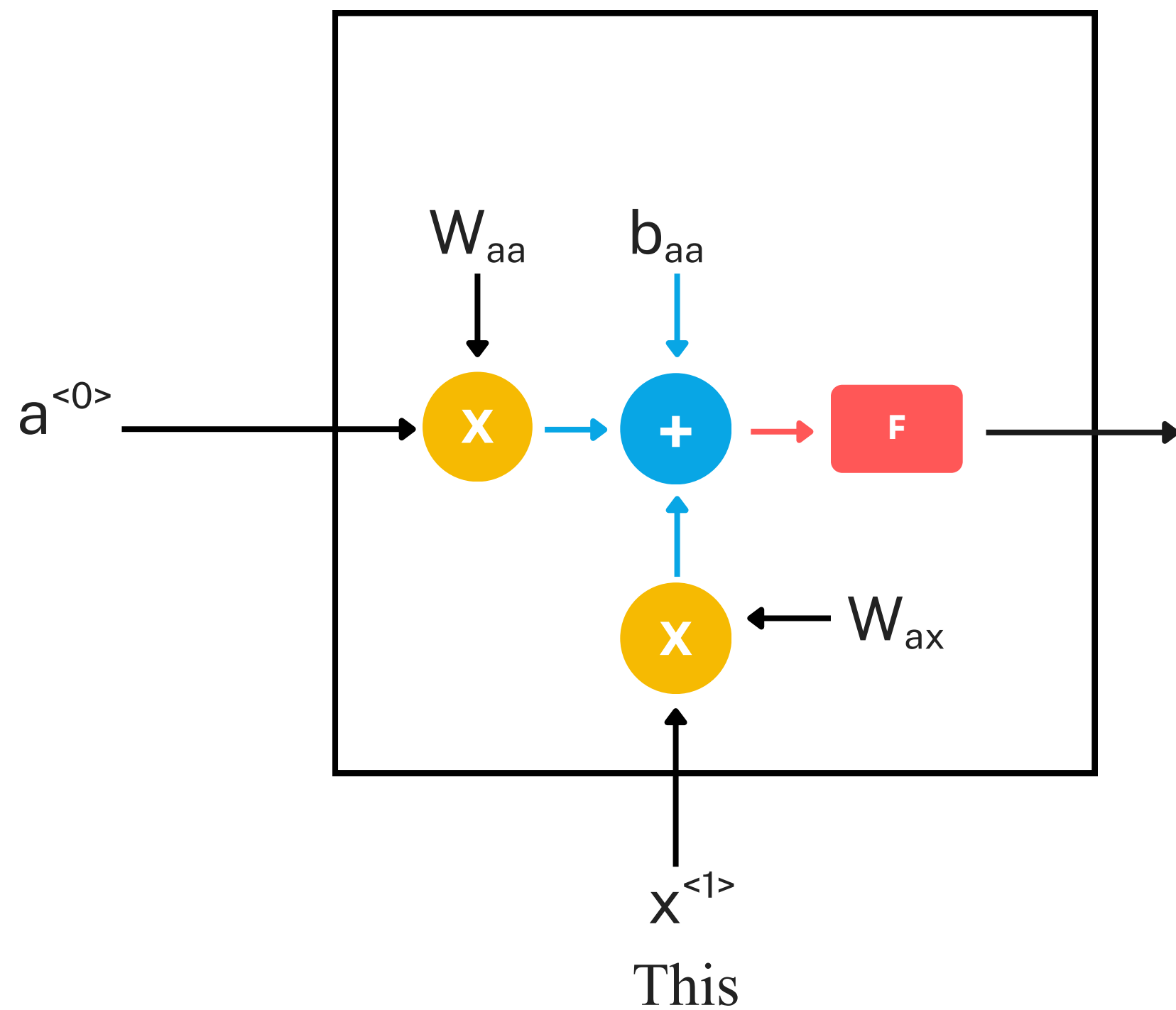


many to one



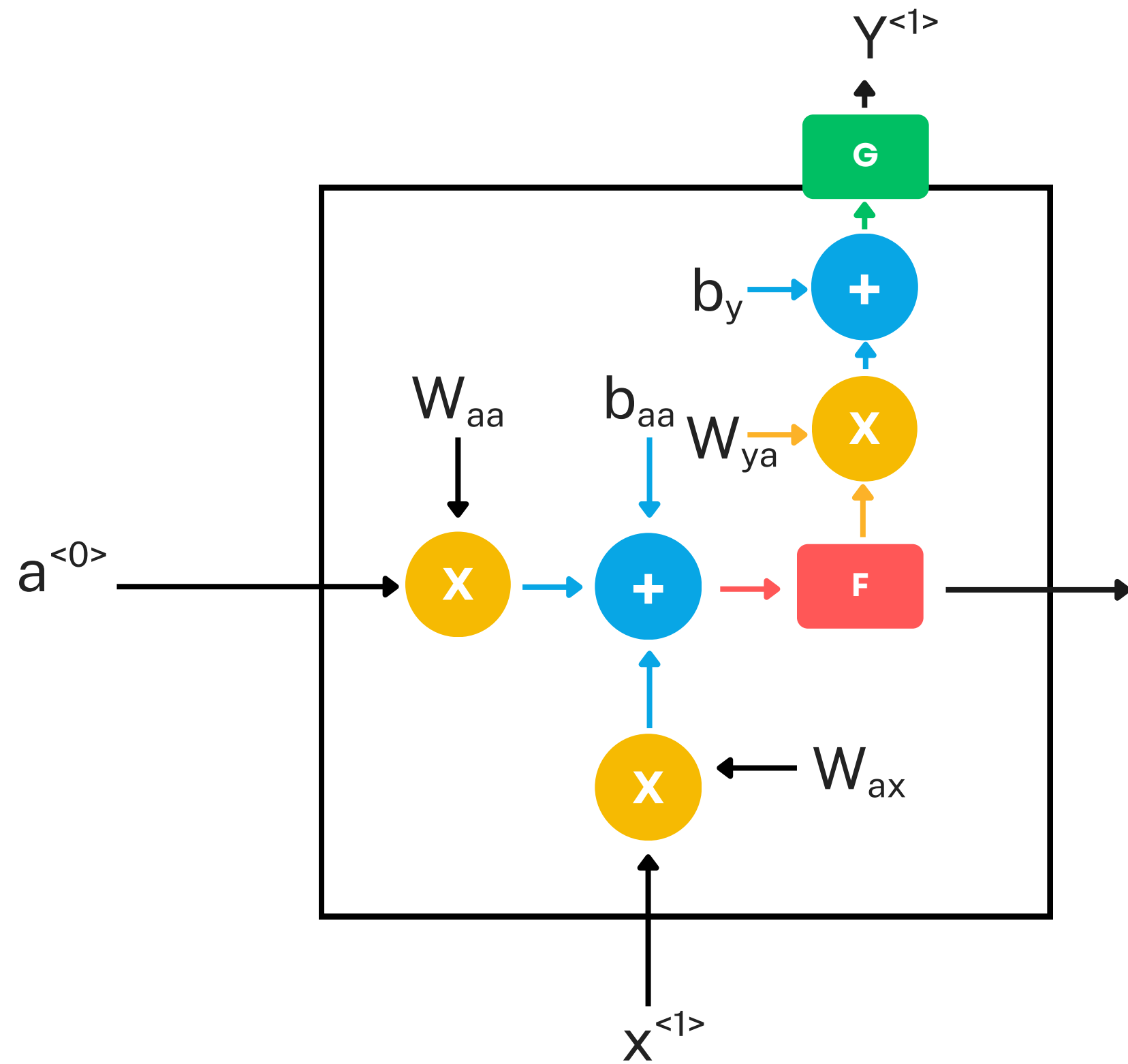
RNN Deep Dive

many to one

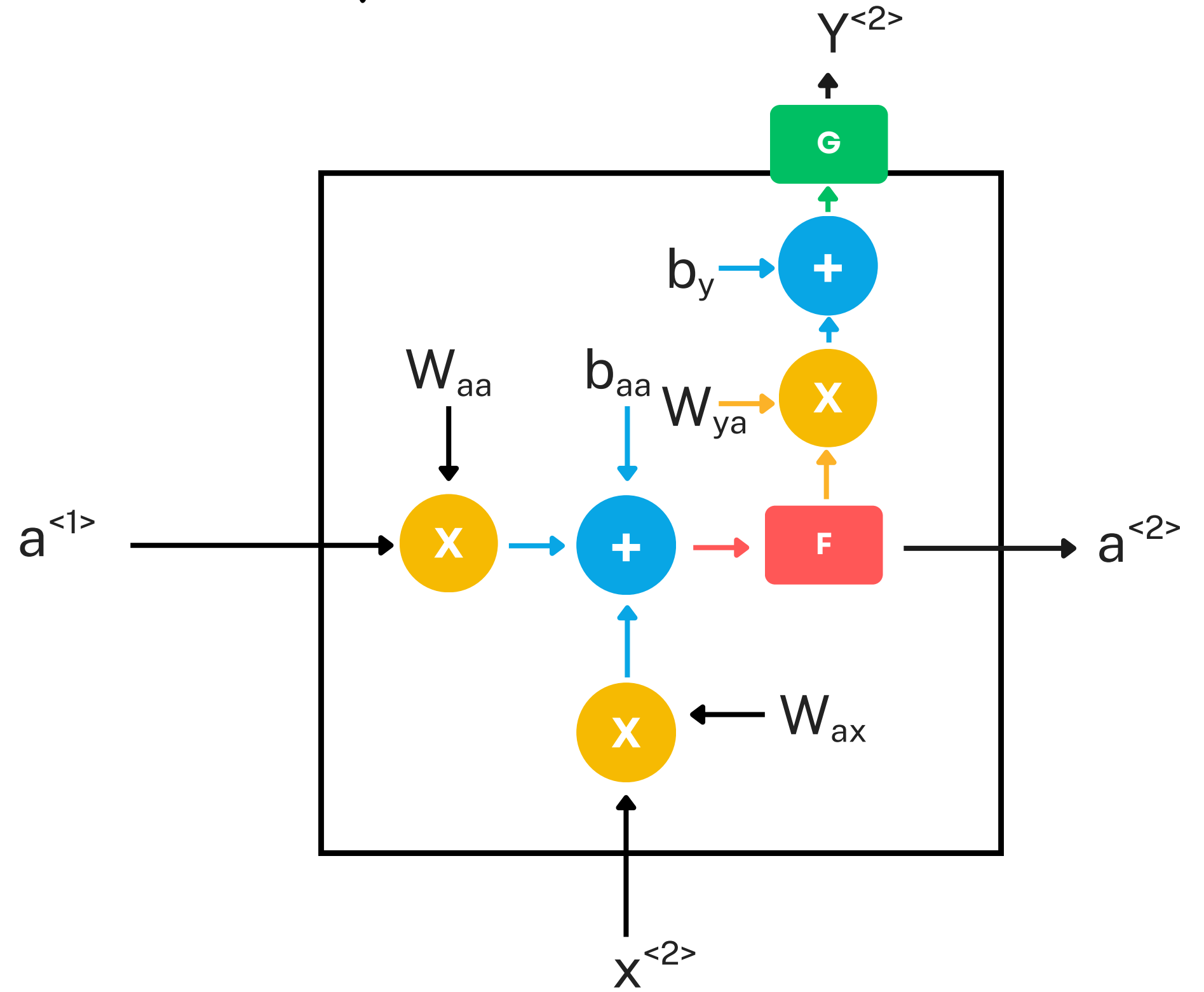


RNN Deep Dive

what if we have 2 cells in each layer?



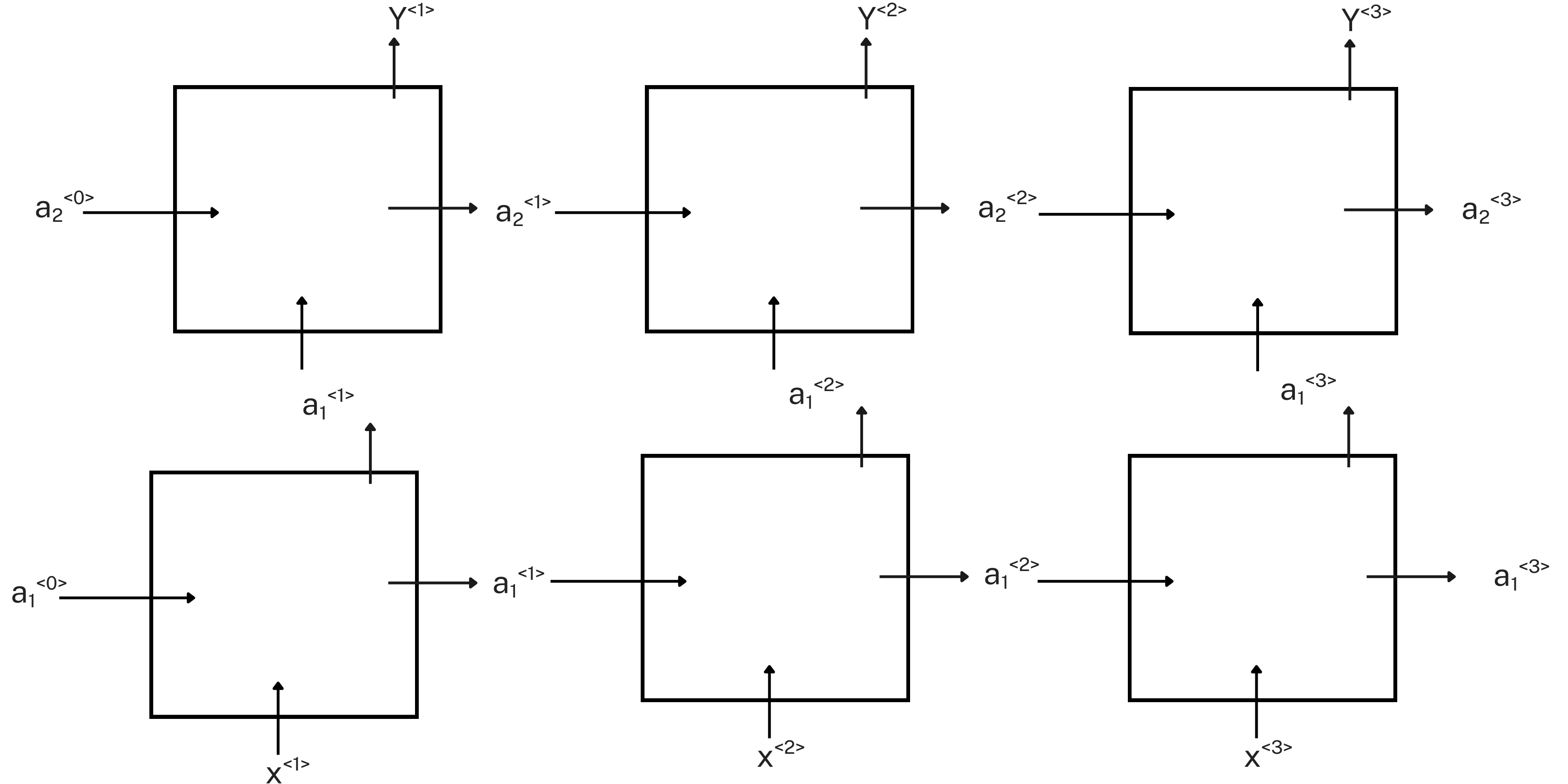
Shayan



worked

RNN Deep Dive

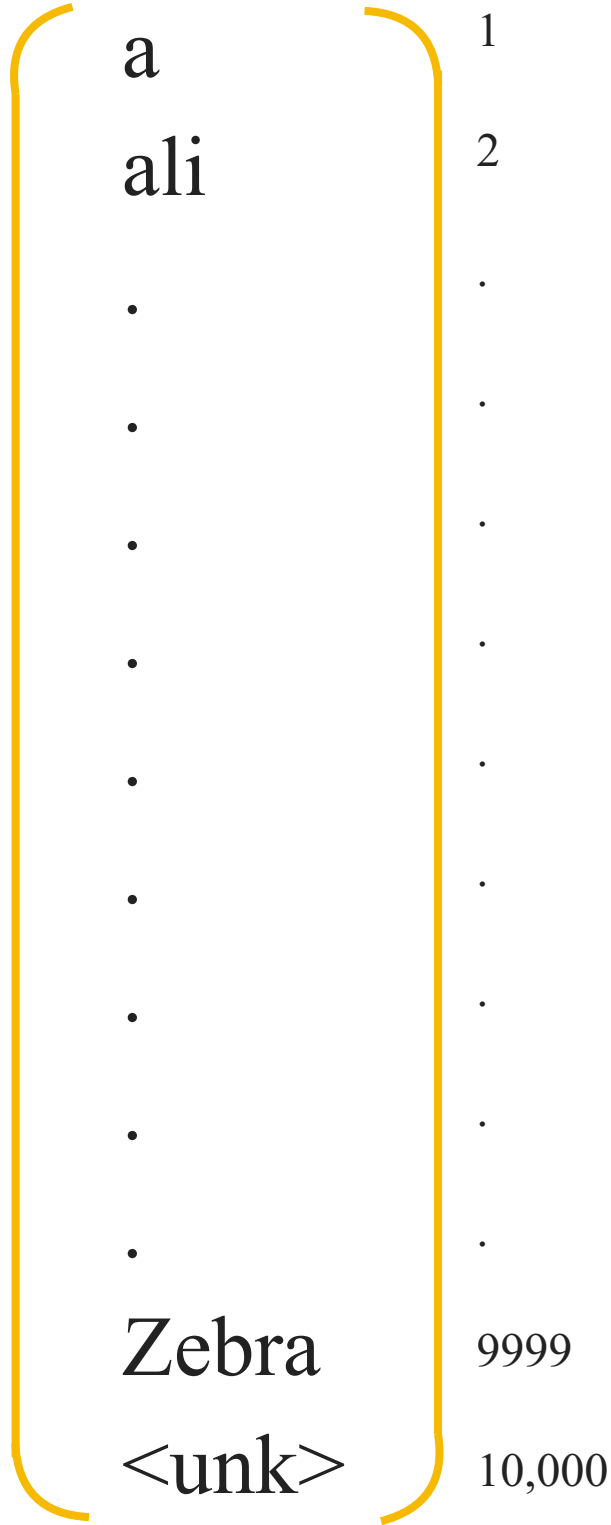
what if we have 2 cells in each layer?



Tutorial: Word Embedding

The King is Royal

Tutorial: Word Embedding



VOCABULARY

The King is Royal

Tutorial: Word Embedding

a	1
ali	2
.	.
.	.
. King	. 3456
.	.
.	.
.	.
. The	. 8989
.	.
Zebra	9999
<unk>	10,000

VOCABULARY

The King is Royal

Tutorial: Word Embedding



VOCABULARY

8989
The

3456
King

is

Royal

Tutorial: Word Embedding

a	1	0
ali	2	0
.	.	.
.	.	.
. King	. 3456	0
.	.	.
.	.	.
.	.	.
.	.	.
. The	. 8989	1
.	.	.
Zebra	9999	0
<unk>	10,000	0

VOCABULARY

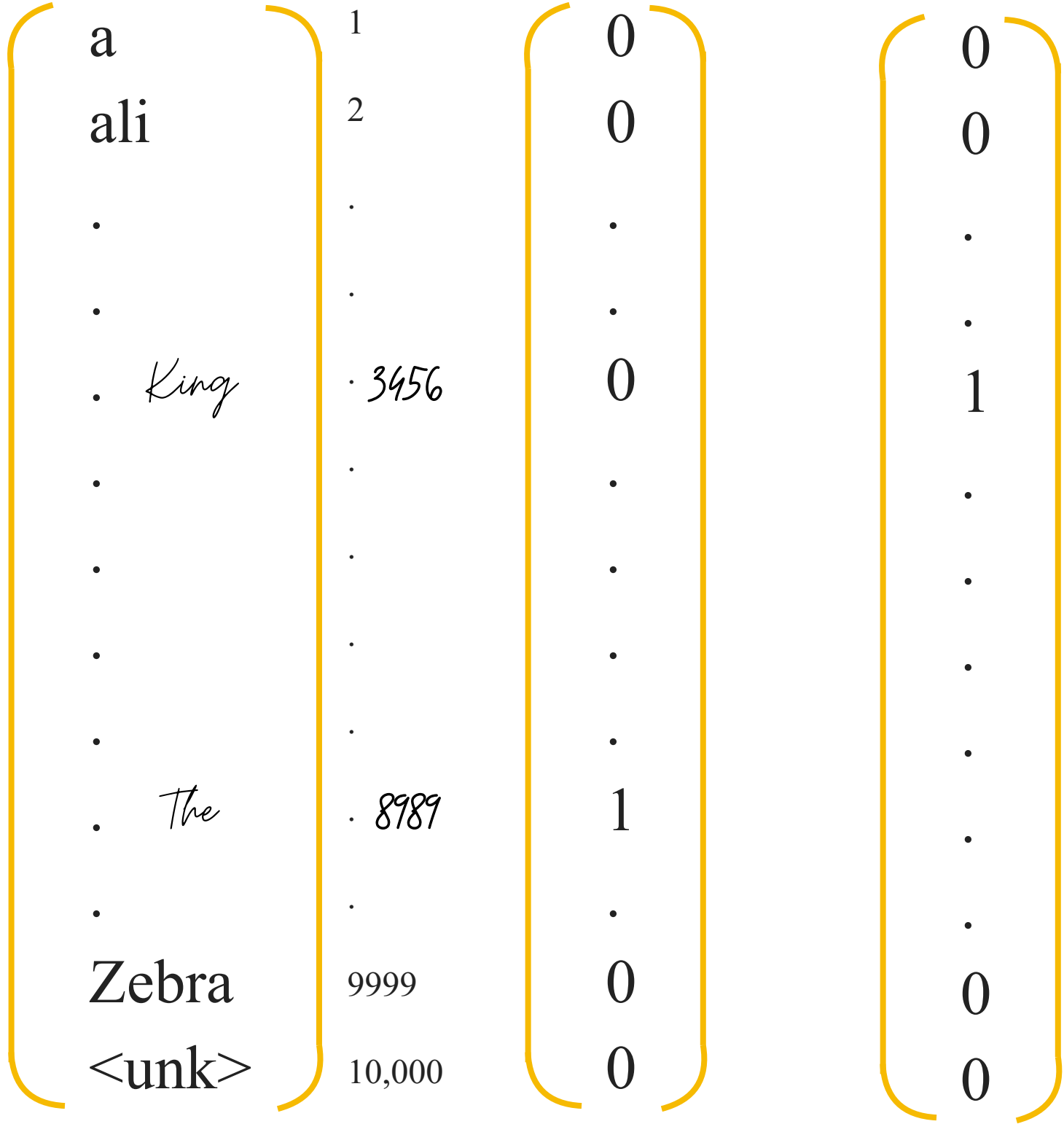
8989
The

3456
King

is

Royal

Tutorial: Word Embedding



VOCABULARY

8989
The

3456
King

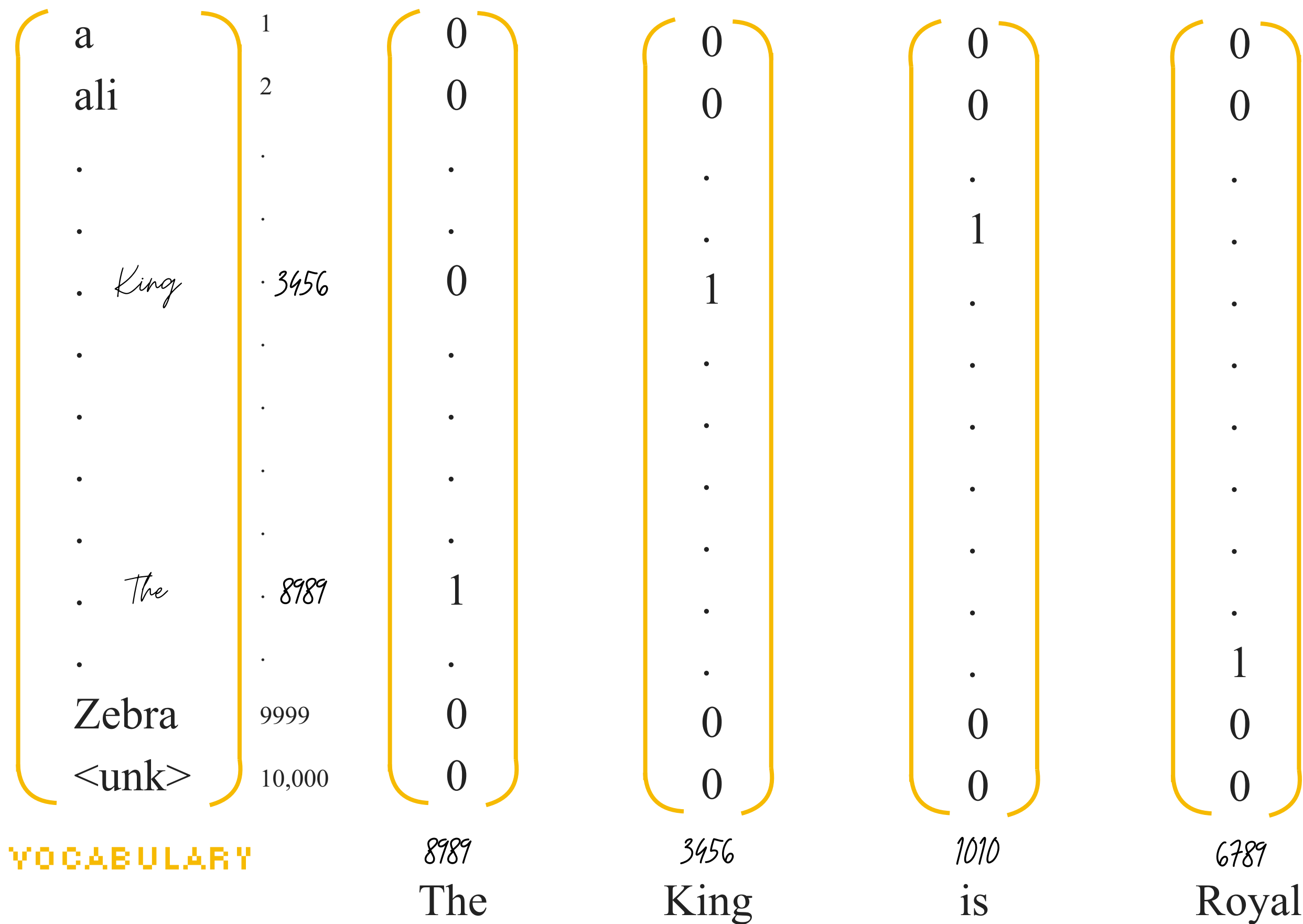
is

Royal

Tutorial: Word Embedding



Tutorial: Word Embedding



Limitation 01
No relationship between words

Tutorial: Word Embedding

Limitation 01

No relationship between words

“She is a King of the country.”

Tutorial: Word Embedding

Limitation 01

No relationship between words

“**She** is a King of the country.”

Tutorial: Word Embedding

Limitation 01

No relationship between words

He

“~~She~~ is a King of the country.”

Tutorial: Word Embedding

Limitation 01

No relationship between words

He

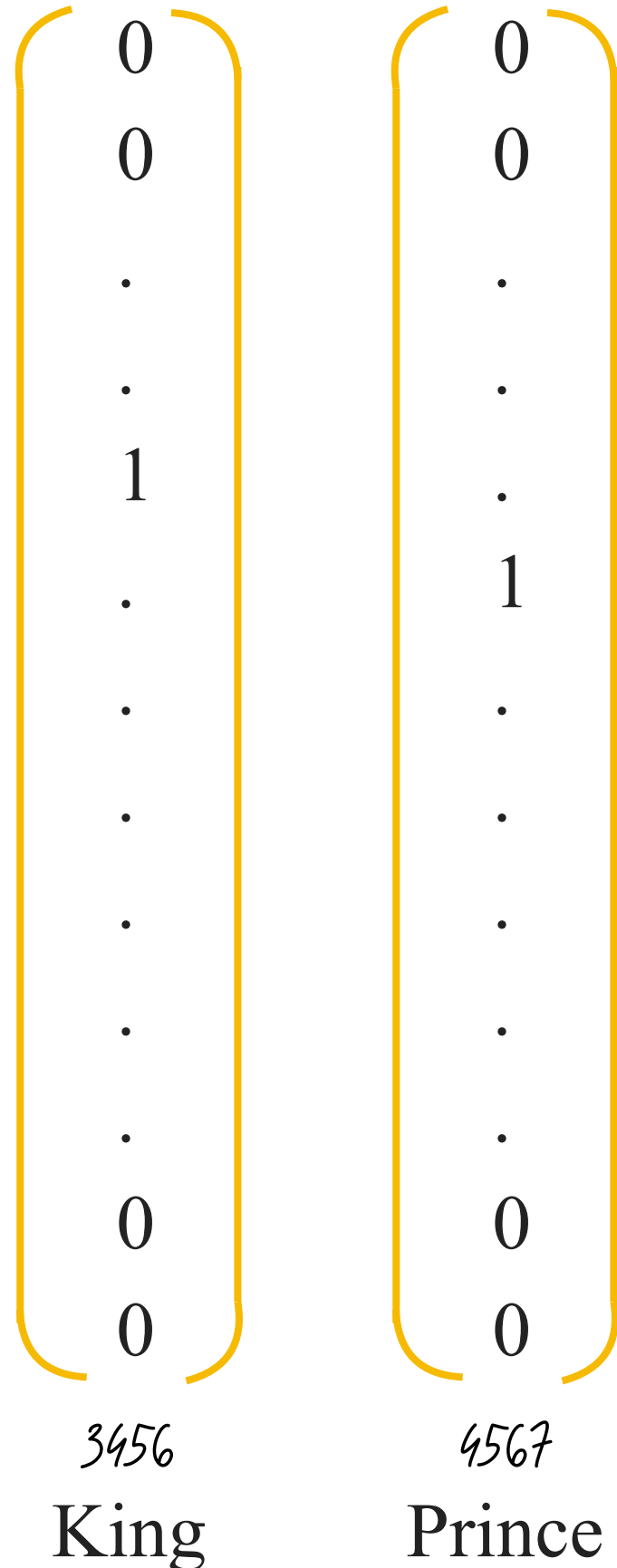
“~~She~~ is a King of the country.”

“She is a Prince of the country.”

Tutorial: Word Embedding

Limitation 01

No relationship between words



He

“~~She~~ is a King of the country.”

“She is a Prince of the country.”

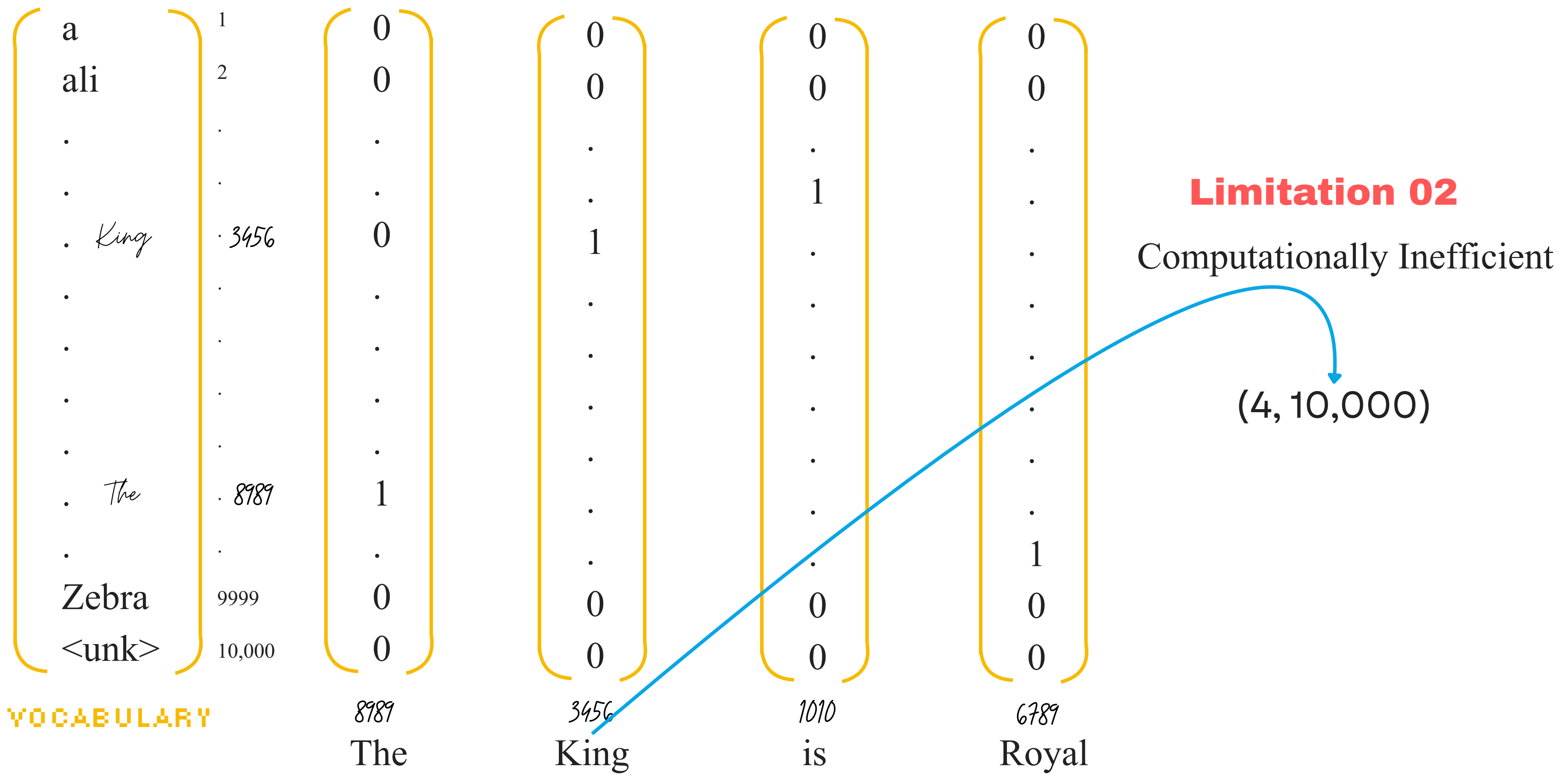
Tutorial: Word Embedding

VOCABULARY	8989 The	3456 King	1010 is	6789 Royal
a	0	0	0	0
ali	0	0	0	0
.	.	.	.	.
.	.	.	1	.
. <i>King</i>	0	1	.	.
.	.	.	.	.
.	.	.	.	.
.	.	.	.	.
. <i>The</i>	1	.	.	.
.	.	.	.	1
Zebra	0	0	0	0
<unk>	0	0	0	0

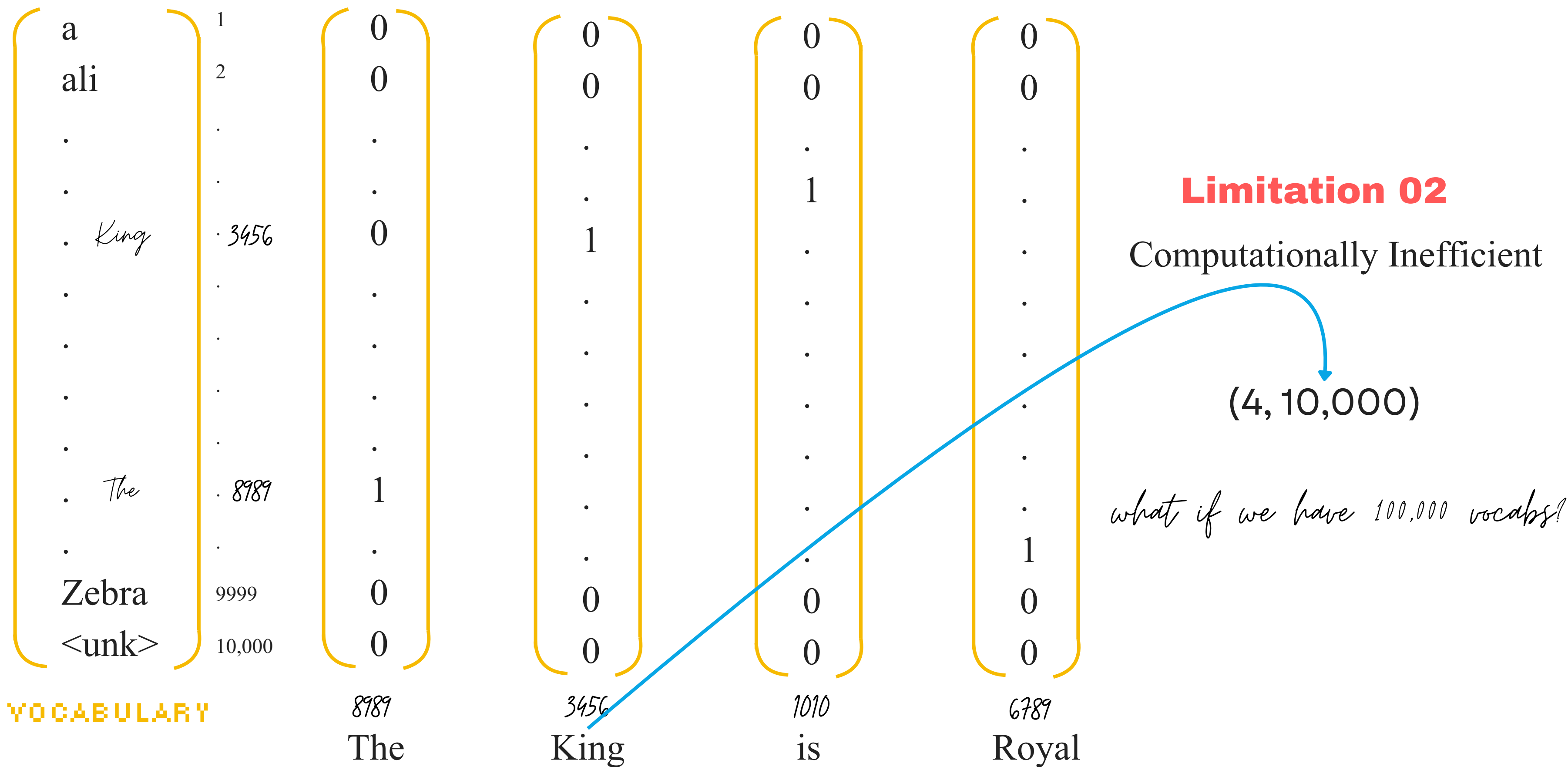
Limitation 02

Computationally Inefficient

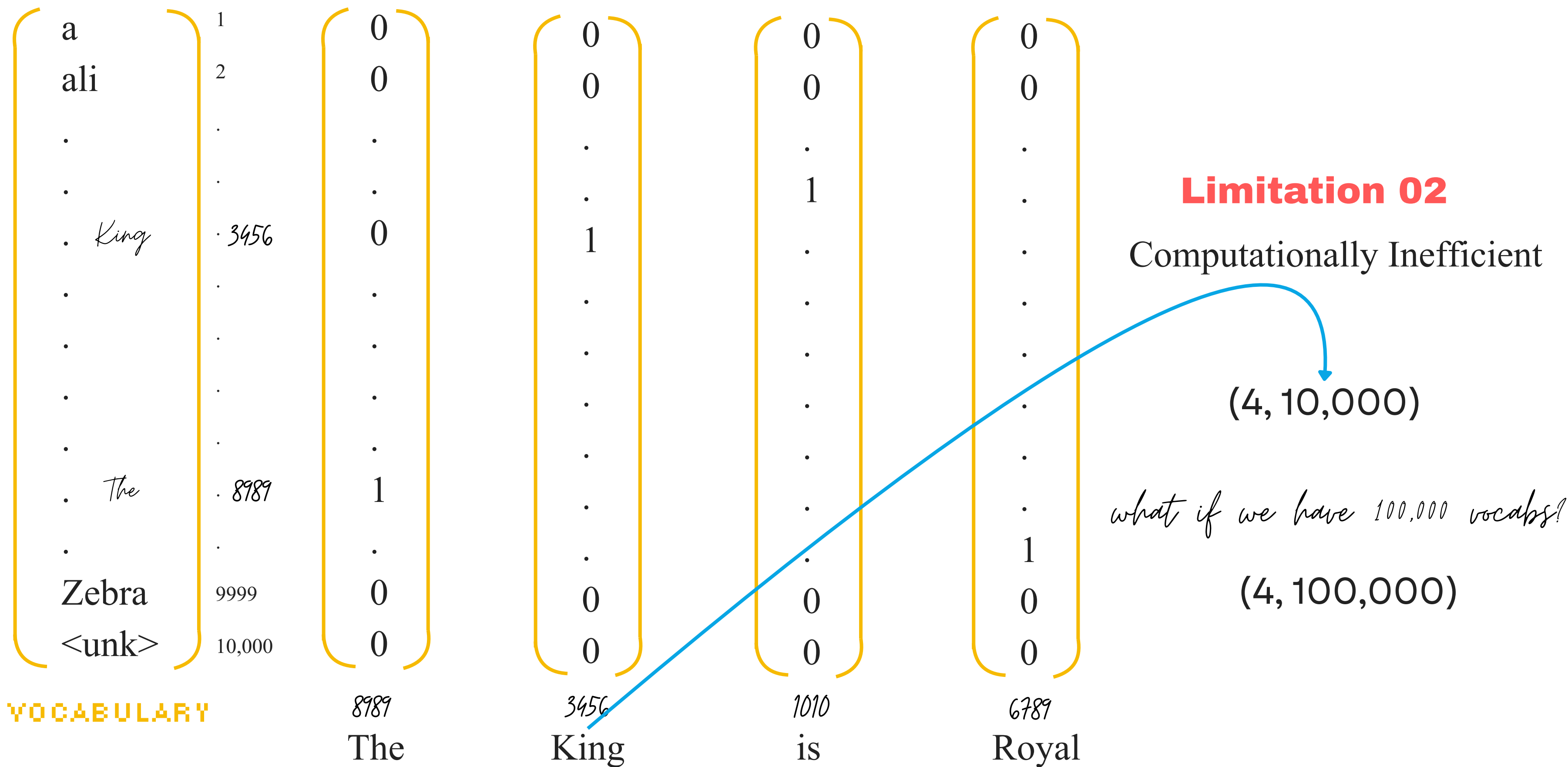
Tutorial: Word Embedding



Tutorial: Word Embedding



Tutorial: Word Embedding



Tutorial: Word Embedding

Featurized Representation: Word Embeddings

	King	Prince	Queen	Man	Woman	Apple	Orange
Gender	0.99	0.99	-0.98	1	-1	0.1	0
Age	0.75	0.4	0.7	0.7	0.6	0.1	0.1
Size	0.8	0.75	0.75	0.8	0.75	0.05	0.05
Weight	0.8	0.75	0.7	0.8	0.7	0.05	0.07
Color	0.6	0.58	0.58	0.6	0.58	-0.4	-0.2
Is_royal	1	1	1	-1	-1	-0.95	-0.99
Is_food	-1	-1	-1	-1	-1	1	1

Tutorial: Word Embedding

Featurized Representation: Word Embeddings

Man - Woman : King - ?

Tutorial: Word Embedding

Featurized Representation: Word Embeddings

Man - Woman : King - **Queen**

Tutorial: Word Embedding

Featurized Representation: Word Embeddings

Man - Woman : King - **Queen**

King
0.99
0.75
0.8
0.8
0.6
1
-1

Tutorial: Word Embedding

Featurized Representation: Word Embeddings

Man - Woman : King - **Queen**

King		Man
0.99		1
0.75		0.7
0.8		0.8
0.8	-	0.8
0.6		0.6
1		-1
-1		-1

Tutorial: Word Embedding

Featurized Representation: Word Embeddings

Man - Woman : King - **Queen**

King		Man		Woman
0.99		1		-1
0.75		0.7		0.6
0.8		0.8		0.75
0.8	-	0.8	+	0.7
0.6		0.6		0.58
1		-1		-1
-1		-1		-1

Tutorial: Word Embedding

Featurized Representation: Word Embeddings

Man - Woman : King - **Queen**

King		Man		Woman		
0.99		1		-1		-1.01
0.75		0.7		0.6		0.65
0.8		0.8		0.75		0.75
0.8	-	0.8	+	0.7	=	0.7
0.6		0.6		0.58		0.58
1		-1		-1		1
-1		-1		-1		-1

Tutorial: Word Embedding

Featurized Representation: Word Embeddings

Man - Woman : King - **Queen**

King		Man		Woman				Queen
0.99		1		-1		-1.01		-0.98
0.75		0.7		0.6		0.65		0.7
0.8		0.8		0.75		0.75		0.75
0.8	-	0.8	+	0.7	=	0.7		0.7
0.6		0.6		0.58		0.58		0.58
1		-1		-1		1		1
-1		-1		-1		-1		-1

Tutorial: Word Embedding

Featurized Representation: Word Embeddings

Islamabad - Pakistan : Washington - ?

Tutorial: Word Embedding

Featurized Representation: Word Embeddings

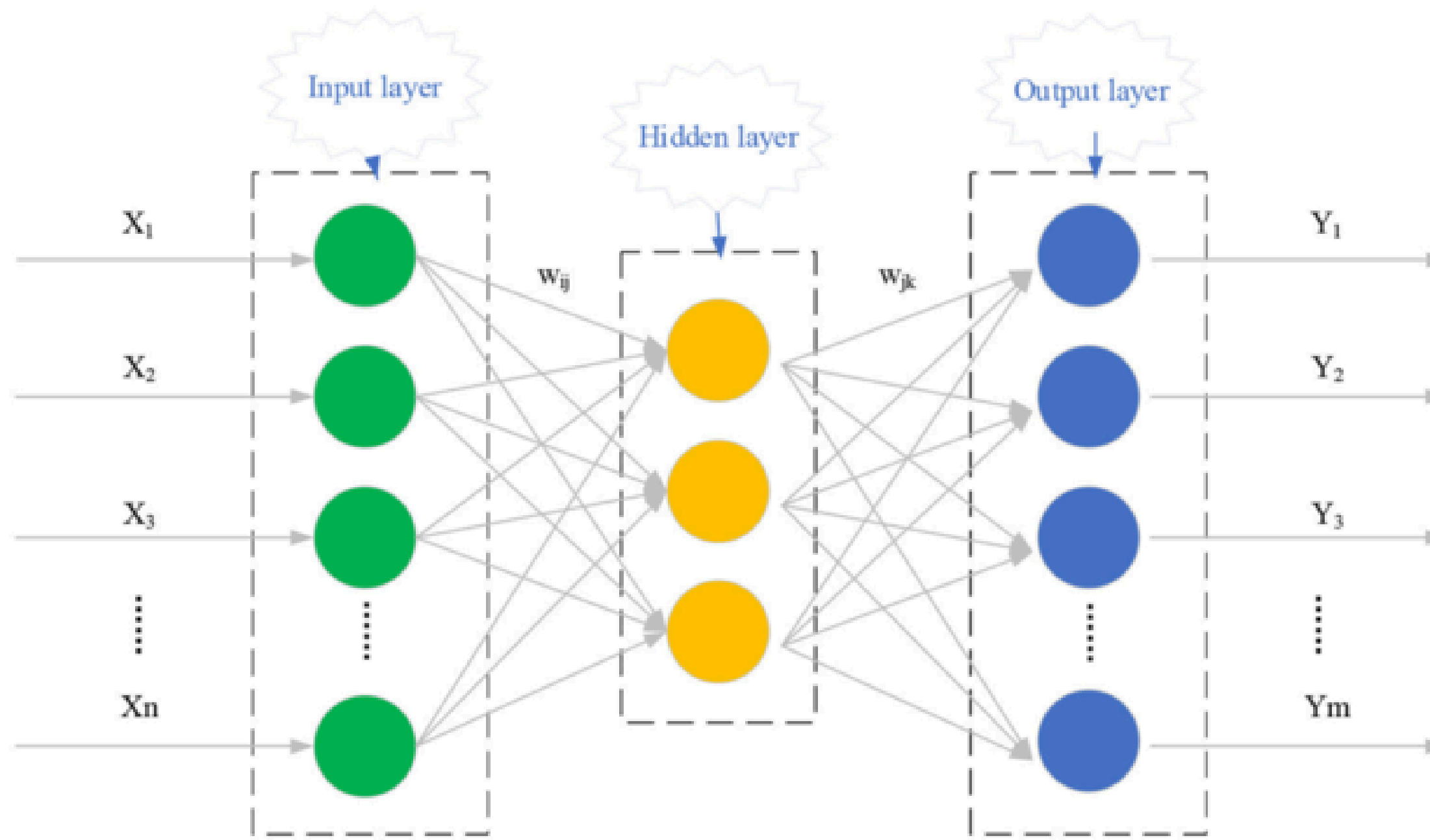
Islamabad - Pakistan : Washington - **USA**

Tutorial: Word Embedding

Featurized Representation: Word Embeddings



Tutorial: Word Embedding



Tutorial: Word Embedding

The future king is the prince

Daughter is the princess

Son is the prince

Only a man can be a king

Only a woman can be a queen

The princess will be a queen

The prince is a strong man

The princess is a beautiful woman

Prince is only a boy now

Prince will be king

A boy will be a man

Tutorial: Word Embedding

Source



The future king is the prince
Daughter is the princess
Son is the prince
Only a man can be a king
Only a woman can be a queen
The princess will be a queen
The prince is a strong man
The princess is a beautiful woman
Prince is only a boy now
Prince will be king
A boy will be a man

Tutorial: Word Embedding

Source Target

The future king is the prince
Daughter is the princess
Son is the prince
Only a man can be a king
Only a woman can be a queen
The princess will be a queen
The prince is a strong man
The princess is a beautiful woman
Prince is only a boy now
Prince will be king
A boy will be a man

Tutorial: Word Embedding

Remove Stop Words

```
future king prince  
Daughter princess  
Son prince  
man king  
woman queen  
princess queen  
prince strong man  
princess beautiful woman  
Prince only boy  
Prince king  
boy man
```

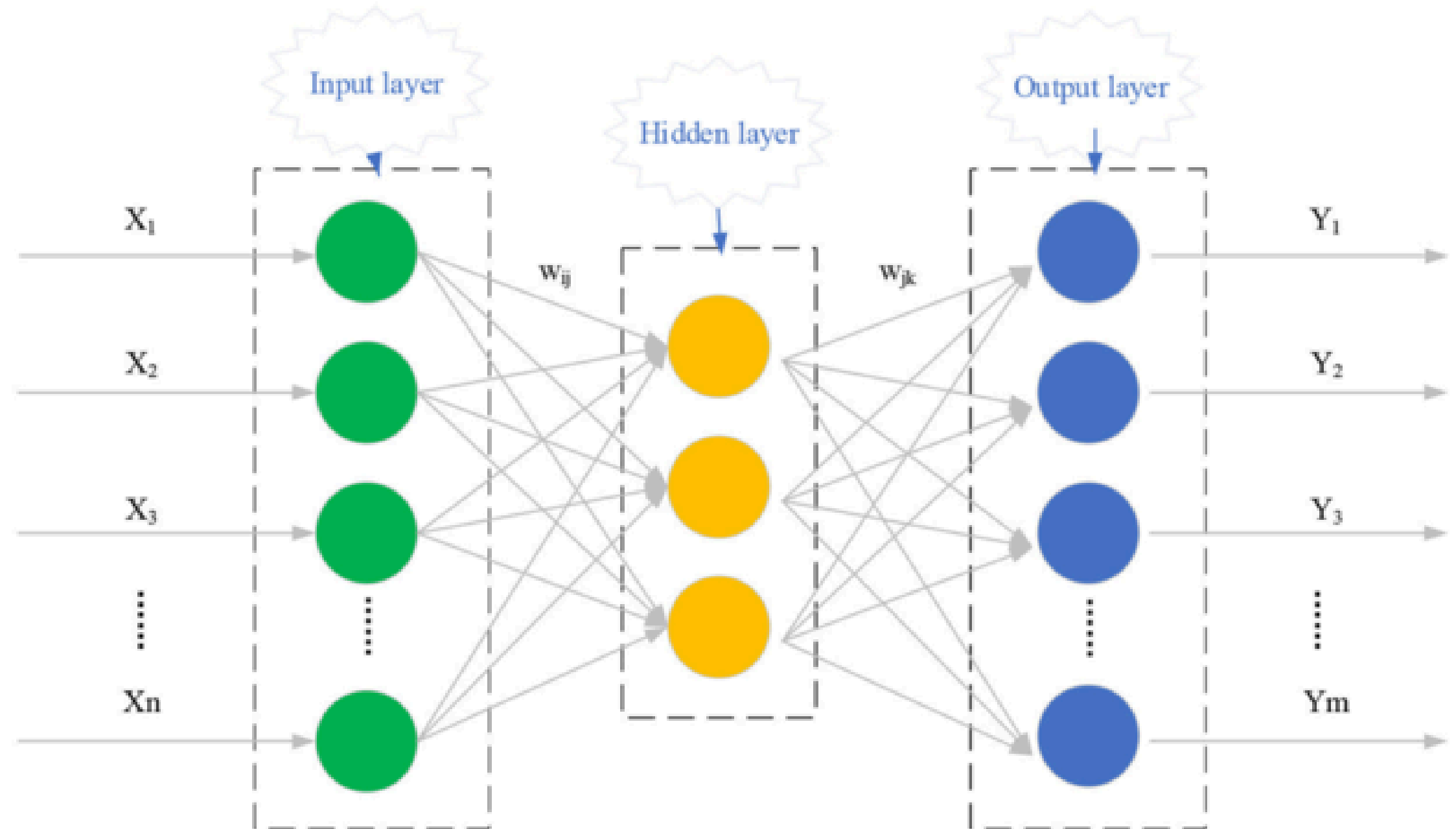
Bigrams

The quick brown fox jumps over

“the quick”, “quick brown”,
“brown fox”, “fox jumps”

Tutorial: Word Embedding

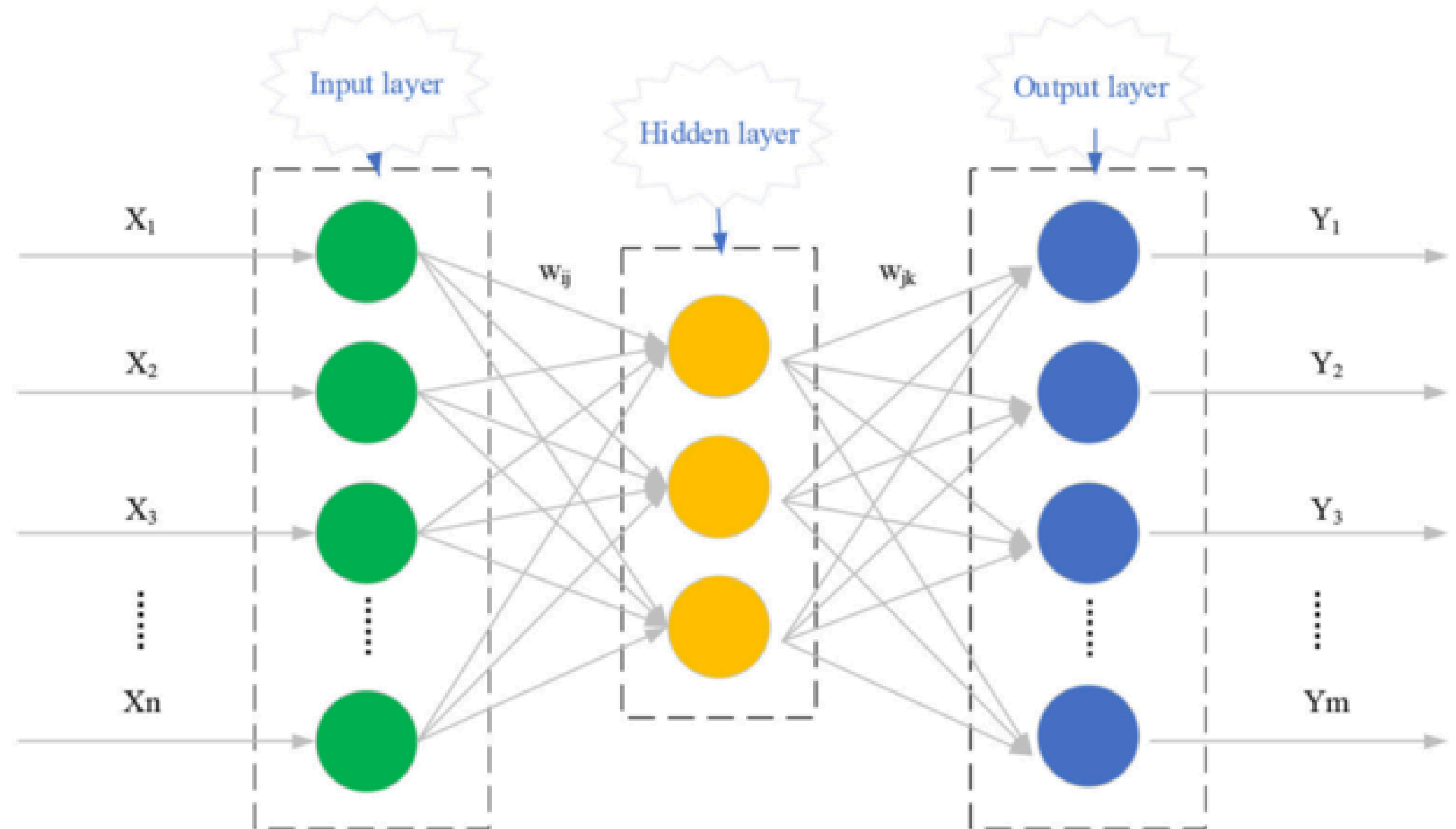
['future', 'king']
['king', 'future']
['future', 'prince']
['prince', 'future']
['king', 'prince']
['prince', 'king']
['daughter', 'princess']
['princess', 'daughter']
['son', 'prince']
['prince', 'son']
['man', 'king']
['king', 'man']
['woman', 'queen']
['queen', 'woman']
['princess', 'queen']
['queen', 'princess']
['prince', 'strong']
['strong', 'prince']
['prince', 'man']
['man', 'prince']
['strong', 'man']
['man', 'strong']
['princess', 'beautiful']
['beautiful', 'princess']



Tutorial: Word Embedding

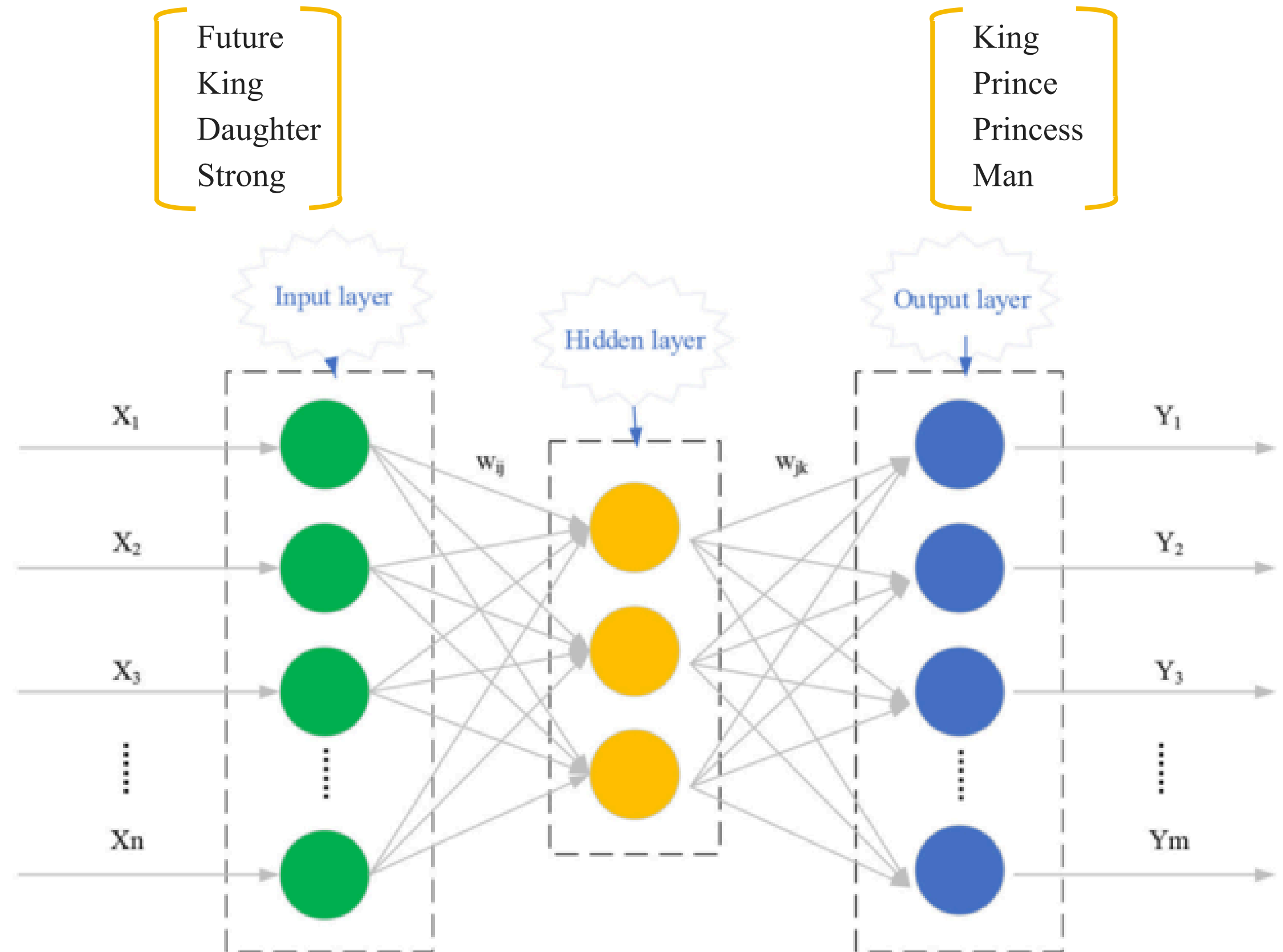
['future', 'king']
['king', 'future']
['future', 'prince']
['prince', 'future']
['king', 'prince']
['prince', 'king']
['daughter', 'princess']
['princess', 'daughter']
['son', 'prince']
['prince', 'son']
['man', 'king']
['king', 'man']
['woman', 'queen']
['queen', 'woman']
['princess', 'queen']
['queen', 'princess']
['prince', 'strong']
['strong', 'prince']
['prince', 'man']
['man', 'prince']
['strong', 'man']
['man', 'strong']
['princess', 'beautiful']
['beautiful', 'princess']

Future
King
Daughter
Strong

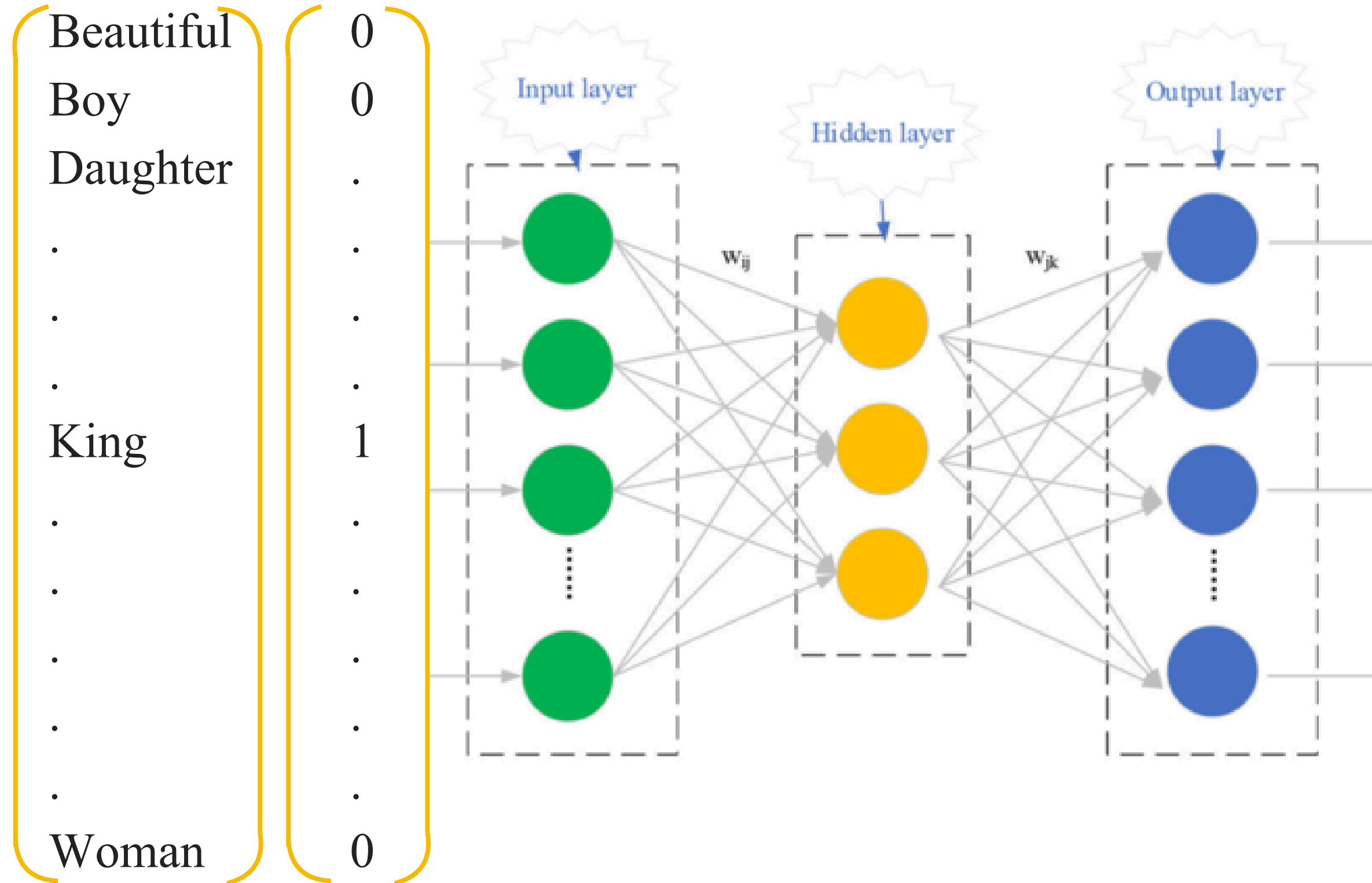


Tutorial: Word Embedding

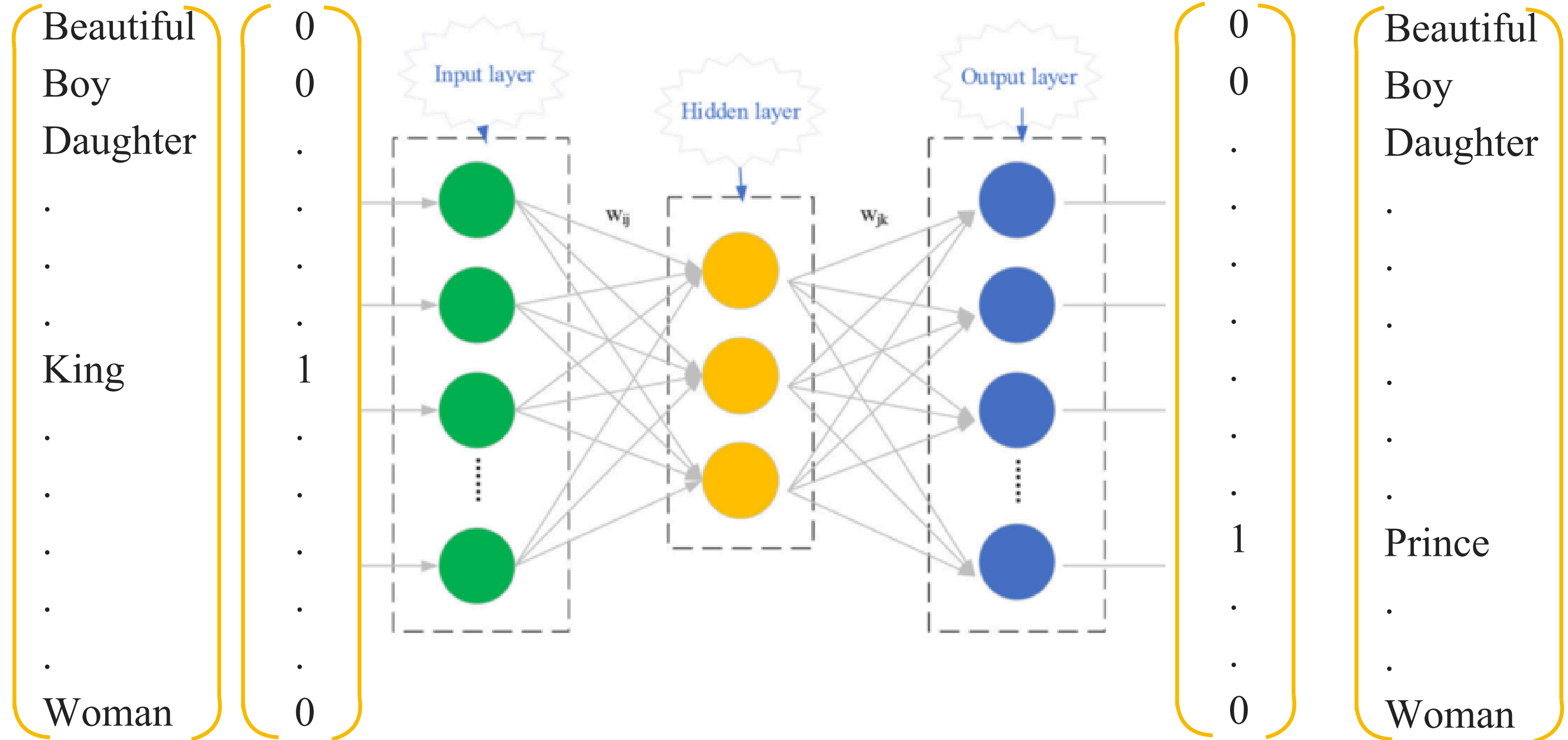
['future', 'king']
['king', 'future']
['future', 'prince']
['prince', 'future']
['king', 'prince']
['prince', 'king']
['daughter', 'princess']
['princess', 'daughter']
['son', 'prince']
['prince', 'son']
['man', 'king']
['king', 'man']
['woman', 'queen']
['queen', 'woman']
['princess', 'queen']
['queen', 'princess']
['prince', 'strong']
['strong', 'prince']
['prince', 'man']
['man', 'prince']
['strong', 'man']
['man', 'strong']
['princess', 'beautiful']
['beautiful', 'princess']



Tutorial: Word Embedding

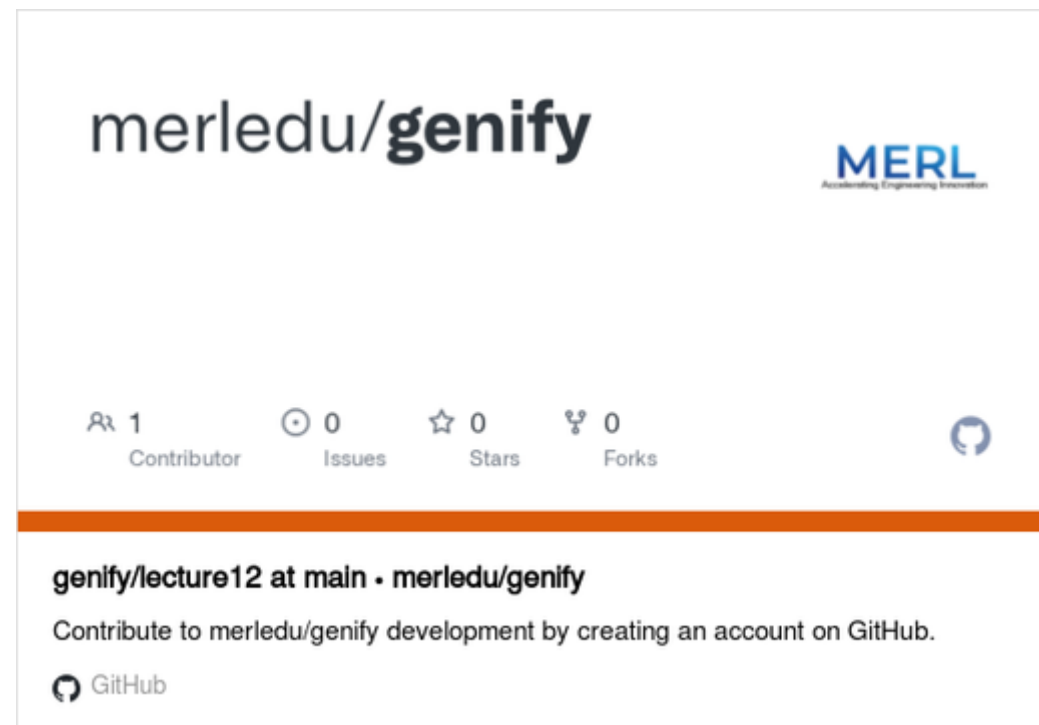


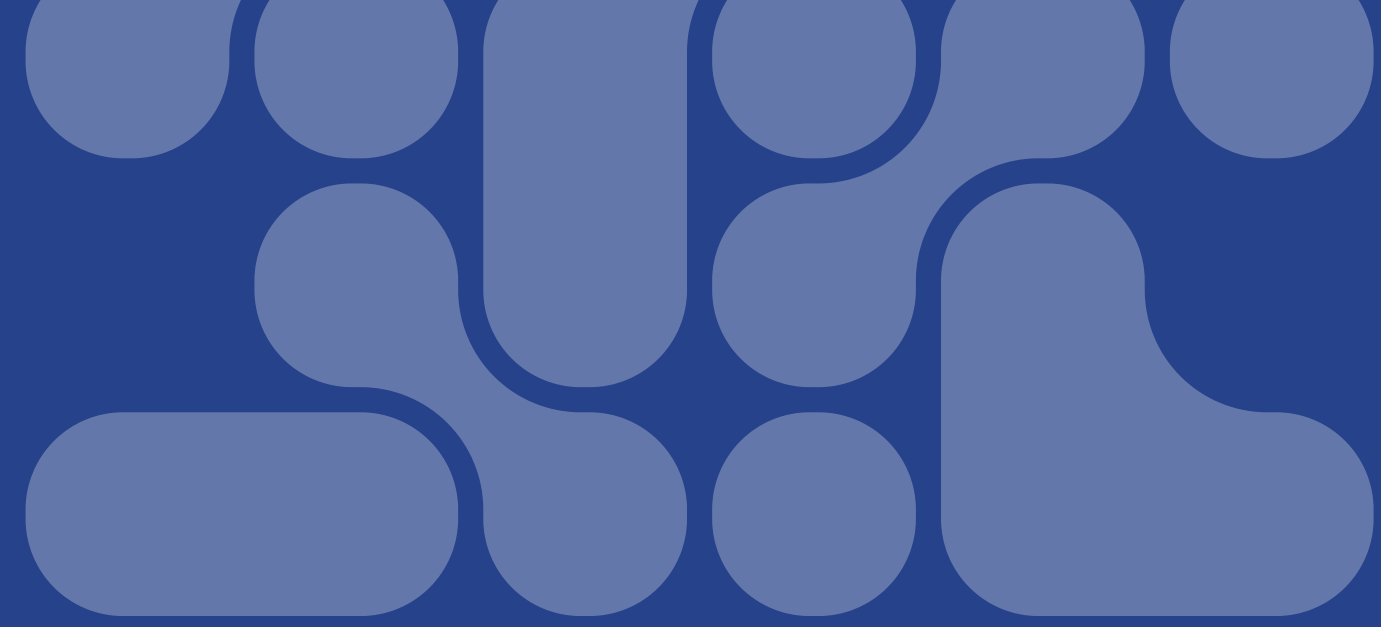
Tutorial: Word Embedding



Python Walkthrough

<https://github.com/merledu/genify/tree/main/lecture12>





**Thank you for
your attention!**

Open floor for discussion.

